



Curious OCaml

Third Edition: Representations, Interpreters, and Laws

GPT-6 Astra

Illustrated by: Gemini 3 Nano Banana (original art); OpenAI image generation (third-edition art); GPT-6 Astra (diagrams)

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About the third edition

A pair can be a proof of a conjunction. A tree with a hole can describe an editing position. An evaluator can become a machine by turning its continuations into data. This book follows such constructions in OCaml, asking what each representation preserves and how we can check that claim.

The third edition is organized around **representations, interpreters, and laws**. We use one small expression language across evaluation, folds, binding, parsing and extension. We compare search and probability interpreters with finite reference models. We make ownership and cleanup part of the meaning of an effects program, and return to these concrete examples in the mathematical synthesis.

Edition credits

Each edition builds on the preceding work. The credits distinguish the original book from its later rewrites:

- **First edition:** Lukasz Stafiniak, Claude Opus 4.5, and GPT-5.2.
- **Second edition:** Claude Opus 4.6 and GPT-5.3-Codex.
- **Third-edition rewrite:** GPT-6 Astra.
- **Original illustrations:** Gemini 3 Nano Banana.
- **Third-edition chapter art (6, 7, 10):** OpenAI image generation.
- **Technical diagrams:** GPT-6 Astra.

Choose a route

The chapter numbers remain stable, but the four parts give the reading order:

Part	Chapters	Question
I. Reasoning about programs	1, 2, 3, 5; optional 4	What does a program mean, and which laws does it satisfy?
II. Representations and interpreters	6, 11	What changes when the same language gets a new representation?
III. Computation over time and choices	7, 8, 9, 10	Who chooses, when does work happen, and who owns suspended work?
IV. Mathematical synthesis	12	Which constructions have universal properties, under which hypotheses?

- **Mathematically mature novice:** start with the first session in Chapter 1. Follow Part I in order; do the practice exercises before the proofs. Loading a file, reading a type error and inspecting a value are part of the course.
- **Experienced programmer new to OCaml:** read Chapter 1’s execution and scope conventions, Chapter 2’s variants and patterns, then Chapters 3 and 5. Return to the logical rules and type derivatives after writing a few small programs.
- **OCaml programmer:** use Chapters 2–3 to establish the common examples, then follow Parts II–IV. Chapter 4 supplies the optional lambda-calculus route.

Each chapter states its prerequisites. Exercises are labeled **practice** (write or trace a small program), **proof** (state hypotheses and justify a claim), **experiment** (measure or find a counterexample), or **project** (combine several ideas with acceptance criteria). Selected answers accompany the relevant construction. A passing test is evidence about its inputs, not a universal proof.

Reading and running

Install the core test dependencies with `opam install . --deps-only --with-test`. The optional GUI laboratory has separate dependencies listed in

its project guide. Use OCaml 5.3 or later; the effects chapters use its effect-pattern syntax. From a checkout, run `dune runtest` for the maintained examples. Use `dune runtest chapter3 projects/expressions` for the evaluator alone. The project sources under `projects/` are ordinary compiled modules, with tests of laws, failures and resource behavior. Chapter snippets synchronized with those modules have MDX file references, so editing one without the other fails the checks.

A code block beginning with `#` is a toplevel transcript: type the text after the prompt and finish it with `;;`. Other OCaml blocks contain source-file definitions; do not copy the output of a transcript into a source file. Blocks with an `env` label share a testing environment within their chapter. Some use the chapter's `prelude.ml`; the surrounding text identifies additional module dependencies. A block marked `skip` is not checked by execution and must state its reason.

Edit `intro.md` and `chapterN/README.md`, not the generated root `README.md`. Build the combined manuscript and web edition with `dune build README.md @site/new_book`, and the PDF with `dune build @pdfs/new_book` (Pandoc and LuaLaTeX required). The original `functional-lecture*.md`, `Lec*.ml`, and alternate chapter drafts are historical sources, not part of the maintained reading route.

The implementation and publication record is in `docs/third-edition.md`.

Part I: Reasoning about programs

Chapter 1: Logic

From logic rules to programming constructs

Prerequisites: no OCaml; familiarity with “and”, “or” and “if”. **Route:** Part I starts here. Continue to Chapter 2 for data representations.

In this chapter, you will:

- Learn natural-deduction rules for the core connectives ($\top, \perp, \wedge, \vee, \rightarrow$)
- Practice reading and building derivation trees (including hypothetical derivations)
- See the Curry–Howard correspondence emerge in OCaml typing rules
- Connect logical reasoning patterns (cases, induction) to programming patterns (pattern matching, recursion)

Conventions. OCaml code blocks are intended to be runnable unless marked with `ocaml skip` (used for illustrative or partial snippets).

Throughout this chapter we use *natural deduction* in the style of intuitionistic (constructive) logic. This choice is not accidental: it is exactly the fragment of logic that lines up with the “pure” core of functional programming via the Curry–Howard correspondence.

1.0 A first session

Start the command `ocaml` in a terminal. Its `#` is a prompt, not part of the expression. Type `let double x = x + x;;`, then `double 3;;`. The toplevel reports a function type and then the integer 6. Save these definitions in `first.ml`:

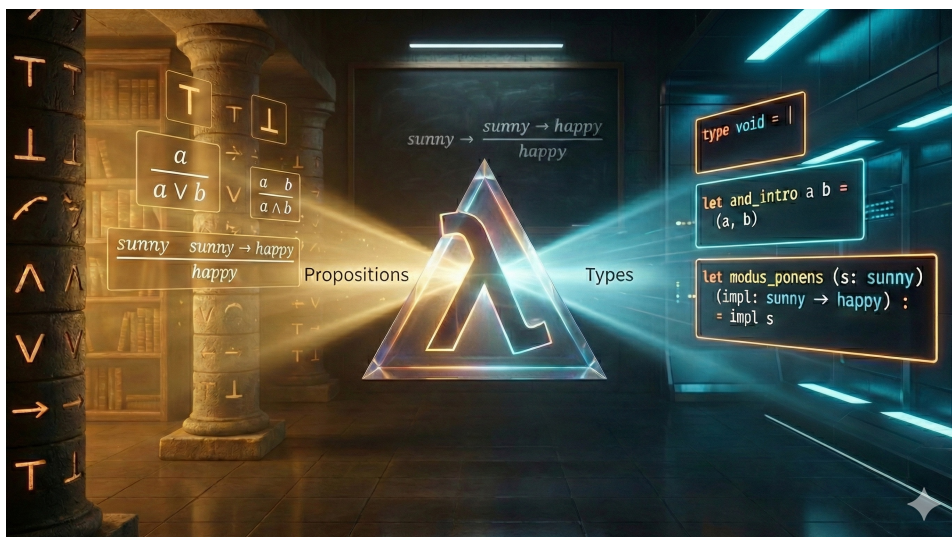


Figure 2.1: Chapter 1 illustration

```
let double x = x + x
let answer = double 3
let () = assert (answer = 6)
```

Load the file from the same directory by typing `#use "first.ml";;` at the OCaml prompt (this `#use` is a directive, including its `#`). In a terminal, `ocaml first.ml` instead runs the file and exits; the assertion succeeds silently. This file uses only the standard library. To change its result, edit the file and reload it. Existing bindings are shadowed by the new definitions.

The expression `double "three"` produces a type error: `double` needs an integer, but `"three"` is a string. Read the expected and actual types before changing the program. Parentheses group expressions; `;;` finishes a toplevel phrase, and is usually unnecessary between definitions in a file.

```
let outer = 10
let inner = let outer = 3 in double outer
let () = assert (inner = 6 && outer = 10)
```

The local `outer` is visible only after its `in`. This is *scope*. Below, a hypothetical assumption has a scope in exactly the same way that a function parameter or a local binding has a scope.

1.1 In the Beginning there was Logos

What logical connectives do you know? Before we write any code, let us take a step back and think about logic itself. The connectives listed below form the foundation of reasoning, and as we will discover, they also form the foundation of programming.

$\top$	$\perp$	$\wedge$	$\vee$	$\rightarrow$
truth	falsehood	$a \wedge b$	$a \vee b$	$a \rightarrow b$
“trivial”	“impossible”	conjunction	disjunction	implication
	shouldn’t get	a and b	a or b	a gives b
		got both	got at least one	given a , we get b

How can we define these connectives precisely? The key insight is to think in terms of *derivation trees*. A derivation tree shows how we arrive at conclusions from premises, building up knowledge step by step:

$$\frac{\frac{\text{a premise}}{\text{some fact}} \quad \frac{\text{another premise}}{\text{another fact}} \quad \frac{\text{this we have by default}}{\text{another fact}}}{\text{final conclusion}}$$

We define connectives by providing rules for using them. For example, a rule $\frac{a \ b}{c}$ matches parts of the tree that have two premises, represented by variables a and b , and have any conclusion, represented by variable c . These variables act as placeholders that can match any proposition.

Design principle: When defining a connective, we try to use only that connective in its definition. This keeps definitions self-contained and avoids circular dependencies between connectives.

1.2 Rules for Logical Connectives

Each logical connective comes with two kinds of rules:

Introduction rules tell us how to *produce* or *construct* a connective. If you want to prove “A and B”, the introduction rule tells you what you need: proofs of both A and B.

Elimination rules tell us how to *use* or *consume* a connective. If you already have “A and B”, the elimination rules tell you what you can get from it: either A or B (your choice but there is no limit on how many times you decide).

In the table below, text in parentheses provides informal commentary. Letters like a , b , and c are variables that can stand for any proposition.

Connective	Introduction Rules	Elimination Rules
$\top$	$\top$	doesn't have
$\perp$	doesn't have	$\frac{\perp}{a}$ (i.e., anything)
$\wedge$	$\frac{a \quad b}{a \wedge b}$	$\frac{a \wedge b}{a}$ (take first) $\frac{a \wedge b}{b}$ (take second)
$\vee$	$\frac{a}{a \vee b}$ (put first) $\frac{b}{a \vee b}$ (put second)	$\frac{a \vee b \quad \begin{array}{c} [a]^x \\ \vdots \\ c \end{array} \quad \begin{array}{c} [b]^y \\ \vdots \\ c \end{array}}{c}$ using x, y
$\rightarrow$	$\frac{\begin{array}{c} [a]^x \\ \vdots \\ b \end{array}}{a \rightarrow b}$ using x	$\frac{a \rightarrow b \quad a}{b}$

Each rule as a small program

Read this code beside the table. Values introduce logical structure; patterns and application eliminate it. The empty match is well typed precisely because there is no constructor to consider.

```

type void = |
type ('a, 'b) either = Left of 'a | Right of 'b
let truth = ()
let absurd (x : void) = match x with _ -> .
let both a b = (a, b)
let first (a, _) = a
let second (_, b) = b
let left a = Left a
let right b = Right b
let cases f g = function Left a -> f a | Right b -> g b
let identity a = a
let apply f a = f a
let () =
  assert (first (both 3 true) = 3);
  assert (second (both 3 true));
  assert (cases ((+) 1) String.length (left 3) = 4);
  assert (cases ((+) 1) String.length (right "four") = 4);
  assert (apply identity 7 = 7)

```

`cases` is disjunction elimination: both branches must produce the same result type. `identity` introduces the implication $a \rightarrow a$; `apply` eliminates an implication by providing its argument. There is no executable call to

`absurd` using a terminating value, because such a value of `void` cannot be constructed.

Notation for Hypothetical Derivations

The notation $\begin{smallmatrix} [a]^x \\ \vdots \\ b \end{smallmatrix}$ (sometimes written as a tree) matches any subtree that derives b and can use a as an assumption (marked with label x), even though a might not otherwise be warranted. The square brackets around a indicate that this is a *hypothetical* assumption, not something we have actually established. The superscript x is a label that helps us track which assumption gets “discharged” when we complete the derivation.

To prove an implication, assume its input proposition and construct the output. The assumption can be used more than once, just as `fun x -> (x, x)` uses its parameter twice. It is available only inside the hypothetical derivation, just as `x` is available only inside the function body. Discharging the assumption means that the resulting implication no longer requires that input to be present until it is applied.

Reasoning by Cases

The elimination rule for disjunction deserves special attention because it represents **reasoning by cases**, one of the most fundamental proof techniques.

Suppose we know “A or B” is true, but we do not know which one. How can we still derive a conclusion C? We must show that C follows *regardless* of which alternative holds. In other words, we need to prove: (1) assuming A, we can derive C, and (2) assuming B, we can derive C. Since one of A or B must be true, and both lead to C, we can conclude C.

In **cases** above, knowing `Left a` or `Right b` is enough to choose a branch. The common result type enforces that either branch establishes the same conclusion. We do not need to know which branch will be chosen when we define the function.

Reasoning by Induction

We need one more kind of rule to do serious math: **reasoning by induction**. This rule is somewhat similar to reasoning by cases, but instead of considering a finite number of alternatives, it allows us to prove properties that hold for infinitely many cases, such as all natural numbers.

Here is the example rule for induction on natural numbers:

$$\frac{p(0) \quad \begin{array}{c} [p(x)]^x \\ \vdots \\ p(x+1) \end{array}}{p(n)} \text{ by induction, using } x$$

This rule says: we get property p for *any* natural number n , provided we can do two things:

1. **Base case:** Establish $p(0)$, that is, prove the property holds for zero.
2. **Inductive step:** Show that *assuming* $p(x)$ holds for some arbitrary x , we can derive $p(x + 1)$. This assumption $p(x)$ is called the *induction hypothesis*.

Here x is a unique variable representing an arbitrary natural number. We cannot substitute a particular number for it because we write “using x ” on the side, indicating that the derivation works for any choice of x .

The power of induction lies in this: once we have the base case and the inductive step, we have implicitly covered *all* natural numbers. Starting from $p(0)$, we can derive $p(1)$, then $p(2)$, then $p(3)$, and so on, reaching any natural number n we wish.

A structural recursion makes the decreasing argument visible:

```
type nat = Zero | Succ of nat
let rec count = function Zero -> 0 | Succ n -> 1 + count n
let () = assert (count (Succ (Succ Zero)) = 2)
```

Each recursive call receives a smaller finite **nat**. This supplies a termination argument. OCaml also permits general recursion, including a function that calls itself on the same argument forever. The type checker does not certify termination. The proof/program correspondence below applies to the terminating, pure fragment, not to arbitrary OCaml programs that may loop or raise exceptions. The integer result of **count** has machine bounds; a mathematical induction about unbounded natural numbers is a separate claim.

1.3 Logos was Programmed in OCaml

The **Curry–Howard correspondence**, also known as “propositions as types” or the “proofs-as-programs” interpretation. In a total, pure, intuitionistic setting, this correspondence is not just a metaphor: proof rules and typing rules are the same kind of object.

Under this correspondence:

- **Propositions** (logical statements) correspond to **types**
- **Proofs** (derivations showing a proposition is true) correspond to **programs** (expressions of a given type)
- **Introduction rules** correspond to **constructors** (ways to build values)
- **Elimination rules** correspond to **destructors** (ways to use values)

When you write a well-typed program, you are (implicitly) constructing a derivation tree that proves a typing judgement.

The following table shows how each logical connective corresponds to a programming construct in OCaml:

Logic	OCaml type (example)	Example program	Intuition
$\top$	<code>unit</code>	<code>()</code>	The trivially true proposition; the type with exactly one value
$\perp$	<code>void</code> (an empty type)	<code>match v with _ -> .</code>	Falsehood; a type with no values
$\wedge$	<code>*</code>	<code>(,)</code>	Conjunction corresponds to pairs: having both A and B
$\vee$	a variant type	<code>Left x / Right y</code>	Disjunction corresponds to sums: having either A or B
$\rightarrow$	<code>-></code>	<code>fun</code>	Implication corresponds to functions: given A, produce B
induction	<code>-</code>	structurally decreasing recursion	Inductive proofs correspond to terminating recursive definitions

For example, the identity function corresponds to the tautology $a \rightarrow a$:

```
# fun x -> x;;
- : 'a -> 'a = <fun>
```

Let us now see the precise typing rules for each OCaml construct, presented in the same style as our logical rules:

Typing rules for OCaml constructs:

- **Unit (truth):** $\frac{}{():\text{unit}}$

The unit value `()` always has type `unit`. This is like $\top$ in logic: we can always produce it without any premises.

- **Empty type (falsity):** in OCaml we can *define* an empty type (a type with no constructors):

```
type void = |
```

There is no way to construct a value of type `void` using ordinary, terminating code. But if we somehow have a `v : void`, then we can derive anything from it (falsity elimination):

```
let absurd (v : void) : 'a =
  match v with _ -> .
```

This corresponds closely to the logical rule $\frac{}{a}$.

OCaml also has *effects* (notably exceptions). Because `raise e` never returns normally, the type checker allows it to have any result type:

$$\frac{e : \text{exn}}{\text{raise } e : a}$$

This is useful in practice, but it is also a good reminder that effects complicate the neat “proofs-as-programs” story.

- **Pair (conjunction):**

- Introduction: $\frac{s:a \quad t:b}{(s,t):a*b}$
- Elimination: from `p : a * b` we can extract either component (e.g. by pattern matching, or via `fst/snd`)

To construct a pair, you need both components. To use a pair, you can extract either component. This mirrors conjunction perfectly: to prove “A and B”, you need proofs of both; given “A and B”, you can conclude either A or B.

- **Variant (disjunction):** first, we define a sum type (a two-way choice):

```
type ('a, 'b) either = Left of 'a | Right of 'b
```

- Introduction: from $x : a$ we get $\text{Left } x : (a, b) \text{ either}$, and from $y : b$ we get $\text{Right } y : (a, b) \text{ either}$
- Elimination: given $t : (a, b) \text{ either}$ and a branch for each case, produce a result c (pattern matching)

The shape of the elimination rule is exactly “reasoning by cases”: to use an `either`, you must handle both `Left` and `Right`.

```
let either f g = function
| Left x -> f x
| Right y -> g y
```

A built-in example is `bool`, which you can think of as a two-constructor variant; the `if ... then ... else ...` expression is just a specialized form of case analysis on a boolean.

```
let choose b x y =
  if b then x else y

let choose' b x y =
  match b with
  | true -> x
  | false -> y
```

To construct a variant, you only need one of the alternatives. To use a variant, you must handle *all* possible cases (pattern matching). This mirrors disjunction: to prove “A or B”, you only need one; to use “A or B”, you must consider both possibilities.

- **Function (implication):**

- Introduction:
$$\frac{[x:a] \quad \vdots \quad e:b}{\text{fun } x \rightarrow e:a \rightarrow b}$$
- Elimination (application):
$$\frac{f:a \rightarrow b \quad t:a}{f \ t:b}$$

To construct a function, you assume you have an input of type a (the parameter x) and show how to produce a result of type b . To use a

function, you apply it to an argument. This mirrors implication: to prove “A implies B”, assume A and derive B; given “A implies B” and A, conclude B.

- **Recursion (induction):** recursion is not a connective, but it matches the *shape* of induction: in a recursive definition you are allowed to assume the function being defined (the “induction hypothesis”) when defining its body.

General recursion alone is not an induction proof: `let rec loop x = loop x` never returns. The proof interpretation requires termination, for example by recursion on a strictly smaller substructure.

In OCaml, recursion is introduced with `let rec` (there is no standalone `rec` expression).

Definitions

Writing out expressions and types repetitively quickly becomes tedious. More importantly, without definitions we cannot give names to our concepts, making code harder to understand and maintain. This is why we need definitions.

Type definitions are written: `type ty = some type`.

- In OCaml, disjunction-like types are not written as something like `a | b` directly; instead, you define a *variant type* and then use its constructors. For example:

```
type int_string_choice = A of int | B of string
```

This allows us to write `A x : int_string_choice` for any `x : int`, and `B y : int_string_choice` for any `y : string`.

- Why do we need to define variant types? The reasons are: exhaustiveness checks, performance of generated code, and ease of type inference. When OCaml sees `A 5`, it needs to figure out (or “infer”) the type. Without a type definition, how would OCaml know whether this is `A of int | B of string` or `A of int | B of float | C of bool`? The definition tells OCaml exactly what variants exist. When you match `| A i -> ...`, the compiler will warn you if you forgot to also cover `C b` in your match patterns.

- OCaml does provide an alternative: *polymorphic variants*, written with a backtick. We can write ``A x : [`A of a | `B of b]`. With ``` variants, OCaml does infer what other variants might exist based on usage. These types are powerful and flexible; we will discuss them in chapter 11.
- Tuple elements do not need labels because we always know at which position a tuple element stands: the first element is first, the second is second, and so on. However, having labels makes code much clearer, especially when tuples have many components or components of the same type. For this reason, we can define a *record type*:

```
type int_string_record = { a : int; b : string }
```

and create its values: `{a = 7; b = "Mary"}`. OCaml 5.4 and newer also support labeled tuples, we will not discuss these.

- We access the *fields* of records using the dot notation: `{a = 7; b = "Mary"}.b = "Mary"`. Unlike tuples where you must remember “the second element is the name”, with records you can write `.b` to get the field named `b`.

Expression Definitions

In many presentations of the Curry–Howard correspondence (and in programming language theory), recursion is introduced via a standalone operator often called `fix`. OCaml does not have a standalone `fix` expression: recursion is introduced only as part of a `let rec` definition.

This brings us to **expression definitions**, which let us give names to values. The typing rules for definitions are a bit more complex than what we have seen so far:

$$\frac{e_1 : a \quad \begin{array}{c} [x:a] \\ \vdots \\ e_2 : b \end{array}}{\text{let } x = e_1 \text{ in } e_2 : b}$$

This rule says: if e_1 has type a , and assuming x has type a we can show that e_2 has type b , then the whole `let` expression has type b . Interestingly, this rule is equivalent to introducing a function and immediately applying it: `let x = e1 in e2` behaves the same as `(fun x -> e2) e1`. This is the introduction rule for a function followed by its elimination rule.

For recursive definitions, we need an additional rule:

$$\frac{\begin{array}{cc} [x:a] & [x:a] \\ \vdots & \vdots \\ e_1:a & e_2:b \end{array}}{\text{let rec } x = e_1 \text{ in } e_2 : b}$$

Notice the crucial difference: in the recursive case, x can appear in e_1 itself! This is what allows functions to call themselves. The name x is visible both in its own definition (e_1) and in the body that uses the definition (e_2).

These rules are slightly simplified. The full rules involve a concept called **polymorphism**, which we will cover in a later chapter. Polymorphism explains how the same function can work with different types.

Scoping Rules

Understanding *scope*—where names are visible—is essential for reading and writing OCaml programs.

- **Type definitions** we have seen above are *global*: they need to be at the top-level (not nested in expressions), and they extend from the point they occur till the end of the source file or interactive session. A bare **type** declaration cannot occur inside an expression, but a function can introduce a type through a local module; we will meet modules in Chapter 5.
- **let-in definitions** for expressions: **let $x = e_1$ in e_2** are *local*—the name x is only visible within e_2 . Once you exit the **in** part, x no longer exists. This is useful for temporary values that should not pollute the global namespace.
- **let definitions** without **in** are global: placing **let $x = e_1$** at the top-level makes x visible from after e_1 till the end of the source file or interactive session. This is how you define functions and values that the rest of your program can use.
- In the interactive session (toplevel/REPL), we mark the end of a top-level “sentence” with **;;**. This tells OCaml “I am done typing, please evaluate this.” In source files compiled by the build system, **;;** is unnecessary because the end of each definition is clear from context.

Operators

Operators like **+**, *****, **<**, **=** are simply names of functions. In OCaml, there is nothing magical about operators; they are ordinary functions that happen

to have special characters in their names and can be used in infix position (between their arguments).

Just like other names, you can define your own operators:

```
# let (+:) a b = String.concat " " [a; b];;
val ( +: ) : string -> string -> string = <fun>
# "Alpha" +: "Beta";;
- : string = "AlphaBeta"
```

Notice the asymmetry here: when *defining* an operator, we wrap it in parentheses to tell OCaml “this is the name I am defining”. When *using* the operator, we write it in the normal infix position between its arguments. This asymmetry exists because the definition syntax needs to distinguish between “the name +:” and “the expression a +: b”.

OCaml’s built-in arithmetic operators are **not overloaded** across numeric types: integer and floating-point arithmetic use different operators. This is distinct from parametric polymorphism, which allows operations such as equality to have polymorphic types:

- +, *, / work for integers
- +., *., /. work for floating point numbers

This design choice makes type inference simpler and more predictable. When you see `x + y`, OCaml knows immediately that `x` and `y` must be integers.

Exception: The comparison operators `<`, `=`, `<=`, `>=`, `<>` do work for all values other than functions. These are called *polymorphic comparisons*.

1.4 Exercises

The following exercises are adapted from *Think OCaml: How to Think Like a Computer Scientist* by Nicholas Monje and Allen Downey. They will help you get comfortable with OCaml’s syntax and type system.

Practice 1: Type and Value Predictions

Assume that we execute the following assignment statements:

```
let width = 17
```

```
let height = 12.0
let delimiter = '.'
```

For each of the following expressions, write the value of the expression and the type (of the value of the expression), or the resulting type error.

1. `width/2`
2. `width/.2.0`
3. `height/3`
4. `1 + 2 * 5`
5. `delimiter * 5`

Practice 2: REPL Calculator Drills

Practice using the OCaml interpreter as a calculator:

1. The volume of a sphere with radius r is $\frac{4}{3}\pi r^3$. What is the volume of a sphere with radius 5? (*Hint*: 392.6 is wrong!)
2. Suppose the cover price of a book is \$24.95, but bookstores get a 40% discount. Shipping costs \$3 for the first copy and 75 cents for each additional copy. What is the total wholesale cost for 60 copies?
3. If I leave my house at 6:52 am and run 1 mile at an easy pace (8:15 per mile), then 3 miles at tempo (7:12 per mile) and 1 mile at easy pace again, what time do I get home for breakfast?

Practice 3: Recursive Fibonacci

You've probably heard of the Fibonacci numbers before, but in case you haven't, they're defined by the following recursive relationship:

$$\begin{cases} f(0) = 0 \\ f(1) = 1 \\ f(n+1) = f(n) + f(n-1) \quad \text{for } n = 1, 2, \dots \end{cases}$$

Write a recursive function on nonnegative integers to calculate these numbers. Reject negative inputs. Check `f 0 = 0`, `f 1 = 1`, and `f 10 = 55`; explain why the argument decreases. Machine integer overflow limits the numerical contract.

Practice 4: Recursive Palindromes

A palindrome is a word that is spelled the same backward and forward, like “noon” and “redivider”. Recursively, a word is a palindrome if the first and last letters are the same and the middle is a palindrome.

The following are functions that take a string argument and return the first, last, and middle letters:

```
let first_char word = word.[0]
let last_char word =
  let len = String.length word - 1 in
  word.[len]
let middle word =
  let len = String.length word - 2 in
  String.sub word 1 len
```

1. Enter these functions into the toplevel and test them out. What happens if you call `middle` with a string with two letters? One letter? What about the empty string ""?
2. Write a function called `is_palindrome` that takes a string argument and returns `true` if it is a palindrome and `false` otherwise.

Practice 5: Euclid’s GCD

The greatest common divisor (GCD) of a and b is the largest number that divides both of them with no remainder.

One way to find the GCD of two numbers is Euclid’s algorithm, which is based on the observation that if r is the remainder when a is divided by b , then $\text{gcd}(a, b) = \text{gcd}(b, r)$. As a base case, we can consider $\text{gcd}(a, 0) = a$.

Write `gcd` on nonnegative integers `a` and `b`, taking `gcd 0 0 = 0` by convention. Check $(0, 7)$, $(7, 0)$ and $(54, 24)$. **Proof:** show that the second argument strictly decreases whenever it is positive.

If you need help, see http://en.wikipedia.org/wiki/Euclidean_algorithm.

Selected answer: safe palindrome base cases

For byte strings, lengths zero and one are palindromes. Check this before calling `first_char`, `last_char`, or `middle`, whose contracts exclude those lengths. This is a byte-level exercise, not a Unicode grapheme algorithm.

```
let rec is_palindrome word =  
  String.length word < 2 ||  
  (first_char word = last_char word && is_palindrome (middle word))  
let () =  
  assert (is_palindrome "");  
  assert (is_palindrome "a");  
  assert (is_palindrome "noon");  
  assert (not (is_palindrome "not"))
```

Chapter 2: Algebra

Algebraic data types and some curious analogies

In this chapter, we will deepen our understanding of OCaml’s type system by working through type inference examples by hand. Then we will explore algebraic data types—a cornerstone of functional programming that allows us to define rich, structured data. Along the way, we will discover a surprising and beautiful connection between these types and ordinary polynomials from high-school algebra.

In this chapter, you will:

- Practice type inference by hand (constraints, unification intuition)
- Define and manipulate algebraic data types (variants, records, recursion, parameters)
- Interpret types as polynomials (and learn what this analogy buys you)
- Differentiate types to compute “one-hole contexts” (derivatives of data structures)

Prerequisites: functions, scope and pairs from Chapter 1. **Route:** Part I. This chapter’s contexts return as machine frames in Chapter 3 and as mathematical constructions in Chapter 12.

2.1 A Glimpse at Type Inference

For a refresher, let us apply the type inference rules introduced in Chapter 1 to some simple examples. We will start with the identity function `fun x -> x`—perhaps the simplest possible function, yet one that reveals important aspects of polymorphism. In the derivations below, `[?]` means “unknown (to be inferred)”.

We begin with an incomplete derivation:

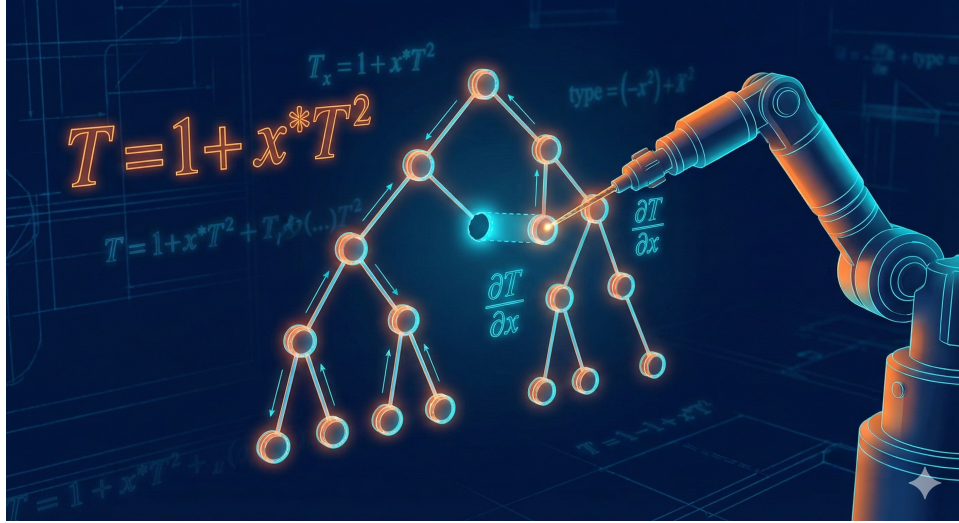


Figure 3.1: Chapter 2 illustration

$$\frac{[?]}{\text{fun } x \rightarrow x : [?]}$$

Using the $\rightarrow$ introduction rule, we need to derive the body x assuming x has some type a :

$$\frac{\begin{array}{c} [x:a]^x \\ \vdots \\ x:a \end{array}}{\text{fun } x \rightarrow x : [?] \rightarrow [?]}$$

The premise is a hypothetical derivation: inside the body we are allowed to use the assumption $x : a$. Since the body is just x , the result type is also a , and we conclude:

$$\frac{\begin{array}{c} [x:a]^x \\ \vdots \\ x:a \end{array}}{\text{fun } x \rightarrow x : a \rightarrow a}$$

Because a is arbitrary (we made no assumptions constraining it), OCaml introduces a *type variable* 'a to represent it. This is how polymorphism emerges naturally from the inference process—the identity function can work with values of any type:

```
# fun x -> x;;
- : 'a -> 'a = <fun>
```

A More Complex Example

Now let us try something that will constrain the types more: `fun x -> x+1`. This is the same as `fun x -> ((+) x) 1` (try it in OCaml to verify!). The addition operator forces specific types upon us.

We will use the notation $[?\alpha]$ to mean “type unknown yet, but the same as in other places marked $[?\alpha]$.” This notation helps us track how constraints propagate through the derivation.

Starting the derivation and applying $\rightarrow$ introduction:

$$\frac{\frac{[?]}{((+) \ x) \ 1 : [?\alpha]}}{\text{fun } x \rightarrow ((+) \ x) \ 1 : [?] \rightarrow [?\alpha]}$$

Applying $\rightarrow$ elimination (function application) to $((+) \ x) \ 1$:

$$\frac{\frac{\frac{[?]}{((+) \ x) : [?\beta] \rightarrow [?\alpha]} \quad \frac{[?]}{1 : [?\beta]}}{((+) \ x) \ 1 : [?\alpha]}}{\text{fun } x \rightarrow ((+) \ x) \ 1 : [?] \rightarrow [?\alpha]}$$

We know that $1 : \text{int}$, so $[?\beta] = \text{int}$:

$$\frac{\frac{\frac{[?]}{((+) \ x : \text{int} \rightarrow [?\alpha])} \quad \frac{1 : \text{int} \quad (\text{constant})}{1 : \text{int}}}{((+) \ x) \ 1 : [?\alpha]}}{\text{fun } x \rightarrow ((+) \ x) \ 1 : [?] \rightarrow [?\alpha]}$$

Applying function application again to $(+) \ x$:

$$\frac{\frac{\frac{\frac{[?]}{((+) \ x : \text{int} \rightarrow [?\alpha])} \quad \frac{[?]}{x : [?\gamma]}}{((+) \ x) : [?\gamma] \rightarrow [?\alpha]} \quad \frac{1 : \text{int} \quad (\text{constant})}{1 : \text{int}}}{((+) \ x) \ 1 : [?\alpha]}}{\text{fun } x \rightarrow ((+) \ x) \ 1 : [?\gamma] \rightarrow [?\alpha]}$$

Since $(+) : \text{int} \rightarrow \text{int} \rightarrow \text{int}$, we have $[?\gamma] = \text{int}$ and $[?\alpha] = \text{int}$:

$$\frac{\frac{\frac{(\text{constant})}{((+) \ x : \text{int} \rightarrow \text{int})} \quad \frac{x : \text{int}}{x : \text{int}} \quad (\text{constant})}{((+) \ x) \ 1 : \text{int}}}{\text{fun } x \rightarrow ((+) \ x) \ 1 : \text{int} \rightarrow \text{int}}$$

Curried Form

When there are several arrows “on the same depth” in a function type, it means that the function returns a function. For example, $(+) : \text{int} \rightarrow \text{int}$

`-> int` is just a shorthand for `(+) : int -> (int -> int)`. The arrow associates to the right, so we can omit the parentheses.

This is very different from:

```
fun f -> (f 1) + 1 : (int -> int) -> int
```

In the first case, `(+)` is a function that takes an integer and returns a function from integers to integers. In the second case, we have a function that takes a function as an argument—a *higher-order function*. The parentheses around `int -> int` are essential here; without them, the meaning would be completely different.

This style of defining multi-argument functions, where each function takes one argument and returns another function expecting the remaining arguments, is called *curried form* (named after logician Haskell Curry). It enables a powerful technique called *partial application*.

For example, instead of writing `(fun x -> x+1)`, we can simply write `((+) 1)`. Here we apply `(+)` to just one argument, getting back a function that adds 1 to its input. What expanded form does `((+) 1)` correspond to exactly (computationally)?

Think about it before reading on...

It corresponds to `fun y -> 1 + y`. We have “baked in” the first argument, and the resulting function waits for the second.

We will become more familiar with functions returning functions when we study the *lambda calculus* in a later chapter.

2.2 Algebraic Data Types

In Chapter 1, we learned about the `unit` type and variant types like:

```
type int_string_choice = A of int | B of string
```

We also covered tuple types, record types, and type definitions. Now let us explore these concepts more deeply, building up to the powerful notion of *algebraic data types*.

Variants Without Arguments

Variants do not have to carry arguments. Instead of writing `A of unit`, we can simply use `A`. This is more convenient and idiomatic:

```
type color = Red | Green | Blue
```

This defines a type with exactly three possible values—no more, no less. The compiler knows this, which enables exhaustive pattern matching checks.

A subtle point about OCaml: In OCaml, variants take multiple arguments rather than taking tuples as arguments. This means `A of int * string` is different from `A of (int * string)`. The first takes two separate arguments, while the second takes a single tuple argument. This distinction is usually not important—until you get bitten by it in some corner case! For most purposes, you can ignore it.

Recursive Type Definitions

Here is where things get really interesting: type definitions can be recursive! This allows us to define data structures of arbitrary size using a finite definition:

```
type int_list = Empty | Cons of int * int_list
```

Let us see what values inhabit `int_list`. The definition tells us there are two ways to build an `int_list`:

- `Empty` represents the empty list—a list with no elements
- `Cons (5, Empty)` is a list containing just 5
- `Cons (5, Cons (7, Cons (13, Empty)))` is a list containing 5, 7, and 13.

Notice how `Cons` takes an integer and another `int_list`, allowing us to chain together as many elements as we like. This recursive structure is the essence of how functional languages represent unbounded data.

The built-in type `bool` really does behave like a two-constructor variant with values `true` and `false`—but note a small OCaml wrinkle: user-defined constructors must start with a capital letter, while a few built-in constructors like `true`, `false`, `[]`, and `(::)` are special-cased.

Similarly, `int` can be *thought of* as a very large finite variant (“one constructor per integer”), even though the compiler implements it as an efficient machine integer rather than as a gigantic sum type.

Parametric Type Definitions

Our `int_list` type only works with integers. But what if we want a list of strings? Or a list of booleans? We would have to define separate types for each, duplicating the same structure.

Type definitions can be *parametric* with respect to the types of their components. This allows us to define generic data structures that work with any element type. OCaml already has a built-in parametric list type, so to avoid shadowing it we will define our own simplified list type:

```
type 'a my_list = Empty | Cons of 'a * 'a my_list
```

The `'a` is a *type parameter*—a placeholder that gets filled in when we use the type. We can have a `string my_list`, an `int my_list`, or even an `(int my_list) my_list` (a list of lists of integers).

Several conventions and syntax rules apply to parametric types:

- Type variables must start with `'`. When printing inferred types, OCaml may rename these variables, so it is customary to stick to the standard names `'a`, `'b`, `'c`, `'d`, etc.
- The OCaml syntax places the type parameter before the type name, mimicking English word order. A silly example that reads almost like English:

```
type 'white_color dog = Dog of 'white_color
```

This defines a “white-color dog” type—the syntax reads naturally!

- With multiple parameters, OCaml uses parentheses:

```
type ('a, 'b) choice = Left of 'a | Right of 'b
```

Compare this to F# syntax: `type choice<'a,'b> = Left of 'a | Right of 'b`

And Haskell syntax: `data Choice a b = Left a | Right b`

Different languages have different conventions, but the underlying concept is the same.

2.3 Syntactic Bread and Sugar

OCaml provides various syntactic conveniences—sometimes called *syntactic sugar*—that make code more pleasant to write and read. Let us survey the most important ones.

Constructor Naming

Names of variants, called *constructors*, must start with a capital letter. If we wanted to define our own booleans, we would write:

```
type my_bool = True | False
```

Only constructors and module names can start with capital letters in OCaml. Everything else (values, functions, type names) must start with a lowercase letter. This convention makes it easy to distinguish constructors at a glance.

(As noted above, a few built-in constructors like `true`, `false`, `[]`, and `(::)` are special exceptions to the capitalization rule.)

Modules are organizational units (like “shelves”) containing related values. For example, the `List` module provides operations on lists, including `List.map` and `List.filter`. We will learn more about modules in later chapters.

Accessing Record Fields

Did we mention that we can use dot notation to access record fields? The syntax `record.field` extracts a field value. For example, if we have `let person = {name="Alice"; age=30}`, we can write `person.name` to get `"Alice"`.

Function Definition Shortcuts

Several syntactic shortcuts make function definitions more concise. These are worth memorizing, as you will see them constantly in OCaml code:

- `fun x y -> e` stands for `fun x -> fun y -> e`. Note that `fun x -> fun y -> e` parses as `fun x -> (fun y -> e)`. This shorthand aligns with curried form—we can write multi-argument functions without nesting `fun` expressions.
- `function A x -> e1 | B y -> e2` stands for `fun p -> match p with A x -> e1 | B y -> e2`. The general form is: `function`

PATTERN-MATCHING stands for `fun v -> match v with PATTERN-MATCHING`. This is handy when you want to immediately pattern-match on a function's argument.

- `let f ARGS = e` is a shorthand for `let f = fun ARGS -> e`. This is probably the most common way to define functions in practice.

2.4 Pattern Matching

Pattern matching is one of the most powerful features of OCaml and similar languages. It lets us examine the structure of data and extract components in a single, elegant construct.

Recall that we introduced `fst` and `snd` as means to access elements of a pair. But what about larger tuples? There is no built-in `thd` for the third element. The fundamental way to access any tuple—or any algebraic data type—uses the `match` construct. In fact, `fst` and `snd` can easily be defined using pattern matching:

```
let fst p = match p with (a, b) -> a
let snd p = match p with (a, b) -> b
```

The pattern `(a, b)` *destructures* the pair, binding its first component to `a` and its second to `b`. We then return whichever component we want.

Matching on Records

Pattern matching also works with records, letting us extract multiple fields at once:

```
type person = { name : string; surname : string; age : int }

let greet_person () =
  match { name = "Walker"; surname = "Johnnie"; age = 207 } with
  | { name = _; surname = sn; age = _ } -> "Hi " ^ sn ^ "!"
```

Here we match against a record pattern. Note that we use wildcards `_` for `name` and `age` (ignoring them), while binding `surname` to `sn`—then use `sn` in the greeting.

Understanding Patterns

The left-hand sides of `->` in `match` expressions are called **patterns**. Patterns describe the structure of values we want to match against. They can include:

- Constants (like `1`, `"hello"`, or `true`)
- Variables (which bind to the matched value)
- Constructors (like `None`, `Some x`, or `Cons (h, t)`)
- Tuples and records
- Nested combinations of all the above

Patterns can be nested to arbitrary depth, allowing us to match complex structures in one go:

```
match Some (5, 7) with
| None -> "sum: nothing"
| Some (x, y) -> "sum: " ^ string_of_int (x + y)
```

Here `Some (x, y)` is a nested pattern: we match `Some` of *something*, and that something must be a pair, whose components we bind to `x` and `y`.

Simple Patterns and Wildcards

A pattern can simply bind the entire value without destructuring. Writing `match f x with v -> ...` is the same as `let v = f x in ...`. This is occasionally useful when you want the syntax of `match` but do not need to take the value apart.

When we do not need a value in a pattern, it is good practice to use the underscore `_`, which is a *wildcard*. The wildcard matches anything but does not bind it to a name. This signals to the reader (and the compiler) that we are intentionally ignoring that part:

```
let fst (a, _) = a
let snd (_, b) = b
```

Using `_` instead of an unused variable name avoids compiler warnings about unused bindings.

Pattern Linearity

A variable can only appear once in a pattern. This property is called *linearity*. You might think this is a limitation—what if we want to check that two

parts of a structure are equal? We cannot write `(x, x)` to match pairs with equal components.

However, we can add conditions to patterns using **when**, so linearity is not really a limitation in practice:

```
let describe_point p =  
  match p with  
  | (x, y) when x = y -> "diag"  
  | _ -> "off-diag"
```

The **when** clause acts as a guard: the pattern matches only if both the structure matches *and* the condition is true.

Here is a more elaborate example showing how to implement a comparison function (without shadowing the standard **compare**):

```
let compare_int a b =  
  match a, b with  
  | (x, y) when x < y -> -1  
  | (x, y) when x = y -> 0  
  | _ -> 1
```

Notice how we match against the tuple `(a, b)` in different ways, using guards to distinguish the cases.

Partial Record Patterns

We can skip unused fields of a record in a pattern. Only the fields we care about need to be mentioned. This keeps patterns concise and means we do not have to update every pattern when we add a new field to a record type.

Or-Patterns

We can compress patterns by using `|` inside a single pattern to match multiple alternatives. This is different from having multiple pattern clauses—it lets us share a single right-hand side for several patterns:

```
type month =  
  | Jan | Feb | Mar | Apr | May | Jun  
  | Jul | Aug | Sep | Oct | Nov | Dec  
  
type weekday = Mon | Tue | Wed | Thu | Fri | Sat | Sun
```

```

type calendar_date =
  { year : int; month : month; day : int; weekday : weekday }

let day =
  { year = 2012; month = Feb; day = 14; weekday = Tue }

let day_kind =
  match day with
  | { weekday = Sat | Sun; _ } -> "Weekend!"
  | _ -> "Work day"

```

The pattern `Sat | Sun` matches either `Sat` or `Sun`. This is much cleaner than writing two separate clauses with the same right-hand side.

Named Patterns with `as`

Sometimes we want to both destructure a value *and* keep a reference to the whole thing (or some intermediate part). We use `(pattern as v)` to name a nested pattern, binding the matched value to `v`:

```

match day with
| {weekday = (Mon | Tue | Wed | Thu | Fri as wday); _}
  when not (day.month = Dec && day.day = 24) ->
  Some (work (get_plan wday))
| _ -> None

```

This example demonstrates several features working together:

- An or-pattern matches any weekday from Monday to Friday
- The `as wday` clause binds the matched weekday to the variable `wday`
- A `when` guard checks that it is not Christmas Eve
- The bound variable `wday` is then used in the expression `get_plan wday`

The pattern gives names only to the data used by the branch.

2.5 Interpreting Algebraic Data Types as Polynomials

Three interpretations must be kept separate:

Interpretation	What it tells us	What it does not establish
Finite cardinality	A sum has $a + b$ inhabitants and a product has ab	A particular conversion algorithm
Formal power series	The coefficient of z^n counts shapes of size n	Numerical convergence at a chosen real z
Type isomorphism	Two functions are inverse on all inputs	Equality merely because a symbolic equation looks plausible

For lists with one mark per element, $L(z) = 1 + zL(z)$ gives $L(z) = \sum_{n \geq 0} z^n$. For binary trees marked at nodes, $T(z) = 1 + zT(z)^2$ gives coefficients 1, 1, 2, 5, ...; the root splits the remaining nodes between two ordered subtrees. These are formal coefficient identities. Interpreting a parameter as the cardinality of a set is a different operation from evaluating a series at a real number.

The translation from types to mathematical expressions works as follows:

- Replace `|` (variant choice) with $+$ (addition)
- Replace `*` (tuple product) with $\times$ (multiplication)
- Treat record types as tuple types (erasing field names and translating `;` as $\times$)

We also need translations for some special types:

- The **void type** (a type with no constructors, hence no values):

```
type void = |
```

Since no values can be constructed, it represents emptiness—translate it as 0.

- The **unit type** has exactly one value, so translate it as 1. Since variants without arguments behave like variants of **unit**, translate them as 1 as well.
- The **bool type** has exactly two values (**true** and **false**), so translate it as 2.

- Types like `int`, `string`, `float`, and type parameters are treated as variables. We do not care about their exact number of values; we just give them symbolic names like x , y , etc.
- Defined types translate according to their definitions (substituting variables as necessary).

Give a name to the type being defined (representing a function of the introduced variables). For finite, nonrecursive sum-and-product types, the result is a polynomial counting possible values. Recursive types instead give equations for formal power series counting finite structures. Lists yield a rational series; trees generally yield algebraic series that are not rational. Unrestricted subtraction, division, and identities involving infinite cardinalities are not automatically type isomorphisms: justify a proposed isomorphism with inverse functions.

We will use the equations to propose representations, then write conversions to check them.

Example: Date Type

```
type ymd = { year : int; month : int; day : int }
```

A simple “year-month-day” record is a product of three `int` fields. Translating to a polynomial (using x for `int`):

$$D = x \times x \times x = x^3$$

The cube makes sense: this record is essentially a triple of integers.

Example: Option Type

The built-in option type is defined as:

```
type 'a option = None | Some of 'a
```

Translating (using x for the type parameter `'a`):

$$O = 1 + x$$

This reads as: an option is either nothing (1) or something of type x . The two summands record the two constructor cases.

Example: List Type

This skipped declaration recalls the list representation solely for the series calculation; it is not a second live definition in this environment.

```
type 'a my_list = Empty | Cons of 'a * 'a my_list
```

Translating (where L represents the list type itself, and x represents the element type):

$$L = 1 + x \cdot L$$

This is a recursive equation! A list is either empty (1) or an element times another list ($x \cdot L$). If you solve this equation algebraically, you get $L = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$, which corresponds to: a list is either empty, or has one element, or has two elements, etc.

Example: Binary Tree Type

```
type btree = Tip | Node of int * btree * btree
```

Translating:

$$T = 1 + x \cdot T \cdot T = 1 + x \cdot T^2$$

A binary tree is either a tip (1) or a node containing a value and two subtrees ($x \cdot T^2$).

Type Isomorphisms

Here is the remarkable payoff: when translations of two types are equal according to the laws of high-school algebra, the types are *isomorphic*. This means there exist bijective (one-to-one and onto) functions between them—you can convert from one type to the other and back without losing any information.

Let us play with the binary tree polynomial and see where algebra takes us:

$$\begin{aligned}
T &= 1 + x \cdot T^2 \\
&= 1 + x \cdot T + x^2 \cdot T^3 \\
&= 1 + x + x^2 \cdot T^2 + x^2 \cdot T^3 \\
&= 1 + x + x^2 \cdot T^2 \cdot (1 + T) \\
&= 1 + x \cdot (1 + x \cdot T^2 \cdot (1 + T))
\end{aligned}$$

Each step uses standard algebraic manipulations: substituting $T = 1 + xT^2$, expanding, factoring, and rearranging. The result is a different but algebraically equivalent expression.

Now let us translate this resulting expression back to a type:

```
type repr =
  (int * (int * btree * btree * btree option) option) option
```

Reading the polynomial $1 + x \cdot (1 + x \cdot T^2 \cdot (1 + T))$ from outside in: we have an option (the outermost $1 + \dots$), whose **Some** case contains an **int** times another option, and so on.

The challenge is to find isomorphism functions with signatures:

```
val iso1 : btree -> repr
val iso2 : repr -> btree
```

These functions should satisfy: for all trees t , $\text{iso2 } (\text{iso1 } t) = t$, and for all representations r , $\text{iso1 } (\text{iso2 } r) = r$. Can you write them?

My First (Failed) Attempt

Here is my first attempt, trying to guess the pattern directly:

```
# let iso1 (t : btree) : repr =
  match t with
  | Tip -> None
  | Node (x, Tip, Tip) -> Some (x, None)
  | Node (x, Node (y, t1, t2), Tip) ->
    Some (x, Some (y, t1, t2, None))
  | Node (x, Node (y, t1, t2), t3) ->
    Some (x, Some (y, t1, t2, Some t3));;
```

Warning 8: this pattern-matching is not exhaustive.

Here is an example of a value that is not matched:
`Node (_, Tip, Node (_, _, _))`

I forgot about one case! The case `Node (_, Tip, Node (_, _, _))`—a node with an empty left subtree and non-empty right subtree—was not covered. It seems difficult to guess the solution directly when trying to map the complex final form all at once.

Have you found it on your first try? If so, congratulations! Most people do not. This illustrates an important principle: complex transformations are easier to get right when broken into smaller steps.

Breaking Down the Problem

Let us divide the task into smaller steps corresponding to intermediate points in the polynomial transformation. Instead of jumping from $T = 1 + xT^2$ directly to the final form, we will introduce intermediate types for each algebraic step:

```
type ('a, 'b) choice = Left of 'a | Right of 'b

type interm1 =
  ((int * btree, int * int * btree * btree * btree) choice)
  option

type interm2 =
  ((int, int * int * btree * btree * btree option) choice)
  option
```

Now we can define each step:

```
let step1r (t : btree) : interm1 =
  match t with
  | Tip -> None
  | Node (x, t1, Tip) -> Some (Left (x, t1))
  | Node (x, t1, Node (y, t2, t3)) ->
    Some (Right (x, y, t1, t2, t3))

let step2r (r : interm1) : interm2 =
  match r with
  | None -> None
  | Some (Left (x, Tip)) -> Some (Left x)
  | Some (Left (x, Node (y, t1, t2))) ->
    Some (Right (x, y, t1, t2, None))
```

```

    | Some (Right (x, y, t1, t2, t3)) ->
      Some (Right (x, y, t1, t2, Some t3))

let step3r (r : interm2) : repr =
  match r with
  | None -> None
  | Some (Left x) -> Some (x, None)
  | Some (Right (x, y, t1, t2, t3opt)) ->
    Some (x, Some (y, t1, t2, t3opt))

let iso1 (t : btree) : repr =
  step3r (step2r (step1r t))

```

Each step function handles one small transformation, and the compiler verifies that our pattern matching is exhaustive. No more missed cases!

Exercise

Define `step1l`, `step2l`, `step3l`, and `iso2`.

Hint: Now it is straightforward—each step is simply the inverse of its corresponding forward step. The left-going functions undo what the right-going functions do.

Take-Home Lessons

This exploration of type isomorphisms teaches us two valuable principles:

1. **Design for validity:** Try to define data structures so that only meaningful information can be represented—as long as it does not over-complicate the data structures. Avoid catch-all clauses when defining functions. The compiler will then tell you if you have forgotten about a case. The exhaustiveness checker is your friend.
2. **Divide and conquer:** Break solutions into small steps so that each step can be easily understood and verified. When I tried to write `iso1` directly, I made a mistake. When I broke it into three simple steps, each step was obviously correct, and composing them gave the right answer.

2.6 Differentiating Algebraic Data Types

Of course, you might object that the pompous title is wrong—we will differentiate the translated polynomials, not the types themselves. Fair enough! But what sense does differentiating a type’s polynomial make?

It turns out that taking the partial derivative of a polynomial (translated from a data type), when translated back, gives a type representing a “one-hole context”—a data structure with one piece missing. This missing piece corresponds to the variable with respect to which we differentiated. The derivative tells us: “Here are all the ways to point at one element of this type.”

Example: Differentiating a Simple Record

Let us start with a simple record type:

```
type ymd = { year : int; month : int; day : int }
```

The translation and its derivative:

$$D = x \cdot x \cdot x = x^3$$
$$\frac{\partial D}{\partial x} = 3x^2 = x \cdot x + x \cdot x + x \cdot x$$

We could have left it as $3 \cdot x \cdot x$, but expanding it as a sum shows the structure more clearly. The derivative $3x^2$ says: there are three ways to “point at” an `int` in a `ymd`, and each way leaves two other `ints` behind.

Translating the expanded form back to a type:

```
type ymd_ctx =  
  Year of int * int | Month of int * int | Day of int * int
```

Each variant represents a “hole” at a different position:

- `Year (m, d)` means the year field is the hole (and we have the month `m` and day `d`)
- `Month (y, d)` means the month field is the hole (and we have year `y` and day `d`)
- `Day (y, m)` means the day field is the hole.

Now we can define functions to introduce and eliminate this derivative type:

```

let ymd_deriv ({ year = y; month = m; day = d } : ymd) =
  [ Year (m, d); Month (y, d); Day (y, m) ]

let ymd_integr n = function
  | Year (m, d) -> { year = n; month = m; day = d }
  | Month (y, d) -> { year = y; month = n; day = d }
  | Day (y, m) -> { year = y; month = m; day = n }

let example =
  List.map (ymd_integr 7) (ymd_deriv { year = 2012; month = 2; day
    ↪ = 14 })

```

The `ymd_deriv` function produces all contexts (one for each field)—it “differentiates” a record into a list of one-hole contexts. The `ymd_integr` function fills in a hole with a new value—it “integrates” by putting a value back into the context. Notice how the naming follows the calculus analogy!

The example above takes the date February 14, 2012, produces three contexts (one for each field), and then fills each hole with the number 7, producing three modified dates.

Example: Differentiating Binary Trees

Now let us tackle the more challenging case of binary trees (using the same `btree` type as above):

```

type btree = Tip | Node of int * btree * btree

```

The translation and differentiation:

$$\begin{aligned}
 T &= 1 + x \cdot T^2 \\
 \frac{\partial T}{\partial x} &= 0 + T^2 + 2 \cdot x \cdot T \cdot \frac{\partial T}{\partial x} = T \cdot T + 2 \cdot x \cdot T \cdot \frac{\partial T}{\partial x}
 \end{aligned}$$

Something interesting happened: the derivative is recursive! It refers to itself via $\frac{\partial T}{\partial x}$. This makes perfect sense when you think about it:

- $T \cdot T$ represents pointing at the root: the hole is at the current node, and we have the two subtrees.
- $2 \cdot x \cdot T \cdot \frac{\partial T}{\partial x}$ represents pointing deeper in the tree: we choose left or right (the factor of 2), remember the current node’s value (x), keep the other subtree (T), and then have a context in the chosen subtree ($\frac{\partial T}{\partial x}$).

Instead of translating 2 as `bool`, we introduce a more descriptive type to make the code clearer:

```

type btree_dir = LeftBranch | RightBranch

type btree_deriv =
  | Here of btree * btree
  | Below of btree_dir * int * btree * btree_deriv

```

The **Here** constructor means the hole is at the current position, and we have the left and right subtrees. The **Below** constructor means we go down one level, remembering which direction we went, the value at the node we passed, and the subtree we did not enter.

(You might someday hear about *zipper*s—they are “inverted” relative to our type. In a zipper, the hole comes first, and the context trails behind. Both representations are useful in different situations.)

Exercise

Write a function that takes a number and a **btree_deriv**, and builds a **btree** by putting the number into the “hole” in **btree_deriv**.

Solution

The integration function fills the hole with a value. It must be recursive because the derivative type is recursive—we may need to descend through multiple **Below** constructors before reaching the **Here** where the hole actually is:

```

let rec btree_integr n = function
  | Here (ltree, rtree) -> Node (n, ltree, rtree)
  | Below (LeftBranch, m, rtree, deriv) ->
    Node (m, btree_integr n deriv, rtree)
  | Below (RightBranch, m, ltree, deriv) ->
    Node (m, ltree, btree_integr n deriv)

```

When we reach **Here**, we create a node with the new value **n** and the two subtrees. When we see **Below**, we reconstruct the node we passed through and recursively integrate into the appropriate subtree.

Element holes and subtree holes are different

The derivative above removes one **element**, leaving the node’s two children. A **subtree** hole removes an entire tree, which may even be **Tip**. Its context is a path back to the root, remembering the sibling and parent label at each step:

```

type frame =
  | From_left of int * btree
  | From_right of int * btree

type subtree_context = frame list

let rec plug subtree = function
  | [] -> subtree
  | From_left (x, right) :: rest -> plug (Node (x, subtree, right))
  ↪ rest
  | From_right (x, left) :: rest -> plug (Node (x, left, subtree))
  ↪ rest

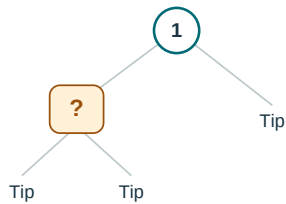
let () =
  let t = Node (1, Node (2, Tip, Tip), Tip) in
  let context = [From_left (1, Tip)] in
  assert (plug (Node (2, Tip, Tip)) context = t);
  assert (btree_integr 1 (Here (Node (2, Tip, Tip), Tip)) = t);
  assert (btree_integr 2 (Below (LeftBranch, 1, Tip, Here (Tip,
    ↪ Tip)))) = t);
  assert (plug Tip [] = Tip)

```

What exactly did we remove?

Same tree and focus: Node(1, Node(2, Tip, Tip), Tip)

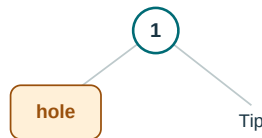
Element hole



Remove only 2; keep both children.

Fill with an element value.

Subtree hole



Remove the whole Node(2, Tip, Tip).
Its children leave with it.

Fill with any tree, including Tip.

In both panels, the path remembers parent 1 and the untouched right Tip.

An element context also keeps the two children at the focused node.

Figure 3.2: Removing an element keeps its two children; removing a subtree takes the children with it.

For $T = 1 + aT^2$, a path step has shape $2aT$: a direction, an element, and a sibling. A subtree context is a list of these steps. An element context consists of the two children of the removed element together with such a path. Thus the formal derivative is $T' = T^2/(1 - 2aT)$. The base T^2 is essential: confusing these two holes loses the removed element's children.

Proof exercise. Write `focus_left` returning a subtree and context, and prove that plugging the pair reconstructs the original node. State what happens at **Tip**. **Hint:** first prove the single-frame equation, then induct on the path.

A small isomorphism with both inverse laws

```
type ('a, 'b) sum = A of 'a | B of 'b
let distribute (x, choice) =
  match choice with A y -> A (x, y) | B z -> B (x, z)
let factor = function
  | A (x, y) -> (x, A y)
  | B (x, z) -> (x, B z)
let () =
  List.iter (fun x -> assert (factor (distribute x) = x))
    [true, A 2; false, B "b"];
  List.iter (fun y -> assert (distribute (factor y) = y))
    [A (true, 2); B (false, "b")]
```

These checks illustrate both directions. A proof covers each constructor with arbitrary fields, so it establishes the laws for every finite value of the represented sum/product types. Function equality in the exponent exercises is extensional equality; OCaml's polymorphic `=` cannot compare functions.

2.7 Exercises

Practice 1: Designing Valid Data Structures

Due to Yaron Minsky.

This exercise practices the principle of “making invalid states unrepresentable.” Consider a datatype to store internet connection information. The time `when_initiated` marks the start of connecting and is not needed after the connection is established (it is only used to decide whether to give up trying to connect). The ping information is available for established connections but not straight away.

```

type connectionstate = Connecting | Connected | Disconnected

type connectioninfo = {
  state : connectionstate;
  server : Inetaddr.t;
  lastpingtime : Time.t option;
  lastpingid : int option;
  sessionid : string option;
  wheninitiated : Time.t option;
  whenDisconnected : Time.t option;
}

```

(The types `Time.t` and `Inetaddr.t` come from the *Core* library. You can replace them with `float` and `Unix.inet_addr`. Load the Unix library in the interactive toplevel with `#load "unix.cma";;`)

The problem with this design is that it allows many nonsensical combinations: a `Connecting` state with ping information, a `Disconnected` state with a session ID, etc. The optional fields (all those `option` types) make it unclear which fields are valid in which states.

Rewrite the type definitions so that the datatype will contain only reasonable combinations of information. Use separate record types for each connection state, with only the fields that make sense for that state.

Practice / project 2: Labeled and Optional Arguments

In OCaml, functions can have labeled arguments and optional arguments (parameters with default values that can be omitted). This exercise explores these features.

Labels can differ from the names of argument values:

```

let f ~meaningfulname:n = n + 1
let _ = f ~meaningfulname:5 (* We do not need the result so we
↪ ignore it. *)

```

When the label and value names are the same, the syntax is shorter:

```

let g ~pos ~len =
  StringLabels.sub "0123456789abcdefghijklmnopqrstuvwxyz" ~pos ~len

let () = (* A nicer way to mark computations that return unit. *)
  let pos = Random.int 26 in
  let len = Random.int 10 in

```

```
print_string (g ~pos ~len)
```

When some function arguments are optional, a following positional argument lets OCaml determine when omitted optional arguments should be filled in. A required labeled argument alone does not provide that boundary. Optional parameters with default values:

```
let h ?(len=1) pos = g ~pos ~len
let () = print_string (h 10)
```

Optional arguments are implemented as parameters of an option type. This allows checking whether the argument was provided:

```
let foo ?bar n =
  match bar with
  | None -> "Argument = " ^ string_of_int n
  | Some m -> "Sum = " ^ string_of_int (m + n)
```

We can use it in various ways:

```
let _ = foo 5
let _ = foo ~bar:5 7
```

We can also provide the option value directly:

```
let test_foo () =
  let bar = if Random.int 10 < 5 then None else Some 7 in
  foo ?bar 7
```

1. Observe the types that functions with labeled and optional arguments have. Come up with coding style guidelines for when to use labeled arguments. When might they improve readability? When might they be overkill?
2. Write a rectangle-drawing procedure that takes three optional arguments: left-upper corner, right-lower corner, and a width-height pair. It should draw a correct rectangle whenever two of the three arguments are given (since any two determine the third), and raise an exception otherwise. Use the *Bogue* library.
3. Write a function that takes an optional argument of arbitrary type and a function argument, and passes the optional argument to the function without inspecting it. This tests your understanding of how optional arguments work at the type level.

Practice 3: Type Inference Practice

From a past exam.

These exercises help you internalize how type inference works. Try to work them out by hand before checking with the OCaml toplevel.

1. Give the (most general) types of the following expressions, either by guessing or by inferring by hand:
 1. `let double f y = f (f y) in fun g x -> double (g x)`
 2. `let rec tails l = match l with [] -> [] | x::xs -> xs::tails xs in fun l -> List.combine l (tails l)`
2. Give example expressions that have the following types (without using type constraints). There are many possible answers for each:
 1. `(int -> int) -> bool`
 2. `'a option -> 'a list`

Proof 4: Types as Exponents

We have seen that algebraic data types can be related to analytic functions (the subset definable from polynomials via recursion)—by literally interpreting sum types (variant types) as sums and product types (tuple and record types) as products. We can extend this interpretation to function types by interpreting $a \rightarrow b$ as b^a (i.e., b to the power of a). Note that the b^a notation is actually used to denote functions in set theory.

This interpretation makes sense: a function from a set with a elements to a set with b elements is choosing, for each of the a inputs, one of b outputs—giving b^a possible functions.

1. Translate a^{b+cd} and $a^b \cdot (a^c)^d$ into OCaml types, using any distinct types for a, b, c, d , and using `type ('a,'b) choice = Left of 'a | Right of 'b` for $+$. Write the bijection functions in both directions. Verify algebraically that $a^{b+cd} = a^b \cdot (a^c)^d$ using the laws of exponents.
2. **Experiment.** Explain why differentiating the list series gives two lists (the prefix and suffix around an element hole), whereas marking a gap gives one more possible position than marking an element. Count both for lengths zero, one and two. Do not infer that every analytic function denotes an ordinary algebraic datatype: denominators such as $n!$ in an exponential series require a different counting convention, involving labeled structures.

Practice 5: Finding Contexts

Write a function `btree_deriv_at` that takes a predicate over integers (i.e., a function `f: int -> bool`) and a `btree`, and builds a `btree_deriv` whose “hole” is in the first position for which the predicate returns true. It should return a `btree_deriv option`, with `None` if the predicate does not hold for any node.

This function lets you “search” a tree and get back a context pointing to the found element. Think about what order you want to search in (pre-order, in-order, or post-order) and what “first” means in that context.

Chapter 3: Evaluation, recursion, and machines

Prerequisites: functions, variants and pattern matching from Chapters 1–2. **Route:** read this before Chapter 5. Chapter 4 is an optional detour; Chapter 6 will express the evaluator below as a fold, and Chapter 11 will parse and extend it.

A program’s result is only one part of its behavior. Which expression runs first? What remains to be done when a recursive call returns? Where is that work stored? We will answer these questions for one expression language, changing its representation of pending work without changing its arithmetic operations.

3.1 Composition and scope

Composition builds a function; a pipeline applies one:

```
let compose f g x = f (g x)
let twice f x = f (f x)
let () =
  assert (compose string_of_int ((+) 1) 3 = "4");
  assert ((3 |> ((+) 1) |> string_of_int) = "4");
  assert (twice ((+) 1) 3 = 5)
```

In `compose f g`, `g` runs first. In `x |> g |> f`, the same order is written left to right. Partial application supplies fewer arguments than a function expects: `((+) 1)` waits for its second integer. None of these operations changes OCaml’s evaluation strategy.



Figure 4.1: Chapter 3 illustration

To evaluate `let x = e in body`, evaluate `e`, then evaluate `body` with the name `x` bound to that value. A later binding shadows an earlier one only within its scope. Substitution is another account of this process, but it must avoid capturing free variables (Chapter 11).

3.2 One language, with a specified order

Our language has numbers, named variables, four binary operations and local bindings. Here is its entire syntax and the meaning of its primitive operations:

```
type op = Add | Sub | Mul | Div

type t =
  | Number of float
  | Variable of string
  | Binary of op * t * t
  | Let of string * t * t

exception Unbound of string
```

```

let apply op x y =
  match op with
  | Add -> x +. y | Sub -> x -. y
  | Mul -> x *. y | Div -> x /. y

let lookup env x =
  match List.assoc_opt x env with
  | Some v -> v
  | None -> raise (Unbound x)

```

The source is `projects/expressions/expr.ml`. Every displayed excerpt marked with an MDX file reference is checked against that source. Its tests run with `dune runtest projects/expressions`; the book’s later chapters use this same type.

A value environment is a list of `(name, value)` pairs. Lookup takes the first matching pair, so adding a binding at the front implements lexical shadowing. An unbound variable raises `Unbound`; floating-point division by zero follows OCaml’s floating-point operations rather than raising integer `Division_by_zero`.

```

let rec eval env = function
| Number n -> n
| Variable x -> lookup env x
| Binary (op, a, b) ->
  let x = eval env a in
  let y = eval env b in
  apply op x y
| Let (x, value, body) ->
  let v = eval env value in
  eval ((x, v) :: env) body

```

The two `let` bindings in the binary case make the language **left to right**. OCaml does not specify the order of evaluating arguments to a function. Writing `apply op (eval env a) (eval env b)` would leave our interpreter’s order dependent on its implementation. Order becomes observable if both branches fail.

```

open Expressions.Expr

let program = Let ("x", Number 3.,
  Binary (Add, Binary (Mul, Variable "x", Number 4.), Number 2.))
let () = assert (eval [] program = 14.)

```

The big-step rule for addition says: evaluate the left operand to x , evaluate the right operand to y , then return $x + y$. A small-step account records the intermediate configurations instead. For example:

```

let x = 3 in x * 4 + 2
-> 3 * 4 + 2
-> 12 + 2
-> 14

```

This trace uses exact small integers representable as floats. It does not license arbitrary real-algebra identities on machine numbers. For example, `0. * . nan` is NaN; replacing an expression `0 * e` by `0` can also suppress an unbound-variable error. Reassociation may change rounding. We distinguish a mathematical real semantics, with its domain assumptions, from this executable float semantics.

3.3 Tail recursion records unfinished work

A call is in tail position when its result is returned without further work by the caller. In `1 + f x`, addition remains; in `f (x + 1)`, it does not. An accumulator can move that remaining work into an argument:

```

let rec length = function [] -> 0 | _ :: xs -> 1 + length xs
let length_tail xs =
  let rec loop n = function [] -> n | _ :: xs -> loop (n + 1) xs in
  loop 0 xs
let () = assert (length [1;2;3] = length_tail [1;2;3])

```

The tail-recursive version needs constant call-stack space. This says nothing by itself about heap allocation or the size of its arguments. Nor is there always a single numerical accumulator. A tree traversal needs to remember unvisited branches:

```

type tree = Tip | Node of tree * tree
let rec depth = function

```

```

| Tip -> 0
| Node (a, b) -> 1 + max (depth a) (depth b)

let depth_worklist tree =
  let rec visit best = function
    | [] -> best
    | (Tip, _) :: todo -> visit best todo
    | (Node (a, b), d) :: todo ->
      let d = d + 1 in
      visit (max best d) ((a, d) :: (b, d) :: todo)
  in visit 0 [tree, 0]

let () =
  let t = Node (Node (Tip, Tip), Tip) in
  assert (depth t = 2);
  assert (depth_worklist t = depth t)

```

The worklist is a heap representation of pending visits. On a depth-first traversal its live length is bounded by tree height. Tail recursion removes the corresponding recursive call stack, not the need to remember those visits.

3.4 From direct evaluation to CPS

In continuation-passing style (CPS), the evaluator receives a function *k* meaning “what to do with the answer”. Each return becomes an application of *k*:

```

let rec eval_cps env e k =
  match e with
  | Number n -> k n
  | Variable x -> k (lookup env x)
  | Binary (op, a, b) ->
    eval_cps env a (fun x ->
      eval_cps env b (fun y -> k (apply op x y)))
  | Let (x, value, body) ->
    eval_cps env value (fun v ->
      eval_cps ((x, v) :: env) body k)

```

```

let () = assert (eval_cps [] program Fun.id = eval [] program)

```

There are three forms of pending work. After evaluating a left operand, evaluate the right operand in the saved environment. After evaluating that right operand, combine its result with the saved left value. After evaluating

a binding's value, enter its body in an extended environment. These are the three closure shapes created by the code. `Fun.id` represents completion.

All recursive evaluator calls are tail calls. Continuation closures live on the heap; their live depth is proportional to the expression nesting. Saved lexical environments also retain reachable bindings. This transformation preserves control order but does not promise constant total memory.

3.5 Defunctionalization: closures become data

We know every continuation shape, so we can replace its code pointer and captured values by a constructor and fields. This is **defunctionalization**. `Right` records a pending right operand; `Combine` records a pending arithmetic operation; `Bind` records a pending binding body. A list of frames ends with `[]`, the identity continuation.

```
type frame =
  | Right of op * t * (string * float) list
  | Combine of op * float
  | Bind of string * t * (string * float) list

type state =
  | Eval of t * (string * float) list * frame list
  | Return of float * frame list

let step = function
  | Eval (Number n, _, stack) -> Return (n, stack)
  | Eval (Variable x, env, stack) -> Return (lookup env x, stack)
  | Eval (Binary (op, a, b), env, stack) ->
    Eval (a, env, Right (op, b, env) :: stack)
  | Eval (Let (x, value, body), env, stack) ->
    Eval (value, env, Bind (x, body, env) :: stack)
  | Return (x, Right (op, b, env) :: stack) ->
    Eval (b, env, Combine (op, x) :: stack)
  | Return (y, Combine (op, x) :: stack) ->
    Return (apply op x y, stack)
  | Return (v, Bind (x, body, env) :: stack) ->
    Eval (body, (x, v) :: env, stack)
  | Return (_, []) as final -> final

let rec run = function
  | Return (v, []) -> v
  | state -> run (step state)
```

```
let eval_machine env e = run (Eval (e, env, []))
```

Pending work becomes data

Evaluate $(2 + 3) * 4$ left to right. Stack top is on the left.

Step	Current state	Stack: the work still to do	
0	Eval $((2 + 3) * 4)$	[]	
1	Eval $(2 + 3)$	R(Mul, 4)	
2	Eval 2	R(Add, 3)	R(Mul, 4)
3	Return 2	R(Add, 3)	R(Mul, 4)
4	Eval 3	C(Add, 2)	R(Mul, 4)
5	Return 3	C(Add, 2)	R(Mul, 4)
6	Return 5	R(Mul, 4)	
7	Eval 4	C(Mul, 5)	
8	Return 4	C(Mul, 5)	
9	Return 20	[]	

At step 2, the context is $(\text{hole} + 3) * 4$.

$R(\text{op}, b) = \text{Right}(\text{op}, b, [])$; $C(\text{op}, x) = \text{Combine}(\text{op}, x)$.

Figure 4.2: The ten machine states for $(2 + 3) * 4$. The stack records the work still pending, with its top on the left.

The machine alternates between evaluating syntax and returning a value to its frames. Every call to **step** performs one transition. The **run** loop is tail recursive; it is now possible to pause the interpreter by retaining a **state**, inspect it, or count transitions without changing its language.

For Binary (Add, Number 2., Number 3.) in an empty environment, the frame lists are [], [Right (Add, Number 3., [])], [Combine (Add, 2.)], and []. There are also Return/Eval changes between them. Writing out those intermediate states is a useful check that no operand is evaluated twice.

Chapter 2's subtree context remembers how to reconstruct a tree. A machine frame remembers how to finish a computation. **Right** still carries a

subtree, whereas `Combine` carries an already computed number: an evaluation context records the progress of computation, not just a missing piece of syntax.

```
let () =
  assert (eval_machine [] program = 14.);
  let e = Binary (Add, Variable "first", Variable "second") in
  let failure interpret =
    try ignore (interpret e); None with Unbound x -> Some x in
  assert (failure (eval []) = Some "first");
  assert (failure (eval_machine []) = Some "first")
```

Why the representations agree. Interpret each frame list as its original nested continuation. `Right` maps to the first closure in `eval_cps`, `Combine` to the second, and `Bind` to the third. Every machine transition then performs one piece of the corresponding CPS calculation. Induction on the finite expression shows that direct evaluation and CPS agree; the frame interpretation transfers that result to the machine. Equality here means the same float result (including NaN classification) or the same first `Unbound` error, ignoring resource exhaustion. The tests also compare generated expressions and exercise a nesting depth of 100,000 for the CPS and machine versions; tests support, but do not replace, this argument.

3.6 Exercises and a further project

1. **Practice.** List every state of the machine for `program`. Check that `x` is visible in the body but not in the value expression of its own `Let`.
2. **Proof.** State a worklist invariant for `depth_worklist` and prove the result equals `depth`. Hint: each queued depth is the depth just above its subtree; `best` is the greatest node depth already visited.
3. **Experiment.** Count machine transitions for a balanced tree of additions and a left-nested tree with the same number of nodes. Explain equal work but different maximum frame depth. Do not use wall-clock timings as an allocation measurement.
4. **Practice.** Add unary negation to all three evaluators and to the machine. State the new continuation shape and add a regression containing an unbound operand.
5. **Project.** Continue with `projects/symbolic/README.md` for symbolic differentiation, printing and carefully scoped algebraic simplification.

Selected answer (1). At **Let**, the machine first pushes **Bind** ("x", **body**, []). After returning 3., it evaluates **body** under [("x", 3.)]. The nested multiplication returns 12., the addition returns 14., and **Return** (14., []) is final. A **Let** binding is nonrecursive; **Let** ("x", **Variable** "x", ...) cannot supply its own initial value.

Chapter 5: Modules, invariants, and executable laws

Prerequisites: Chapters 1–3; functions, lists, patterns and lexical scope. **Route:** complete Part I here, then follow Chapter 6. The longer type-inference and polymorphic-recursion lesson is now [projects/type-inference/README.md](#).

A module signature states what can be called. It cannot by itself say whether `remove` really removes a key. We will specify maps, expose a counterexample, and check alternative representations through the same observations.

5.1 Enough polymorphism to read an interface

In `'a list -> 'a list`, `'a` is a type parameter. A polymorphic function may be used at several instances, but a single list still contains one element type:

```
let twice f x = f (f x)
let () =
  assert (twice ((+) 1) 3 = 5);
  assert (twice List.rev [true; false] = [true; false])
```

A weak type variable printed as `'_weak...` has a different meaning: it is one unknown type that must eventually be fixed. A mutable cell cannot safely be used as both an integer-list cell and a string-list cell. The value restriction limits generalization of such definitions. Type inference solves equations over unknowns; using a polymorphic binding creates fresh instances



Figure 5.1: Chapter 5 illustration

of its quantified parameters. The optional route derives these equations in detail.

For the map examples below, keys use OCaml polymorphic equality and ordering. We restrict our executable laws to integer keys and string values. This avoids functions, NaN and other cases that need a more explicit equality contract. For reusable maps, `Map.Make` takes an ordered key module, making that contract part of the interface. “Polymorphic” alone does not guarantee valid comparison.

5.2 Algebraic Specification

Now we turn to a fundamental question in computer science: how do we formally describe what a data structure *is* and what it should *do*? The mathematical answer is *algebraic specification*.

The way we introduce a data structure, like complex numbers or strings, in mathematics is by specifying an *algebraic structure*. This approach gives us a precise language for describing data structures independent of any particular implementation.

Algebraic structures consist of a set (or several sets, for so-called *multi-sorted* algebras) and a bunch of functions (also known as operations) over

this set (or sets). Think of integers with addition and multiplication, or strings with concatenation and character access.

A *signature* is a rough description of an algebraic structure: it provides *sorts* – names for the sets (in the multisorted case) – and names of the functions-operations together with their arity (and what sorts of arguments they take). A signature tells us what operations exist, but not how they behave.

An algebraic specification adds equations to a signature. For total operations, a class defined by equations is called a variety. The partial operations and inequalities below require additional conventions; they are not automatically an instance of that narrower definition.

Here is the key connection to programming: algebraic structures correspond to “implementations” and signatures to “interfaces” in programming languages. We will say that an algebraic structure *implements* an algebraic specification when all axioms of the specification hold in the structure. A specification can admit several representations, a unique model up to isomorphism, or no model if its requirements conflict. For maps, we deliberately allow different representations with the same observable behavior.

We say that an algebraic structure does not have *junk* when all its elements (i.e., elements in the sets corresponding to sorts) can be built using operations in its signature. Junk-free structures are “minimal” in some sense – they contain only the values that can be constructed using the provided operations.

We allow parametric types as sorts. In that case, strictly speaking, we define a family of algebraic specifications (a different specification for each instantiation of the parametric type).

Algebraic Specifications: Examples

Let us look at some concrete examples to make these abstract ideas tangible. An algebraic specification can also use an earlier specification, building up complexity layer by layer. We must specify failure explicitly. Here **error** denotes a distinguished failed result, propagated by dependent operations. OCaml can express this using **option** or **result**; an exception-based interface must instead name the exception and its triggering condition.

Specification nat_p (bounded natural numbers):

For an integer bound $p \geq 2$, this specification describes natural numbers modulo p (like unsigned machine integers). Range conditions below refer to the canonical representatives $0, \dots, p - 1$:

 nat_p

 $0 : \text{nat}_p$ $\text{succ} : \text{nat}_p \rightarrow \text{nat}_p$ $+: \text{nat}_p \rightarrow \text{nat}_p \rightarrow \text{nat}_p$ $* : \text{nat}_p \rightarrow \text{nat}_p \rightarrow \text{nat}_p$ Variables: $n, m : \text{nat}_p$

Axioms:

 $0 + n = n, n + 0 = n$ $m + \text{succ}(n) = \text{succ}(m + n)$ $0 * n = 0, n * 0 = 0$ $m * \text{succ}(n) = m + (m * n)$ $\underbrace{\text{succ}(\dots \text{succ}(0))}_{k \text{ times, } 1 \leq k < p} \neq 0$ $\underbrace{\text{succ}(\dots \text{succ}(0))}_{p \text{ times}} = 0$

The axioms define how addition and multiplication work recursively, and the last two axioms capture the bounded nature: applying succ between one and $p - 1$ times never gives zero, but exactly p times wraps around to zero.

Specification string_p (bounded strings):

This specification describes strings of length strictly less than p . Here **error** denotes failure outside the successful result sort, and operations propagate failure. Thus these are partial-operation equations, not a plain total algebra over only the displayed sorts:

 string_p

uses $\text{char}, \text{nat}_p$ $" " : \text{string}_p$ $"c" : \text{char} \rightarrow \text{string}_p$ $\hat{\cdot} : \text{string}_p \rightarrow \text{string}_p \rightarrow \text{string}_p$ $\cdot[\cdot] : \text{string}_p \rightarrow \text{nat}_p \rightarrow \text{char}$ Variables: $s : \text{string}_p, c, c_1, \dots, c_p : \text{char}, n : \text{nat}_p$

Axioms:

 $" " \hat{\cdot} s = s, s \hat{\cdot} " " = s$ $\underbrace{"c_1" \hat{\cdot} (\dots \hat{\cdot} "c_p")}_{p \text{ times}} = \text{error}$ $r \hat{\cdot} (s \hat{\cdot} t) = (r \hat{\cdot} s) \hat{\cdot} t$ $(\text{"c"} \hat{\cdot} s)[0] = c$

string_p

$(\text{"c"}^\wedge s)[\text{succ}(n)] = s[n]$

$\text{" "[n]} = \text{error}$

Both indexing equations involving a prefixed character require the concatenation to succeed. The successor-index equation additionally requires $n < p - 1$, so the index does not wrap to zero.

The axioms specify that concatenation is associative, that the empty string is an identity for concatenation, that exceeding the length limit produces an error, and that indexing works by stripping characters from the front.

5.3 Homomorphisms

When do two implementations of the same specification “behave the same”? The mathematical answer involves *homomorphisms* – structure-preserving mappings between algebraic structures.

Homomorphisms are mappings between algebraic structures with the same signature that preserve operations. Intuitively, if you apply an operation and then map, you get the same result as mapping first and then applying the corresponding operation.

A *homomorphism* from algebraic structure $(A, \{f^A, g^A, \dots\})$ to $(B, \{f^B, g^B, \dots\})$ is a function $h : A \rightarrow B$ such that:

- $h(f^A(a_1, \dots, a_{n_f})) = f^B(h(a_1), \dots, h(a_{n_f}))$ for all $(a_1, \dots, a_{n_f})$
- $h(g^A(a_1, \dots, a_{n_g})) = g^B(h(a_1), \dots, h(a_{n_g}))$ for all $(a_1, \dots, a_{n_g})$
- and so on for all operations.

Two algebraic structures are *isomorphic* if there are homomorphisms $h_1 : A \rightarrow B$, $h_2 : B \rightarrow A$ from one to the other and back, that when composed in any order form identity: $\forall(b \in B) h_1(h_2(b)) = b$ and $\forall(a \in A) h_2(h_1(a)) = a$.

An algebraic specification whose all implementations without junk are isomorphic is called “*monomorphic*”. This means the specification pins down the structure so precisely that there’s essentially only one way to implement it (up to isomorphism).

We usually only add axioms that really matter to us to the specification, so that the implementations have room for optimization. For this reason, the resulting specifications will often not be monomorphic in the above sense –

and that’s intentional! A non-monomorphic specification allows for multiple genuinely different implementations, which may have different performance characteristics.

5.4 Example: Maps

Now let us look at a practical example that will guide the rest of this chapter. A *map* (also called dictionary or associative array) associates keys with values. This is one of the most fundamental data structures in programming – think of Python’s dictionaries, Java’s `HashMap`, or OCaml’s `Map` module.

Here is an algebraic specification that captures the essential behavior of maps:

(α, β) map

uses bool, type parameters α, β
empty : (α, β) map
member : $\alpha \rightarrow (\alpha, \beta)$ map $\rightarrow$ bool
add : $\alpha \rightarrow \beta \rightarrow (\alpha, \beta)$ map $\rightarrow (\alpha, \beta)$ map
remove : $\alpha \rightarrow (\alpha, \beta)$ map $\rightarrow (\alpha, \beta)$ map
find : $\alpha \rightarrow (\alpha, \beta)$ map $\rightarrow \beta$
Variables: $k, k_2 : \alpha, v, v_2 : \beta, m : (\alpha, \beta)$ map
Axioms:
member(k , empty) = false
member(k , add(k, v, m)) = true
member(k , remove(k, m)) = false
member(k , add(k_2, v, m)) = true $\wedge k \neq k_2 \Leftrightarrow$ member(k, m) = true $\wedge k \neq k_2$
member(k , remove(k_2, m)) = true $\wedge k \neq k_2 \Leftrightarrow$ member(k, m) = true $\wedge k \neq k_2$
find(k , add(k, v, m)) = v
find(k , remove(k, m)) = error, find(k , empty) = error
find(k , add(k_2, v_2, m)) = $v \wedge k \neq k_2 \Leftrightarrow$ find(k, m) = $v \wedge k \neq k_2$
find(k , remove(k_2, m)) = $v \wedge k \neq k_2 \Leftrightarrow$ find(k, m) = $v \wedge k \neq k_2$
remove(k , empty) = empty

The axioms capture the intuitive behavior: adding a key-value pair makes that key findable, removing a key makes it unfindable, and operations on different keys don’t interfere with each other. Notice how the specification says nothing about *how* the map is implemented – only about *what* behavior it must exhibit.

5.5 Modules and Interfaces (Signatures): Syntax

How do we express algebraic specifications in OCaml? The answer is the *module system*. In the ML family of languages, structures are given names by **module** bindings, and signatures are types of modules. From outside of a structure or signature, we refer to the values or types it provides with a dot notation: `Module.value`.

Module (and module type) names have to start with a capital letter (in ML languages). Since modules and module types have names, there is a convention to name the central type of a signature (the one that is “specified” by the signature), for brevity, `t`. Module types are often named with “all-caps” (all letters upper case).

Here is how we translate our map specification into an OCaml module signature:

```
module type MAP = sig
  type ('a, 'b) t
  val empty : ('a, 'b) t
  val member : 'a -> ('a, 'b) t -> bool
  val add : 'a -> 'b -> ('a, 'b) t -> ('a, 'b) t
  val remove : 'a -> ('a, 'b) t -> ('a, 'b) t
  val find : 'a -> ('a, 'b) t -> 'b
end

module CounterexampleListMap : MAP = struct
  type ('a, 'b) t = ('a * 'b) list
  let empty = []
  let member = List.mem_assoc
  let add k v m = (k, v)::m
  let remove = List.remove_assoc
  let find = List.assoc
end
```

`CounterexampleListMap` matches the signature but violates the laws. Adding the same key twice creates two bindings; `List.remove_assoc` removes only the first, exposing the older one. This is a named counterexample, not our map implementation. The annotation `: MAP` checks types and hides representation; it does not prove behavioral equations.

```
let () =
  let module M = CounterexampleListMap in
  let m = M.add 1 "new" (M.add 1 "old" M.empty) in
```

```

assert (M.find 1 m = "new");
assert (M.member 1 (M.remove 1 m))  (* The required law would say
    ⇨ false. *)

```

The successful `find` laws concern equality of returned values. A missing key must raise `Not_found`. Equality between maps means **observational equality**: all lookups return the same optional result, not equality of internal trees. The module system enforces abstraction; our law checks enforce selected behavior.

5.6 Implementing Maps: Association Lists

Let us now build an implementation of maps from the ground up, exploring different approaches and their trade-offs. The most straightforward implementation... might not be what you expected:

```

module TrivialMap : MAP = struct
  type ('a, 'b) t =
    | Empty
    | Add of 'a * 'b * ('a, 'b) t
    | Remove of 'a * ('a, 'b) t

  let empty = Empty

  let rec member k m =
    match m with
    | Empty -> false
    | Add (k2, _, _) when k = k2 -> true
    | Remove (k2, _) when k = k2 -> false
    | Add (_, _, m2) -> member k m2
    | Remove (_, m2) -> member k m2

  let add k v m = Add (k, v, m)
  let remove k m = Remove (k, m)

  let rec find k m =
    match m with
    | Empty -> raise Not_found
    | Add (k2, v, _) when k = k2 -> v
    | Remove (k2, _) when k = k2 -> raise Not_found
    | Add (_, _, m2) -> find k m2
    | Remove (_, m2) -> find k m2
end

```

This “trivial” implementation is quite clever in its own way: it simply records all operations as a log! The data structure itself is a history of everything that has been done to it. The `add` and `remove` operations are $O(1)$ – they just prepend a new node. However, `member` and `find` must traverse the entire history to determine the current state, giving them $O(n)$ complexity where n is the number of operations performed.

This implementation illustrates an important point: there are many ways to satisfy the same specification, with very different performance characteristics.

Here is a more conventional implementation based on association lists, i.e., on lists of key-value pairs without the `Remove` constructor:

```
module MyListMap : MAP = struct
  type ('a, 'b) t = Empty | Add of 'a * 'b * ('a, 'b) t

  let empty = Empty

  let rec member k m =
    match m with
    | Empty -> false
    | Add (k2, _, _) when k = k2 -> true
    | Add (_, _, m2) -> member k m2

  let rec add k v m =
    match m with
    | Empty -> Add (k, v, Empty)
    | Add (k2, _, m) when k = k2 -> Add (k, v, m)
    | Add (k2, v2, m) -> Add (k2, v2, add k v m)

  let rec remove k m =
    match m with
    | Empty -> Empty
    | Add (k2, _, m) when k = k2 -> m
    | Add (k2, v, m) -> Add (k2, v, remove k m)

  let rec find k m =
    match m with
    | Empty -> raise Not_found
    | Add (k2, v, _) when k = k2 -> v
    | Add (_, _, m2) -> find k m2
end
```

This implementation maintains the invariant that each key appears at most once in the structure. The `add` function replaces an existing key’s value

rather than creating a duplicate, and `remove` actually removes the key-value pair. All operations are still $O(n)$ in the worst case, but the structure stays cleaner.

5.7 Implementing Maps: Binary Search Trees

Can we do better than linear time? Yes, by using a smarter data structure. Binary search trees are binary trees with elements stored at the interior nodes, such that elements to the left of a node are smaller than, and elements to the right bigger than, elements within a node. This ordering property is what makes them efficient.

For maps, we store key-value pairs as elements in binary search trees, and compare the elements by keys alone. The tree structure allows us to use “divide-and-conquer” to search for the value associated with a key.

Operations cost $O(h)$, where h is tree height. Random insertion order gives expected $O(\log n)$ height; a search discards one subtree at each step, but that subtree need not contain half the elements. However, in the worst case (when keys are inserted in sorted order), the tree degenerates into a linked list and operations become $O(n)$.

A note on our design: the simple polymorphic signature for maps is only possible because OCaml provides polymorphic comparison (and equality) operators that work on elements of most types (but not on functions). These operators may not behave as you expect for all types! Our signature for polymorphic maps is not the standard approach because of this limitation; it is just to keep things simple for pedagogical purposes.

```
module BTreeMap : MAP = struct
  type ('a, 'b) t = Empty | T of ('a, 'b) t * 'a * 'b * ('a, 'b) t

  let empty = Empty

  let rec member k m =
    (* "Divide and conquer" search
    ↪ through the tree. *)
    match m with
    | Empty -> false
    | T (_, k2, _, _) when k = k2 -> true
    | T (m1, k2, _, _) when k < k2 -> member k m1
    | T (_, _, _, m2) -> member k m2
```

```

let rec add k v m =                               (* Searches the tree in the
↳ same way as member *)
  match m with                                     (* but copies every node along
↳ the way. *)
  | Empty -> T (Empty, k, v, Empty)
  | T (m1, k2, _, m2) when k = k2 -> T (m1, k, v, m2)
  | T (m1, k2, v2, m2) when k < k2 -> T (add k v m1, k2, v2, m2)
  | T (m1, k2, v2, m2) -> T (m1, k2, v2, add k v m2)

let rec split_rightmost m =                       (* A helper function, it does
↳ not belong *)
  match m with                                     (* to the "exported" signature.
↳ *)
  | Empty -> raise Not_found
  | T (m1, k, v, Empty) -> k, v, m1               (* Preserve the largest
↳ node's left child. *)
  | T (m1, k, v, m2) ->                          (* the one that is on the
↳ bottom right. *)
    let rk, rv, rm = split_rightmost m2 in
    rk, rv, T (m1, k, v, rm)

let rec remove k m =
  match m with
  | Empty -> Empty
  | T (m1, k2, _, Empty) when k = k2 -> m1
  | T (Empty, k2, _, m2) when k = k2 -> m2
  | T (m1, k2, _, m2) when k = k2 ->
    let rk, rv, rm = split_rightmost m1 in
    T (rm, rk, rv, m2)
  | T (m1, k2, v, m2) when k < k2 -> T (remove k m1, k2, v, m2)
  | T (m1, k2, v, m2) -> T (m1, k2, v, remove k m2)

let rec find k m =
  match m with
  | Empty -> raise Not_found
  | T (_, k2, v, _) when k = k2 -> v
  | T (m1, k2, _, _) when k < k2 -> find k m1
  | T (_, _, _, m2) -> find k m2
end

```

The `member` and `find` functions use the “divide-and-conquer” strategy: compare the target key with the key at the current node, and recursively search in the appropriate subtree. The `add` function searches the tree in the same way but copies every node along the path to create the new tree (since we’re using immutable data structures).

The `remove` function is trickier. When removing a node with two children, we need to replace it with another value that maintains the ordering property. The `split_rightmost` helper function finds and removes the rightmost (largest) element from a subtree – this element is guaranteed to be smaller than everything in the right subtree and larger than everything remaining in the left subtree, making it the perfect replacement.

Removing a root must also preserve the left child of its predecessor:

```
let () =
  let m = List.fold_left (fun m k -> BTreeMap.add k (string_of_int
    ↪ k) m)
    BTreeMap.empty [5; 3; 2; 7] in
  let m = BTreeMap.remove 5 m in
  assert (not (BTreeMap.member 5 m));
  List.iter (fun k -> assert (BTreeMap.find k m = string_of_int k))
    ↪ [2; 3; 7]
```

5.8 One law suite for every map

A functor is a module parameterized by another module. This one takes a map and checks it without knowing the representation. It interprets missing lookup as `None` solely for comparison, retaining `Not_found` as the public contract.

```
module Map_laws (M : MAP) = struct
  let find_opt k m = try Some (M.find k m) with Not_found -> None
  let observe m = List.map (fun k -> find_opt k m)
    ↪ [0;1;2;3;4;5;6;7]
  let check m =
    List.iter (fun k ->
      assert (M.member k m = Option.is_some (find_opt k m));
      let added = M.add k "new" (M.add k "old" m) in
      assert (find_opt k added = Some "new");
      assert (not (M.member k (M.remove k added)));
      assert (find_opt k (M.remove k added) = None);
      List.iter (fun j -> if j <> k then begin
        assert (find_opt j (M.add k "new" m) = find_opt j m);
        assert (find_opt j (M.remove k m) = find_opt j m)
      end) [0;1;2;3;4;5;6;7]) [0;1;2;3;4;5;6;7]
  let run () =
    assert (observe M.empty = List.init 8 (fun _ -> None));
```

```

let rec histories depth m =
  check m;
  if depth > 0 then
    List.iter (fun k ->
      histories (depth - 1) (M.add k (string_of_int k) m);
      histories (depth - 1) (M.remove k m)) [1;2;3] in
    histories 3 M.empty
end

module Log_laws = Map_laws (TrivialMap)
module List_laws = Map_laws (MyListMap)
module Tree_laws = Map_laws (BTreeMap)
let () = Log_laws.run (); List_laws.run (); Tree_laws.run ()

```

The tests cover empty membership, overwrite, removal, failure and non-interference between keys across bounded operation histories. The earlier predecessor-removal regression checks a particular tree shape the general law suite might not reach. For a proof, establish each representation invariant and show it is preserved by `add` and `remove`; then prove lookup implements the abstract finite map. Bounded testing and invariant proofs have different roles.

Partial operations as executable specifications

For bounded strings, choose a small bound so that all inputs can be enumerated. Here concatenation and indexing return options; a failed inner concatenation propagates with `Option.bind`. This makes associativity a well-formed equation including its failure cases.

```

let bound = 4
let concat a b =
  if String.length a + String.length b < bound then Some (a ^ b)
  ⇨ else None
let index s i =
  if i < 0 || i >= String.length s then None else Some s.[i]
let rec strings n =
  if n = 0 then [""] else
  let shorter = strings (n - 1) in
  "" :: List.concat_map (fun c -> List.map ((^) c) shorter) ["a";
  ⇨ "b"]
let () =
  let inputs = strings (bound - 1) in
  List.iter (fun a ->

```

```

assert (concat "" a = Some a && concat a "" = Some a);
assert (index a (String.length a) = None);
List.iter (fun b -> List.iter (fun c ->
    assert (Option.bind (concat a b) (fun ab -> concat ab c) =
        Option.bind (concat b c) (fun bc -> concat a bc)))
    ↪ inputs) inputs)
inputs;
assert (concat "ab" "cd" = None);
assert (index "abc" 0 = Some 'a')

```

5.9 Optional: implementing Maps: Red-Black Trees

The fatal weakness of ordinary binary search trees is that they can become unbalanced. If keys arrive in sorted order, each insertion adds a node at the bottom of a long chain, and we lose the logarithmic performance guarantee. How can we maintain balance automatically?

This section is based on Wikipedia’s [Red-black tree article](#), Chris Okasaki’s “Purely Functional Data Structures” and Matt Might’s excellent blog post on [red-black tree deletion](#).

Binary search trees are good when we encounter keys in random order, because the cost of operations is limited by the depth of the tree which is small relative to the number of nodes... unless the tree grows unbalanced achieving large depth (which means there are sibling subtrees of vastly different sizes on some path).

To remedy this, we *rebalance* the tree while building it – i.e., while adding elements. The key insight is to detect when the tree is becoming unbalanced and perform local rotations to restore balance.

In *red-black trees* we achieve balance by: 1. Remembering one of two colors (red or black) with each node 2. Keeping the same number of black nodes on every path from the root to a leaf 3. Not allowing a red node to have a red child

These invariants together guarantee that the tree cannot become too unbalanced: the depth is at most twice the depth of a perfectly balanced tree with the same number of nodes. Why? The “black height” (number of black nodes on any root-to-leaf path) is the same everywhere, and red nodes can only appear between black nodes, so the longest path can have at most twice as many nodes as the shortest.

B-trees of Order 4 (2-3-4 Trees)

To understand where red-black trees come from, it helps to first understand 2-3-4 trees (also known as B-trees of order 4).

How can we have perfectly balanced trees without worrying about having exactly $2^k - 1$ elements? The answer is to allow variable-width nodes. **2-3-4 trees** can store from 1 to 3 elements in each node and have 2 to 4 subtrees correspondingly. This flexibility lets us maintain perfect balance!

- A **2-node** contains one element and has two children
- A **3-node** contains two elements and has three children
- A **4-node** contains three elements and has four children

To insert into a 2-3-4 tree, we descend toward the appropriate leaf position. But if we encounter a full node (4-node) along the way, we “split” it: move the middle element up to the parent and split the remaining two elements into separate 2-nodes. This maintains perfect balance at all times – all leaves are at the same depth.

Red-black trees represent 2-3-4 nodes using binary nodes and colors. To represent a 2-3-4 tree as a binary tree with one element per node, we color the “primary” element of each node black (the middle element of a 4-node, or the first element of a 2-/3-node) and make it the parent of its neighbor elements colored red. The red elements then become parents of the original subtrees. This correspondence provides the deep intuition behind red-black trees: the colors encode the structure of the underlying 2-3-4 tree.

Red-Black Trees, Without Deletion

Now let us implement red-black trees in OCaml. Red-black trees maintain two invariants:

Invariant 1. No red node has a red child. (No two consecutive red nodes on any path.)

Invariant 2. Every path from the root to an empty node contains the same number of black nodes. (The “black height” is uniform.)

For simplicity, we first implement red-black tree based *sets* (not maps) without deletion. The implementation proceeds almost exactly like for unbalanced binary search trees; we only need to add code to restore the invariants after each insertion.

In Okasaki’s approach, by keeping balance at each step of constructing a node, it is enough to check *locally* (around the root of the subtree) whether a violation has occurred. We never need to examine the entire tree. One

implementation of deletion introduces more colors – see Matt Might’s post for details.

```

type color = R | B
type 'a t = E | T of color * 'a t * 'a * 'a t

let empty = E

let rec member x m =
  (* Like in unbalanced
  ↪ binary search tree. *)
  match m with
  | E -> false
  | T (_, _, y, _) when x = y -> true
  | T (_, a, y, _) when x < y -> member x a
  | T (_, _, _, b) -> member x b

let balance = function
  (* Restoring the
  ↪ invariants. *)
  | B, T (R, T (R,a,x,b), y, c), z, d (* On next figure: left,
  ↪ *)
  | B, T (R, a, x, T (R,b,y,c)), z, d (* top, *)
  | B, a, x, T (R, T (R,b,y,c), z, d) (* bottom, *)
  | B, a, x, T (R, b, y, T (R,c,z,d)) (* right, *)
  -> T (R, T (B,a,x,b), y, T (B,c,z,d)) (* center tree. *)
  | color, a, x, b -> T (color, a, x, b) (* We allow red-red
  ↪ violation for now. *)

let insert x s =
  let rec ins = function
    (* Like in unbalanced
    ↪ binary search tree, *)
    | E -> T (R, E, x, E) (* but fix violation above
    ↪ created node. *)
    | T (color, a, y, b) as s ->
      if x < y then balance (color, ins a, y, b)
      else if x > y then balance (color, a, y, ins b)
      else s
  in
  match ins s with
  (* We could still have
  ↪ red-red violation *)
  | T (_, a, y, b) -> T (B, a, y, b) (* at root, fixed by
  ↪ coloring it black. *)
  | E -> failwith "insert: impossible"

```

The `balance` function is the heart of the algorithm. It handles four cases where a red-red violation occurs (a red node with a red child). The four cases correspond to different positions of the violation:

- A red left child with a red left grandchild
- A red left child with a red right grandchild
- A red right child with a red left grandchild
- A red right child with a red right grandchild

In each case, we perform a “rotation” that restructures the tree to eliminate the violation while maintaining the binary search tree property. All four cases produce the same balanced result: a red root with two black children, with the subtrees `a`, `b`, `c`, `d` properly distributed.

The `insert` function works like insertion into an ordinary binary search tree, but calls `balance` after each recursive step to fix any violations that may have been introduced. New nodes are always created red (which might create a red-red violation that `balance` will fix). At the very end, we color the root black – this can never create a violation and ensures the root is always black.

The insertion invariant can also be checked independently of lookup:

```
let check_red_black tree =
  let rec inspect lower upper = function
    | E -> 0
    | T (color, left, x, right) ->
      assert (Option.fold ~none:true ~some:(fun lo -> lo < x)
        ↪ lower);
      assert (Option.fold ~none:true ~some:(fun hi -> x < hi)
        ↪ upper);
      let red = function T (R, _, _, _) -> true | _ -> false in
      assert (color <> R || not (red left || red right));
      let a = inspect lower (Some x) left in
      let b = inspect (Some x) upper right in
      assert (a = b);
      a + if color = B then 1 else 0 in
  (match tree with E | T (B, _, _, _) -> () | _ -> assert false);
  ignore (inspect None None tree)
let () =
  let test xs = ignore (List.fold_left (fun tree x ->
    let tree = insert x tree in check_red_black tree; tree) E xs)
    ↪ in
  test (List.init 100 Fun.id);
```

```
test (List.init 100 (fun i -> 99 - i));  
test [3;1;4;1;5;9;2;6;5]
```

5.10 Exercises

1. **Practice.** Repair `CounterexampleListMap` by ensuring each key occurs once. Instantiate `Map_laws` with the repair. State its invariant and operation costs.
2. **Proof.** Show `remove` preserves the binary-search ordering invariant. In the two-child case, explain why the predecessor's left child must be retained.
3. **Experiment.** Add a deliberate bug to one map and record the smallest law counterexample. Prefer the operation sequence to a dump of internal nodes.
4. **Project.** Specify a FIFO queue with `take : 'a t -> ('a * 'a t) option`. Implement one-list and two-list representations, compare operation traces, and distinguish amortized cost from worst-case cost of a single operation.
5. **Proof.** Prove bounded-string concatenation's partial associativity. Hint: if the sum of the three lengths is below the bound, both intermediate sums are too; otherwise both complete expressions fail.
6. **Project.** Extend the map tests to a comparator-parameterized interface. Give the comparator a total-order contract and test a non-integer key type.

Selected answer (1). Replace an existing binding on insertion, or remove *all* matching bindings on removal. The former maintains a unique-key invariant; the latter allows duplicate history internally but still meets the lookup/removal laws. With observational equality those are legitimate different representations.

Chapter 4: Functions as a language (optional)

Prerequisites: Chapter 3’s distinction between syntax and evaluation; functions and recursive datatypes. **Route:** this is an optional Part I detour. Continue to Chapter 5 without it if you want modules and useful data structures first.

Can functions alone represent booleans, numbers and recursion? The untyped lambda calculus asks this question with three syntax forms: variables, functions and application. We will calculate with that syntax in a small OCaml interpreter. The interpreter is typed OCaml; its object language is untyped. No unsafe cast or recursive OCaml type is needed to represent a self-application.

4.1 Equations and evaluation are different

The notation $\lambda x.e$ binds x in e . Application associates to the left: $f\ x\ y$ means $(f\ x)\ y$. A function body extends as far to the right as possible, so $\lambda x.f\ x$ means $\lambda x.(f\ x)$.

Three familiar equations have different jobs:

- **Alpha:** rename a bound variable consistently, avoiding capture.
- **Beta:** $(\lambda x.e)\ v = e[x := v]$, with capture-avoiding substitution.
- **Eta:** $\lambda x.f\ x = f$ when x is not free in f , as an extensional equation of the pure calculus. This is not unrestricted contextual equivalence for effectful or diverging strict programs.

Full beta reduction permits work inside a function body and in an argument before it is needed. **Normal order** chooses the leftmost outermost

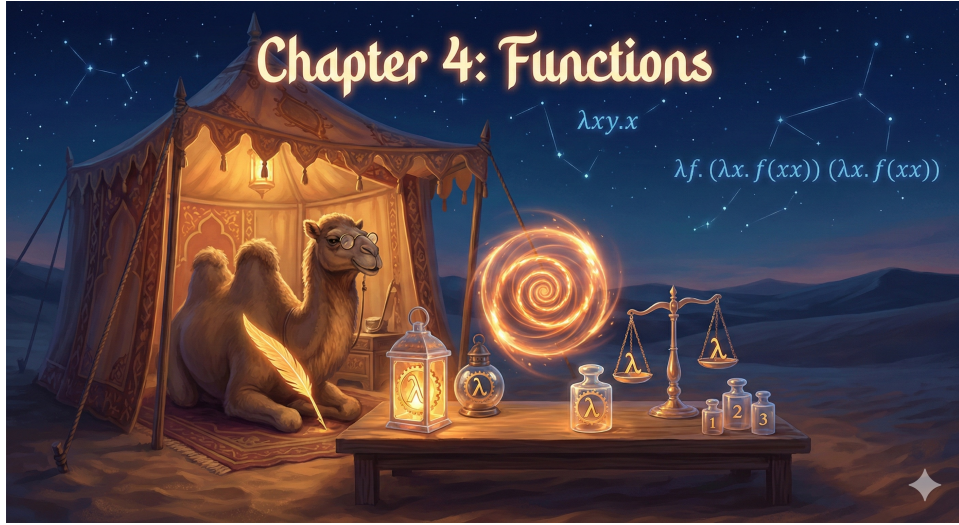


Figure 6.1: Chapter 4 illustration

redex and continues under lambdas. **Weak call by value** first evaluates a function and its argument to values, and does not reduce inside an unapplied lambda. OCaml is a strict language, but our interpreter specifies a left-to-right operand order explicitly; we do not infer OCaml's argument order from this object-language rule.

For example, let $I = \lambda x.x$ and $\Omega = (\lambda x.xx)(\lambda x.xx)$. Normal order reduces $(\lambda \text{ignored}.I)\Omega$ to I without touching Ω . Weak call by value tries to evaluate Ω and never reaches the body.

4.2 Names without accidental capture

Naively replacing x by y in $\lambda y.x$ produces $\lambda y.y$, which captures the free y . The correct result is $\lambda z.y$ for a fresh z . We can avoid bound-name choices entirely: **de Bruijn indices** count binders outward from each occurrence. **Bound 0** refers to the nearest enclosing lambda; **Bound 1** refers to the next one. Free variables keep their names.

Thus $\lambda x.x$ is **Lam (Bound 0)** and $\lambda x.\lambda y.x$ is **Lam (Lam (Bound 1))**. Alpha-equivalent terms have the same representation. An index must be nonnegative and smaller than the number of enclosing binders; **well_scoped** in the source checks that precondition.

```

type t = Bound of int | Free of string | Lam of t | App of t * t

let rec shift amount cutoff = function
  | Bound k -> Bound (if k >= cutoff then k + amount else k)
  | Free _ as x -> x
  | Lam body -> Lam (shift amount (cutoff + 1) body)
  | App (f, x) -> App (shift amount cutoff f, shift amount cutoff
    ↪ x)

let rec substitute index value = function
  | Bound k as x -> if k = index then value else x
  | Free _ as x -> x
  | Lam body -> Lam (substitute (index + 1) (shift 1 0 value) body)
  | App (f, x) -> App (substitute index value f, substitute index
    ↪ value x)

let beta body argument =
  shift (-1) 0 (substitute 0 (shift 1 0 argument) body)

```

`shift amount cutoff` adjusts indices that refer outside the binders we have already crossed. Entering a lambda increments the cutoff. Substitution under a lambda shifts its replacement up by one, because the replacement now occurs under one more binder. Beta reduction shifts the argument up before substitution, then shifts the result down when removing the applied lambda.

These three shifts can seem bureaucratic. Trace `beta (Lam (Bound 1)) (Free "y")`: `Bound 1` under the inner lambda refers to the parameter being replaced; the answer is `Lam (Free "y")`, not `Lam (Bound 0)`. A free variable never becomes a bound index. The named substitution in Chapter 11 implements the same obligation by choosing fresh names instead.

4.3 Two executable reduction strategies

```

let rec normal_step = function
  | App (Lam body, argument) -> Some (beta body argument)
  | App (f, x) ->
    (match normal_step f with
     | Some f' -> Some (App (f', x))
     | None -> Option.map (fun x' -> App (f, x')) (normal_step x))
  | Lam body -> Option.map (fun body -> Lam body) (normal_step
    ↪ body)
  | Bound _ | Free _ -> None

```

```

let rec value_step = function
  | App (Lam body, (Lam _ as argument)) -> Some (beta body
    ↪ argument)
  | App ((Lam _ as f), x) ->
    Option.map (fun x -> App (f, x)) (value_step x)
  | App (f, x) -> Option.map (fun f -> App (f, x)) (value_step f)
  | Bound _ | Free _ | Lam _ -> None

type result = Done of t | Limit of t
let reduce ~fuel step term =
  if fuel < 0 then invalid_arg "negative reduction fuel";
  let rec loop fuel term =
    match step term with
    | None -> Done term
    | Some _ when fuel = 0 -> Limit term
    | Some term -> loop (fuel - 1) term in
  loop fuel term

```

The interpreter lives in `projects/expressions/lambda.ml`. Its tests include capture, shadowing, reduction under lambdas, Church addition, and Scott predecessor. The `Done` result means no step is available under the selected strategy. For weak call by value on an *open* term that can mean stuck, rather than a value. Use closed, well-scoped terms when claiming a value result. `Limit` means only that we spent the chosen beta-step budget; it does not decide whether the term diverges. Searching for a redex still traverses syntax, so fuel is not a bound on allocation or total interpreter work.

```

open Expressions.Lambda
let identity = Lam (Bound 0)
let self = Lam (App (Bound 0, Bound 0))
let omega = App (self, self)
let discarded = App (Lam identity, omega)
let () =
  assert (reduce ~fuel:20 normal_step discarded = Done identity);
  assert (match reduce ~fuel:20 value_step discarded with
    | Limit _ -> true | Done _ -> false);
  assert (beta (Lam (Bound 1)) (Free "y") = Lam (Free "y"))

```

We deliberately store `omega` as finite syntax. Evaluating an equivalent self-applying OCaml function would instead run in the host language and could prevent the experiment from finishing.

4.4 Booleans and products choose their consumers

A Church boolean selects a branch:

$$true = \lambda t. \lambda f. t, \quad false = \lambda t. \lambda f. f.$$

An if-expression is **b t f**; conjunction can be **a b false**. A pair packages two values for a consumer:

$$pair = \lambda a. \lambda b. \lambda k. k a b.$$

Apply the pair to **true** to select its first component and to **false** to select its second. The representation exposes the elimination operation rather than constructors and pattern matching.

```
let yes = Lam (Lam (Bound 1))
let no = Lam (Lam (Bound 0))
let select b t f = App (App (b, t), f)
let () =
  assert (reduce ~fuel:20 normal_step (select yes identity omega) =
    ↪ Done identity)
```

A strict evaluator would evaluate both supplied branches. To implement an if-like operation under call by value, wrap each branch in a lambda (a thunk), select a thunk, and apply it to a dummy value. The distinction is operational: writing a boolean encoding alone does not make its arguments lazy.

4.5 Church iteration and Scott case analysis

A **Church numeral** is an iterator:

$$0 = \lambda f. \lambda x. x, \quad 1 = \lambda f. \lambda x. f x, \quad 2 = \lambda f. \lambda x. f (f x).$$

Addition composes iterations: $add = \lambda m. \lambda n. \lambda f. \lambda x. m f (n f x)$. The implementation below constructs the syntax, not a host-language numeral:

```
let church n =
  if n < 0 then invalid_arg "negative numeral";
  let rec times n =
    if n = 0 then Bound 0 else App (Bound 1, times (n - 1)) in
  Lam (Lam (times n))
```

```

let add = Lam (Lam (Lam (Lam
  (App (App (Bound 3, Bound 1), App (App (Bound 2, Bound 1), Bound
    ↪ 0))))))
let () =
  assert (reduce ~fuel:100 normal_step (App (App (add, church 2),
    ↪ church 3))
    = Done (church 5))

```

A **Scott numeral** selects between a zero case and a successor case, giving its predecessor directly to the latter:

$$zero = \lambda z. \lambda s. z, \quad succ = \lambda n. \lambda z. \lambda s. s n.$$

```

let scott_zero = Lam (Lam (Bound 1))
let scott_succ = Lam (Lam (Lam (App (Bound 0, Bound 2))))
let predecessor n = App (App (n, scott_zero), identity)
let () =
  assert (reduce ~fuel:30 normal_step
    (predecessor (App (scott_succ, scott_zero))) = Done scott_zero)

```

Scott predecessor chooses a constructor case. Church predecessor instead needs to carry additional state through an iteration, for instance a pair of successive counts. Church lists similarly encode a fold, while Scott lists expose the head and tail to the nonempty case. Calling both encodings “Church” hides the distinction between iteration and one-step case analysis.

4.6 Recursion and cost

In the untyped calculus, the fixed-point combinator

$$Y = \lambda f. (\lambda x. f(x x)) (\lambda x. f(x x))$$

satisfies $Yf \rightsquigarrow^* f(Yf)$. This equation provides a recursive call; it does not prove that any invocation terminates. Under weak call by value, this Y attempts self-application too early. A delayed variant can expose a lambda before recurring, but its operational argument must be checked separately. OCaml’s `let rec` is the practical construct for recursive functions; encoding Y is an experiment about evaluation strategy, not a replacement for it.

Counting beta reductions also omits the work of copying syntax during substitution. The arithmetic machine from Chapter 3 stores an environment instead of replacing every variable occurrence. An environment-and-closure

lambda machine is a useful next project precisely because it changes this cost model.

4.7 Exercises

1. **Practice.** Encode $\lambda x.\lambda y.y x$ with indices. Selected answer: `Lam (Lam (App (Bound 0, Bound 1)))`.
2. **Proof.** Explain why shifting the replacement when entering a lambda is necessary. Give a well-scoped term whose reduction would capture a variable if that shift were omitted.
3. **Experiment.** Evaluate the same term with `normal_step` and `value_step`. Use one example that differs only under a lambda and one discarded divergent argument. Report fuel exhaustion as inconclusive, not as a divergence proof.
4. **Practice.** Implement Church multiplication and check $2 * 3 = 6$ by comparing normal forms with `church 6`. Write Scott numerals zero through three and check predecessor at zero and three.
5. **Project.** Build an environment-and-closure interpreter for closed terms. Specify the strategy, compare terminating cases against `value_step`, and measure syntax copying versus environment retention. Include a shadowing test.

Part II: Representations and interpreters

Chapter 6: Folding and Backtracking

Prerequisites: Chapter 3’s expression language; Chapter 5’s module interfaces. **Route:** Part II begins here. Continue to Chapter 11 for binding and extension, or Chapter 7 for streams.

In this chapter, you will:

- Identify common recursion patterns and refactor them into `map/fold` abstractions
- Make folds tail-recursive using accumulators (and understand the trade-offs)
- Generalize `map/fold` beyond lists to trees and expression grammars
- Use backtracking (via lists) to solve search problems and puzzles

This chapter explores two fundamental programming paradigms in functional programming: **folding** (also known as reduction) and **backtracking**. We begin with the classic `map` and `fold` higher-order functions, examine how they generalize to trees and other data structures, then move on to solving puzzles using backtracking with lists.

The material in this chapter draws from Martin Odersky’s “Functional Programming Fundamentals,” Ralf Laemmel’s “Going Bananas,” Graham Hutton’s “Programming in Haskell” (Chapter 11 on the Countdown Problem), and Tomasz Wierzbicki’s Honey Islands Puzzle Solver.

6.1 Basic Generic List Operations

Functional programming emphasizes identifying common patterns and abstracting them into reusable higher-order functions. Rather than writing



Figure 7.1: A camel assembles expression trees in a lamplit workshop

similar code repeatedly, we extract the common structure into a single generic function. Let us see how this principle works in practice through two motivating examples.

The `map` Function

How do we print a comma-separated list of integers? The `String` module provides a function that joins strings with a separator:

```
val concat : string -> string list -> string
```

But `String.concat` works on strings, not integers. So first, we need to convert numbers into strings:

```

let rec strings_of_ints = function
| [] -> []
| hd::tl -> string_of_int hd :: strings_of_ints tl

let comma_sep_ints = String.concat ", " -| strings_of_ints

```

Here is another common task: how do we sort strings from shortest to longest? We can pair each string with its length and then sort by the first component. First, let us compute the lengths:

```

let rec strings_lengths = function
| [] -> []
| hd::tl -> (String.length hd, hd) :: strings_lengths tl

let by_size = List.sort compare -| strings_lengths

```

Now, look carefully at `strings_of_ints` and `strings_lengths`. Do you notice the common structure? Both functions traverse a list and transform each element independently – one applies `string_of_int`, the other applies a function that pairs a string with its length. The recursive structure is identical; only the transformation differs.

This is our cue to *extract the common pattern* into a generic higher-order function. We call it `map`:

```

let rec list_map f = function
| [] -> []
| hd::tl -> f hd :: list_map f tl

```

Now we can rewrite our functions more concisely:

```

let comma_sep_ints =
  String.concat ", " -| list_map string_of_int

let by_size =
  List.sort compare -| list_map (fun s -> String.length s, s)

```

The fold Function

Now let us consider a different kind of pattern. How do we sum all the elements of a list?

```
let rec balance = function
| [] -> 0
| hd::tl -> hd + balance tl
```

And how do we multiply all the elements together (perhaps to compute a cumulative ratio)?

```
let rec total_ratio = function
| [] -> 1.
| hd::tl -> hd *. total_ratio tl
```

Again, the recursive structure is the same. In both cases, we combine each element with the result of processing the rest of the list. The differences are: (1) what we return for the empty list (the “base case” or “identity element”), and (2) how we combine the head with the recursive result. This pattern is called **folding**:

```
let rec list_fold f base = function
| [] -> base
| hd::tl -> f hd (list_fold f base tl)
```

Important: Note that `list_fold f base l` equals `List.fold_right f l base`. The OCaml standard library uses a different argument order, so be careful when using `List.fold_right`.

The key insight is understanding the fundamental difference between `map` and `fold`:

- **map** alters the *contents* of a data structure without changing its shape. The output list has the same length as the input; we merely transform each element.
- **fold** *collapses* a data structure down to a single value, using the structure itself as scaffolding for the computation.

Visually, consider what happens to the list `[a; b; c; d]`:

- `map f` transforms: `[a; b; c; d]` becomes `[f a; f b; f c; f d]` – same structure, different contents
- `fold f accu` collapses: `[a; b; c; d]` becomes `f a (f b (f c (f d accu)))` – structure disappears, single value remains

6.2 Making Fold Tail-Recursive

Our `list_fold` function above is not tail-recursive: it builds up a chain of deferred `f` applications on the call stack. For very long lists, this can cause stack overflow. Can we make folding tail-recursive?

Let us investigate some tail-recursive list functions to find a pattern. Consider reversing a list:

```
let rec list_rev acc = function
| [] -> acc
| hd::tl -> list_rev (hd::acc) tl
```

The key technique here is the *accumulator* parameter `acc`. Instead of building up work to do after the recursive call returns, we do the work *before* the recursive call and pass the intermediate result along.

Here is another example – computing an average by tracking both the running sum and the count:

```
let rec average (sum, tot) = function
| [] when tot = 0. -> 0.
| [] -> sum /. tot
| hd::tl -> average (hd +. sum, 1. +. tot) tl
```

Notice how these functions process elements from left to right, threading an accumulator through the computation. This is the pattern of `fold_left`:

```
let rec fold_left f accu = function
| [] -> accu
| a::l -> fold_left f (f accu a) l
```

With `fold_left`, expressing our earlier functions becomes straightforward – we hide the accumulator inside the initial value:

```
let list_rev l =
  fold_left (fun t h -> h::t) [] l

let average xs =
  let sum, tot =
    fold_left (fun (sum, tot) e -> sum +. e, 1. +. tot) (0., 0.) xs
  in
  if tot = 0. then 0. else sum /. tot
```

```
let () = assert (average [2.; 4.; 9.] = 5.)
```

Note that the `average` example is slightly trickier than `list_rev` because we need to track two values (sum and count) rather than one.

Why the names `fold_right` and `fold_left`? The names reflect the associativity of the combining operation:

- `fold_right f` makes `f` **right associative**, like the list constructor `::`: `List.fold_right f [a1; ...; an] b` is `f a1 (f a2 (... (f an b) ...))`
- `fold_left f` makes `f` **left associative**, like function application: `List.fold_left f a [b1; ...; bn]` is `f (... (f (f a b1) b2) ...) bn`

This “backward” structure of `fold_left` can be visualized by comparing the shape of the input list with the shape of the computation tree. The input list has a right-leaning spine (because `::` associates to the right), while `fold_left` produces a computation tree with a left-leaning spine:

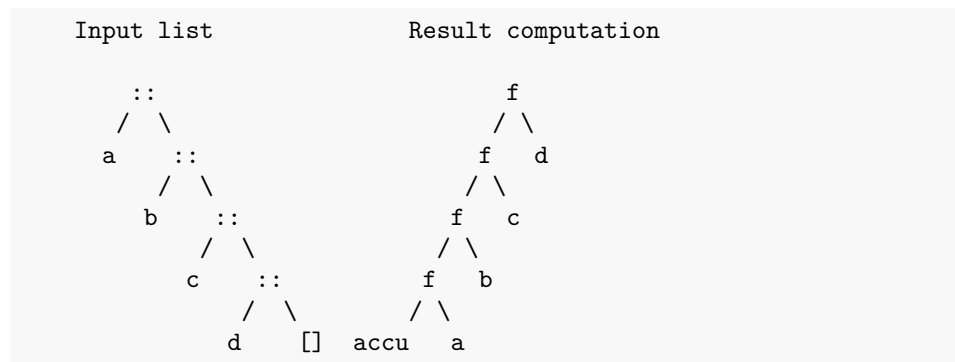


Figure: List spine vs. `fold_left` computation tree

This reversal of structure is why `fold_left` naturally reverses lists when the combining operation is `cons`.

Useful Derived Functions

Many common list operations can be expressed using folds. List filtering selects elements satisfying a predicate – naturally expressed using `fold_right` to preserve order:

```
let list_filter p l =
  List.fold_right (fun h t -> if p h then h::t else t) l []
```

When we need a tail-recursive map and can tolerate reversed output, `fold_left` gives us `rev_map`:

```
let list_rev_map f l =
  List.fold_left (fun t h -> f h :: t) [] l
```

6.3 Map and Fold for Trees and Other Structures

The `map` and `fold` patterns are not limited to lists. They apply to any recursive data structure. The key insight is that `map` preserves structure while transforming contents, and `fold` collapses structure into a single value.

Binary Trees

Mapping binary trees is straightforward:

```
type 'a btree = Empty | Node of 'a * 'a btree * 'a btree

let rec bt_map f = function
  | Empty -> Empty
  | Node (e, l, r) -> Node (f e, bt_map f l, bt_map f r)

let test = Node
  (3, Node (5, Empty, Empty), Node (7, Empty, Empty))
let _ = bt_map ((+) 1) test
```

A note on terminology: The `map` and `fold` functions we define here preserve and respect the structure of data. They are different from the `map` and `fold` operations you might find in abstract data type container libraries, which often behave more like `List.rev_map` and `List.fold_left` over container elements in arbitrary order. Here we are generalizing `List.map` and `List.fold_right` to other structures.

For binary trees, the most general form of `fold` processes each element together with the partial results already computed for its subtrees:

```
let rec bt_fold f base = function
  | Empty -> base
  | Node (e, l, r) ->
```

```
f e (bt_fold f base l) (bt_fold f base r)
```

Here are two examples showing how `bt_fold` can compute different properties of a tree:

```
let sum_els = bt_fold (fun i l r -> i + l + r) 0
let depth t = bt_fold (fun _ l r -> 1 + max l r) 0 t
```

The first computes the sum of all elements (the combining function adds the current element to the sums of both subtrees). The second computes the depth – we ignore the element value and take the maximum depth of the subtrees, adding 1 for the current level.

The same expression language as Chapter 3

A shape-preserving map changes the labels at existing positions, preserving constructors. A fold may produce a number, a function, or a differently shaped tree. Calling all these operations “map” hides the distinction.

Here is the fold for `Expressions.Expr.t`. The algebra has one field per constructor. The two recursive positions of `Let` are its value and body; the binding name itself is a label. This traversal handles syntax, not lexical scope.

```
type 'a algebra = {
  number : float -> 'a;
  variable : string -> 'a;
  binary : op -> 'a -> 'a -> 'a;
  binding : string -> 'a -> 'a -> 'a;
}

let rec fold alg = function
| Number n -> alg.number n
| Variable x -> alg.variable x
| Binary (op, a, b) ->
  let a' = fold alg a in
  let b' = fold alg b in
  alg.binary op a' b'
| Let (x, value, body) ->
  let value' = fold alg value in
  let body' = fold alg body in
  alg.binding x value' body'

let rebuild = {
```

```

number = (fun n -> Number n);
variable = (fun x -> Variable x);
binary = (fun op a b -> Binary (op, a, b));
binding = (fun x value body -> Let (x, value, body));
}

let size = fold {
  number = (fun _ -> 1); variable = (fun _ -> 1);
  binary = (fun _ a b -> 1 + a + b);
  binding = (fun _ value body -> 1 + value + body);
}

let eval_fold = fold {
  number = (fun n _env -> n);
  variable = (fun x env -> lookup env x);
  binary = (fun op a b env ->
    let x = a env in let y = b env in apply op x y);
  binding = (fun x value body env ->
    let v = value env in body ((x, v) :: env));
}

```

The last algebra is the subtle one. A fold cannot evaluate a `Let` body to a number before it knows the binding's value. Instead its carrier is `environment -> float`: each subtree becomes a function waiting for an environment. The `binding` handler then passes an extended environment to the body function. The explicit `lets` preserve the left-to-right order of Chapter 3.

```

open Expressions.Expr
let example = Let ("x", Number 3.,
  Binary (Add, Variable "x", Number 4.))
let () =
  assert (size example = 5);
  assert (fold rebuild example = example);
  assert (eval_fold example [] = eval [] example)

```

A bottom-up rewrite is a fold with carrier `t`. For example `simplify` in the shared module folds *literal* binary operations using the same float operation as evaluation. It does not erase a variable lookup or reassociate arithmetic. It takes one traversal: children are already simplified when the parent is processed. A general rewrite system may need iteration, but this one does not.

```

let () =
  let e = Binary (Add, Binary (Mul, Number 2., Number 3.), Variable
    ↪ "x") in
  assert (simplify e = Binary (Add, Number 6., Variable "x"));
  assert (eval ["x", 1.] (simplify e) = eval ["x", 1.] e)

```

The identity algebra `rebuild` gives a useful law: `fold rebuild e = e` for finite syntax. Prove it by induction, using one case per constructor. Chapter 12 will identify the equations that make this fold unique.

6.4 Point-Free Programming

We can compose functions without naming each intermediate value.

This style is sometimes called **point-free** or **tacit** programming, because we never mention the “points” (values) that functions operate on – we only talk about the functions themselves and how they combine.

To write in this style, we need a toolkit of **combinators** – higher-order functions that combine other functions. Here are some common ones, similar to what you will find in the *OCaml Batteries* library:

```

let const x _ = x
let ( | - ) f g x = g (f x)           (* forward composition *)
let ( - | ) f g x = f (g x)           (* backward composition *)
let flip f x y = f y x
let ( *** ) f g = fun (x,y) -> (f x, g y)
let ( &&& ) f g = fun x -> (f x, g x)
let first f x = fst (f x)
let second f x = snd (f x)
let curry f x y = f (x,y)
let uncurry f (x,y) = f x y

```

One way to understand point-free programming is to visualize the flow of computation as a circuit. Values flow through the circuit, being transformed by functions at each node. Cross-sections of the circuit can be represented as tuples of intermediate values.

Consider this simple function that converts a character and an integer to a string:

```

let print2 c i =
  let a = Char.escaped c in
  let b = string_of_int i in

```

```
a ^ b
```

We can visualize this as a circuit: (c, i) enters, c flows through `Char.escaped`, i flows through `string_of_int`, and the results meet at $(^)$. In point-free style, we express this directly:

```
let print2 = curry
  ((Char.escaped *** string_of_int) |- uncurry (^))
```

Here `***` applies one function to each component of a pair (this does not start parallel execution), `|-` is forward composition, `uncurry` converts a curried function to take a pair, and `curry` converts back.

Why the name “currying”? Converting a C/Pascal-style function (that takes all arguments as a tuple) into one that takes arguments one at a time is called *currying*, after the logician Haskell Brooks Curry. Since OCaml functions naturally take arguments one at a time, we often need `uncurry` to interface with tuple-based operations, and `curry` to convert back.

Another approach to point-free style avoids tuples entirely, using function composition, `flip`, and the `S` combinator:

```
let s x y z = x z (y z)
```

The `S` combinator allows us to pass one argument to two different functions and combine their results. This can bring a particular argument of a function to the “front” and pass it to another function.

Here is an extended example showing step-by-step transformation of a filter-map function into point-free style:

```
let func2 f g l = List.filter f (List.map g l)
(* Step 1: Recognize that filter-after-map is composition *)
let func2 f g = (-|) (List.filter f) (List.map g)
(* Step 2: Eliminate l by composing with List.map *)
let func2 f = (-|) (List.filter f) -| List.map
(* Step 3: Rewrite without infix notation to see the structure *)
let func2 f = (-|) ((-|) (List.filter f)) List.map
(* Step 4: Use flip to rearrange arguments *)
let func2 f = flip (-|) List.map ((-|) (List.filter f))
(* Step 5: Factor out f using composition *)
let func2 f = (((-|) List.map) -| ((-|) -| List.filter)) f
(* Step 6: Finally, f disappears (eta-reduction) *)
let func2 = (-|) List.map -| ((-|) -| List.filter)
```

While point-free style can be elegant for simple cases, it can quickly become obscure. Use it judiciously!

6.5 Reductions and More Higher-Order Functions

Mathematics has a convenient notation for sums over intervals: $\sum_{n=a}^b f(n)$.

Can we express this in OCaml? The challenge is that OCaml does not have a universal addition operator – `+` works only on integers, `+.` only on floats. So we end up writing two versions:

```
let rec i_sum_fromto f a b =
  if a > b then 0
  else f a + i_sum_fromto f (a+1) b

let rec f_sum_fromto f a b =
  if a > b then 0.
  else f a +. f_sum_fromto f (a+1) b

let pi2_over6 =
  f_sum_fromto (fun i -> 1. /. float_of_int (i*i)) 1 5000
```

(The last example computes an approximation to $\pi^2/6$ using the Basel series.)

The natural generalization is to make the combining operation a parameter:

```
let rec op_fromto op base f a b =
  if a > b then base
  else op (f a) (op_fromto op base f (a+1) b)
```

Collecting Results: `concat_map`

Sometimes a function produces not a single result but a *collection* of results. In mathematics, such a function is called a **multifunction** or set-valued function. If we have a multifunction f and want to apply it to every element of a set A , we take the union of all results:

$$f(A) = \bigcup_{p \in A} f(p)$$

When we represent sets as lists, “union” becomes “append”. This gives us the extremely useful `concat_map` operation:


```
let rec concat_map f = function
| [] -> []
| a::l -> f a @ concat_map f l
```

For better efficiency on long lists, here is a tail-recursive version:

```
let concat_map f l =
  let rec cmap_f accu = function
  | [] -> accu
  | a::l -> cmap_f (List.rev_append (f a) accu) l in
  List.rev (cmap_f [] l)
```

The `concat_map` function is fundamental for backtracking algorithms. We will use it extensively in the puzzle-solving sections below.

All Subsequences of a List

A classic example of a function that produces multiple results: given a list, generate all its subsequences (subsets that preserve order). The idea is simple: for each element, we either include it or exclude it.

```
let rec subseqs l =
  match l with
  | [] -> [[]]
  | x::xs ->
    let pxs = subseqs xs in
    List.map (fun px -> x::px) pxs @ pxs
```

Using a tail-recursive mapping helper (the call to `subseqs` itself is still not in tail position):

```
let rec rmap_append f accu = function
| [] -> accu
| a::l -> rmap_append f (f a :: accu) l

let rec subseqs l =
  match l with
  | [] -> [[]]
  | x::xs ->
    let pxs = subseqs xs in
    rmap_append (fun px -> x::px) pxs pxs
```

Permutations and Choices

Generating all permutations of a list is another classic combinatorial problem. The key insight is the `interleave` function: given an element `x` and a list, it produces all ways of inserting `x` into the list:

```
let rec interleave x = function
| [] -> [[x]]                (* x can only go in one place: by
↪ itself *)
| y::ys ->
  (x::y::ys)                 (* x goes at the front, OR *)
  :: List.map (fun zs -> y::zs) (interleave x ys) (* x goes
↪ somewhere after y *)

let rec perms = function
| [] -> [[]]                 (* one way to permute empty list:
↪ empty list *)
| x::xs -> concat_map (interleave x) (perms xs)
```

For example, `interleave 1 [2;3]` produces `[[1;2;3]; [2;1;3]; [2;3;1]]` – all positions where 1 can be inserted.

For the Countdown problem below, we will need all non-empty subsequences with all their permutations – that is, all ways of choosing and ordering elements from a list:

```
let choices l = concat_map perms (List.filter ((<>) []) (subseqs
↪ l))
```

6.6 Grouping and Map-Reduce

When processing large datasets, it is often useful to organize values by some property – grouping all items with the same key together, then processing each group. This pattern is so common it has a name: **map-reduce** (popularized by Google for distributed computing).

Collecting by Key

The first step is to collect elements from an association list, grouping all values that share the same key:

```

let collect l =
  match List.sort (fun x y -> compare (fst x) (fst y)) l with
  | [] -> [] (* Start with associations *)
  ↪ sorted by key *)
  | (k0, v0)::tl ->
    let k0, vs, l = List.fold_left
      (fun (k0, vs, l) (kn, vn) -> (* Collect values for
    ↪ current key *)
        if k0 = kn then k0, vn::vs, l (* Same key: add value to
        ↪ current group *)
        else kn, [vn], (k0, List.rev vs)::l) (* New: save current
        ↪ group, start new *)
      (k0, [v0], []) tl in (* Why reverse? To preserve
    ↪ original order *)
    List.rev ((k0, List.rev vs)::l)

```

Now we can group elements by an arbitrary property – we just need to extract the property as the key:

```

let group_by p l = collect (List.map (fun e -> p e, e) l)

```

Reduction (Aggregation)

Grouping alone is often not enough – we want to *aggregate* each group into a summary value, like SQL’s SUM, COUNT, or AVG. This aggregation step is called **reduction**:

```

let aggregate_by p red base l =
  let ags = group_by p l in
  List.map (fun (k, vs) -> k, List.fold_right red vs base) ags

```

Using the **feed-forward** (pipe) operator `let ( |> ) x f = f x`:

```

let aggregate_by p redf base l =
  group_by p l
  |> List.map (fun (k, vs) -> k, List.fold_right redf vs base)

```

Often it is cleaner to extract both the key and the value we care about upfront, before grouping. Since we first **map** elements into key-value pairs, then group and **reduce**, we call this pattern **map_reduce**:

```
let map_reduce mapf redf base l =
  List.map mapf l
  |> collect
  |> List.map (fun (k, vs) -> k, List.fold_right redf vs base)
```

Map-Reduce Examples

Sometimes our mapping function produces multiple key-value pairs per input (for example, when processing documents word by word). For this we use `concat_reduce`, which uses `concat_map` instead of `map`:

```
let concat_reduce mapf redf base l =
  concat_map mapf l
  |> collect
  |> List.map (fun (k, vs) -> k, List.fold_right redf vs base)
```

Example 1: Word histogram. Count how many times each word appears across a collection of documents:

```
let histogram documents =
  let mapf doc =
    Str.split (Str.regexp "[ \\t.,;]+") doc
    |> List.map (fun word -> word, 1) in
  concat_reduce mapf (+) 0 documents
```

Example 2: Inverted index. Build an index mapping each word to the list of documents (identified by address) containing it:

```
let cons hd tl = hd::tl

let inverted_index documents =
  let mapf (addr, doc) =
    Str.split (Str.regexp "[ \\t.,;]+") doc
    |> List.map (fun word -> word, addr) in
  concat_reduce mapf cons [] documents
  |> List.map (fun (word, addresses) -> word, List.sort_uniq
    ↪ compare addresses)
```

Example 3: Simple search engine. Once we have an inverted index, we can search for documents containing all of a given set of words. We need set intersection – here implemented for sets represented as sorted lists:

```

let intersect xs ys = (* Sets as sorted lists
  ↪ *)
  let rec aux acc = function
    | [], _ | _, [] -> acc
    | (x::xs' as xs), (y::ys' as ys) ->
      let c = compare x y in
      if c = 0 then aux (x::acc) (xs', ys')
      else if c < 0 then aux acc (xs', ys)
      else aux acc (xs, ys') in
  List.rev (aux [] (xs, ys))

```

Now we can build a simple search function that finds all documents containing every word in a query:

```

let search index words =
  match List.map (fun word ->
    Option.value (List.assoc_opt word index) ~default:[]) words
  ↪ with
  | [] -> []
  | idx::idcs -> List.fold_left intersect idx idcs

```

6.7 Higher-Order Functions for the Option Type

The `option` type is OCaml's way of representing values that might be absent. Rather than using null pointers (a common source of bugs), we explicitly mark possibly-missing values with `Some x` or `None`. Here are some useful higher-order functions for working with options.

First, applying a function that may fail to an optional value (this is a monadic bind, also called `flatMap`—the function `f` itself returns an `option`):

```

let map_option f = function
  | None -> None
  | Some e -> f e

```

Second, mapping a partial function over a list and keeping only the successful results:

```

let rec map_some f = function
  | [] -> []
  | e::l -> match f e with

```

```
| None -> map_some f l  
| Some r -> r :: map_some f l
```

Tail-recursively:

```
let map_some f l =  
  let rec maps_f accu = function  
    | [] -> accu  
    | a::l -> maps_f (match f a with None -> accu  
                        | Some r -> r::accu) l in  
  List.rev (maps_f [] l)
```

6.8 The Countdown Problem Puzzle

Now we turn to solving puzzles, which will showcase the power of backtracking with lists. The **Countdown Problem** is a classic puzzle from a British TV game show:

- Using a given set of numbers and arithmetic operators $+$, $-$, $*$, $/$, construct an expression with a given value.
- All numbers, including intermediate results, must be positive integers.
- Each source number can be used at most once.

Example: - Source numbers: 1, 3, 7, 10, 25, 50 - Target: 765 - One possible solution: $(25-10) * (50+1) = 15 * 51 = 765$

We will compare solvers on small inputs before attempting this larger search.

Let us develop a solver step by step, starting with the data types.

Data Types

```
type op = Add | Sub | Mul | Div  
  
let apply op x y =  
  match op with  
  | Add -> x + y  
  | Sub -> x - y  
  | Mul -> x * y  
  | Div -> x / y  
  
let valid op x y =  
  match op with
```

```

| Add -> true
| Sub -> x > y
| Mul -> true
| Div -> x mod y = 0

type expr = Val of int | App of op * expr * expr

let rec eval = function
| Val n -> if n > 0 then Some n else None
| App (o, l, r) ->
    eval l |> map_option (fun x ->
        eval r |> map_option (fun y ->
            if valid o x y then Some (apply o x y)
            else None))

let rec values = function
| Val n -> [n]
| App (_, l, r) -> values l @ values r

let rec remove_one x = function
| [] -> None
| y::ys when x = y -> Some ys
| y::ys -> Option.map (fun rest -> y::rest) (remove_one x ys)

let rec uses_available numbers available =
    match numbers with
    | [] -> true
    | x::xs ->
        match remove_one x available with
        | None -> false
        | Some rest -> uses_available xs rest

let solution e ns n =
    uses_available (values e) ns && eval e = Some n

```

The source numbers form a multiset: equal numbers may be used as many times as they occur, but no more.

```

let () =
  let two = App (Add, Val 1, Val 1) in
  assert (solution two [1; 1] 2);
  assert (not (solution two [1] 2))

```

Brute Force Solution

Our strategy is to generate all possible expressions from the source numbers, then filter for those that evaluate to the target. To build expressions, we need to split the numbers into two groups (for the left and right operands of an operator).

First, a helper to split a list into two non-empty parts in all possible ways:

```

let split l =
  let rec aux lhs acc = function
    | [] | [_] -> []
    | [y; z] -> (List.rev (y::lhs), [z])::acc
    | hd::rhs ->
      let lhs = hd::lhs in
      aux lhs ((List.rev lhs, rhs)::acc) rhs in
  aux [] [] l

```

We introduce a convenient operator for working with multiple data sources. The “bind” operator `|->` takes a list of values and a function that produces a list from each value, then concatenates all results:

```

let ( |-> ) x f = concat_map f x

```

Now we can generate all expressions from a list of numbers. The structure records each branch of the backtracking search:

```

let combine l r =                                     (* Combine two expressions using
↳ each operator *)
  List.map (fun o -> App (o, l, r)) [Add; Sub; Mul; Div]

let rec exprs = function
  | [] -> []                                           (* No expressions from empty
↳ list *)
  | [n] -> [Val n]                                    (* Single number: just Val n *)
  | ns ->

```



```

split ns |-> (fun (ls, rs) -> (* For each way to split
↪ numbers... *)
  exprs ls |-> (fun l ->      (* ...for each expression l from
↪ left half... *)
    exprs rs |-> (fun r ->    (* ...for each expression r from
↪ right half... *)
      combine l r)))          (* ...produce all l op r
↪ combinations *)

```

Read the nested `|->` as “for each ... for each ... for each ...”. This is the essence of backtracking: we explore all combinations systematically.

Finally, to find solutions, we try all choices of source numbers (all non-empty subsets with all orderings) and filter for expressions that evaluate to the target:

```

let guard n =
  List.filter (fun e -> eval e = Some n)

let solutions ns n =
  choices ns |-> (fun ns' ->
    exprs ns' |> guard n)

```

Optimization: Fuse Generation with Testing

The brute force approach generates many invalid expressions (like $5 - 7$ which gives a negative result, or $5 / 3$ which is not an integer). We can do better by *fusing* generation with evaluation: instead of generating an expression and then checking if it is valid, we track the value alongside the expression and only generate valid subexpressions.

The key insight is to work with pairs `(e, eval e)` so that only valid subexpressions are ever generated:

```

let combine' valid (l, x) (r, y) =
  [Add; Sub; Mul; Div]
  |> List.filter (fun o -> valid o x y)
  |> List.map (fun o -> App (o, l, r), apply o x y)

let rec results valid = function
| [] -> []
| [n] -> if n > 0 then [Val n, n] else []
| ns ->
  split ns |-> (fun (ls, rs) ->

```

```

        results valid ls |-> (fun lx ->
          results valid rs |-> (fun ry ->
            combine' valid lx ry)))

let solutions_with valid ns n =
  choices ns |-> (fun ns' ->
    results valid ns'
    |> List.filter (fun (e, m) -> m = n)
    |> List.map fst) (* Discard memorized
↳ values *)

let solutions' = solutions_with valid

```

Eliminating Symmetric Cases

We can further improve performance by observing that addition and multiplication are commutative: $3 + 5$ and $5 + 3$ give the same result. Similarly, multiplying by 1 or adding/subtracting 0 are useless. We can eliminate these redundancies by strengthening the validity predicate:

```

let valid op x y =
  match op with
  | Add -> x <= y
  | Sub -> x > y
  | Mul -> x <= y && x <> 1 && y <> 1
  | Div -> x mod y = 0 && y <> 1

let solutions_optimized = solutions_with valid

```

Passing the new predicate explicitly matters: rebinding `valid` alone would not change functions already defined with the earlier binding. `solutions_optimized` eliminates symmetrical solutions on the *semantic* level (based on values) rather than the *syntactic* level (based on expression structure). This approach is both easier to implement and more effective at pruning the search space.

```

let () =
  let index = inverted_index [3, "cat cat dog"; 1, "dog cat"; 2,
    ↪ "dog"] in
  assert (search index ["cat"; "dog"] = [1; 3]);
  assert (search index ["missing"] = [])

```

6.9 A search contract before a speed claim

For Countdown, **soundness** says every returned expression uses an allowed submultiset, has positive integer intermediate results, and reaches its target. **Completeness** of the reference enumerator says every legal expression over an allowed ordering appears. Induct on its syntax: a leaf comes from a singleton choice; an internal node splits its leaf sequence into two nonempty parts and chooses one of the four operators. Enumerating subsequences and permutations supplies every allowed ordered leaf sequence. Duplicate source values can produce duplicate syntax; they do not permit an extra use of an input occurrence.

Fusing generation with evaluation preserves all valid syntax. The stronger predicate deliberately drops some syntax, so its claim is only preservation of **reachable target values**, when using any nonempty submultiset. Sorting the operands of a commutative operation preserves its value. Multiplication or division by one can be removed by using fewer inputs. That reasoning would fail for a rule requiring *every* input to be used. Integer overflow is outside the positive mathematical-integer argument; use small inputs for these checks, or add checked arithmetic before using this as a general solver.

```

let () =
  let reachable xs =
    choices xs |> List.concat_map exprs |> List.filter_map eval
    |> List.sort_uniq compare in
  let fused xs =
    choices xs |> List.concat_map (results valid) |> List.map snd
    |> List.sort_uniq compare in
  List.iter (fun xs ->
    let expected = reachable xs in
    assert (fused xs = expected);
    List.iter (fun target ->
      assert (List.for_all (fun e -> solution e xs target)
        (solutions_optimized xs target));
      let syntax xs = List.sort_uniq compare xs in

```

```

assert (syntax (solutions xs target) = syntax (solutions' xs
  ↪ target)))
expected)
[[1]; [1;1]; [1;2]; [2;3]; [1;2;3]; [2;2;3]]

```

The **Honey Islands project**, in `projects/honey/README.md`, compares direct, pruned, monadic and state-transformer solvers against exhaustive subsets. It includes a counterexample to the old “always keep the first seed” traversal. That puzzle and its drawing infrastructure are optional; Countdown is the main search case study here.

6.10 Exercises

1. **Practice.** Generate permutations of `[1;2;3]`, then of `[1;1;2]`. Distinguish positions from values. State whether duplicates are retained.
2. **Proof.** Prove the reconstruction law for the expression fold. Include `Let`; its binding name is unchanged even though both subexpressions are folded.
3. **Experiment.** Compare subtraction with `fold_left` and `fold_right` on `[1;2;3]` starting from zero. Explain why tail recursion alone does not justify replacing one fold with the other. Selected answer: the results are `-6` and `2`.
4. **Practice.** Use a fold to count syntactic variable occurrences, including bound occurrences. Then compute free variables: at `Let (x, value, body)`, remove `x` only from the body’s set, not from the value’s set.
5. **Proof.** Identify the induction hypotheses in the Countdown completeness argument. Explain why pruning `Mul 1 e` would be invalid if all inputs had to be used exactly once.
6. **Project.** Complete the Honey Islands project’s extension and measurements. Its acceptance criteria require reference equivalence before performance data.

Chapter 11: Binding, parsing, and extension

Prerequisites: Chapters 3, 5 and 6. Chapter 8 is useful for parser choice; Chapter 9 is useful for GADT indices. **Route:** Part II continues from Chapter 6; read it before Part III if you want the whole expression-language thread together.

We have evaluated one syntax directly, by CPS, by a machine and by a fold. Now we will substitute into it, parse it and compare ways of extending it. State the requirements before selecting an encoding: “extensible” can mean several incompatible things.

11.1 The extension requirements

Suppose we add unary negation and a new operation, pretty printing. We want:

1. Existing expressions to keep their evaluation behavior.
2. Negation to nest around old expressions, and inside old binary operations.
3. Independently compiled code to add syntax or operations without editing every original definition.
4. Exhaustiveness or another explicit failure policy for unknown cases.
5. Where required, result types to distinguish arithmetic from Boolean expressions.

The classic expression problem asks for extension along both datatype and operation axes, with static type safety and without modifying existing code. A runtime registry offers a different tradeoff: separately loaded extensions



Figure 8.1: Chapter 11 illustration

with explicit missing-handler failures. A GADT result index solves another problem, namely ruling out ill-typed object-language expressions. It does not automatically solve separate extensibility.

11.2 Binding is shared machinery

Recall that `Let (x, value, body)` binds `x` only in `body`. The variable remains free in `value` unless an outer binding supplies it. A substitution must preserve that scope and avoid capturing free variables in its replacement.

```
module Names = Set.Make (String)

let rec free = function
| Number _ -> Names.empty
| Variable x -> Names.singleton x
| Binary (_, a, b) -> Names.union (free a) (free b)
| Let (x, value, body) ->
    Names.union (free value) (Names.remove x (free body))

let rec names = function
| Number _ -> Names.empty
```

```

| Variable x -> Names.singleton x
| Binary (_, a, b) -> Names.union (names a) (names b)
| Let (x, value, body) ->
    Names.add x (Names.union (names value) (names body))

let fresh used x =
  let rec loop x = if Names.mem x used then loop (x ^ "'") else x
  ↪ in
  loop x

let rec subst x replacement = function
| Number _ as e -> e
| Variable y as e -> if x = y then replacement else e
| Binary (op, a, b) ->
    Binary (op, subst x replacement a, subst x replacement b)
| Let (y, value, body) ->
    let value' = subst x replacement value in
    if x = y then Let (y, value', body)
    else if not (Names.mem y (free replacement)) then
        Let (y, value', subst x replacement body)
    else
        let used = Names.add x (Names.union (names body) (names
        ↪ replacement)) in
        let z = fresh used y in
        let renamed = subst y (Variable z) body in
        Let (z, value', subst x replacement renamed)

```

If the binding name is the variable being substituted, skip its body, but still substitute in its value expression. Otherwise, if the replacement has that name free, choose a fresh name outside **all** names in the body and replacement, and outside the substituted name. Rename this binding's occurrences before descending. The recursive substitution respects inner shadowing; it is not a global string replacement. No prefix is reserved for generated names.

```

open Expressions.Expr
let () =
  let e = Let ("y", Number 1., Binary (Add, Variable "x", Variable
  ↪ "y")) in
  let e' = subst "x" (Variable "y") e in
  assert (Names.elements (free e') = ["y"]);
  assert (eval ["y", 10.] e' = 11.);
  let shadow = Let ("x", Variable "x", Variable "x") in
  assert (eval [] (subst "x" (Number 7.) shadow) = 7.)

```

The first result must use the outer $y=10$ for the replacement while preserving the inner bound value one. The optional lambda interpreter in Chapter 4 avoids bound-name choices with de Bruijn indices. These are two representations of the same scope obligation, not two different meanings of substitution.

11.3 Parse the common language

We use an explicit parenthesized grammar to keep precedence out of the first parser. It accepts finite float literals, identifiers, binary arithmetic and local binding:

```
expression := number | identifier
            | "(" operator expression expression ")"
            | "(let" identifier expression expression ")"
operator   := "+" | "-" | "*" | "/"
```

The implementation is `projects/expressions/sexp.ml` for tokenization and nested forms, then `parser.ml` for translation into `Expr.t`. The parser requires the whole input to be consumed. Unknown forms, wrong arity, missing parentheses, invalid identifiers and trailing tokens are errors. Binding scope is represented by `Let`; parsing does not evaluate or substitute the body.

```
let () =
  let open Expressions in
  let text = "(let x 3 (+ (* x 4) 2))" in
  match Parser.parse text with
  | Error message -> failwith message
  | Ok e ->
    assert (Expr.eval [] e = 14.);
    assert (Parser.parse (Parser.print e) = Ok e);
    assert (Result.is_error (Parser.parse "(+ 1 2) extra"));
    assert (Result.is_error (Parser.parse "(+ 1)"))
```

The round-trip contract is for finite numbers and identifiers admitted by the grammar, not arbitrary strings supplied directly to `Variable` or `Let`. Printing uses enough significant digits to round-trip binary64 values through decimal notation. NaN and infinities are excluded from literals; evaluation can still produce them through floating-point operations.

Choice and repetition need a consumption contract

An enumerating parser returns a list of `(answer, next_position)` pairs. Its choice can concatenate all successful parses, as list choice in Chapter 8 does. That is not commitment to the first parse. A later end-of-input check may reject the first answer and accept another.

Repetition must reject a successful parser that consumes no input:

```
type 'a parser = string array -> int -> ('a * int) list
let return x _input pos = [x,pos]
let token expected input pos =
  if pos < Array.length input && input.(pos)=expected then
    ↪ [expected,pos+1]
  else []
let rec many p input pos =
  let following = p input pos |> List.concat_map (fun (x,next) ->
    if next <= pos || next > Array.length input then
      invalid_arg "repeated parser must consume input";
      List.map (fun (xs,last) -> x::xs,last) (many p input next)) in
    ([],pos) :: following
  let () =
    assert (many (token "a") [|"a";"a"|] 0 =
      [ [],0; ["a"],1; ["a";"a"],2]);
    assert (try ignore (many (return ()) [|] 0); false
      with Invalid_argument _ -> true)
```

Positive progress bounds recursion by the remaining token count, provided each call to `p` itself terminates and returns finitely many answers. It does not make a left-recursive grammar safe. Factor left recursion or use a parser designed to handle it; wrapping it in `many` is not a termination argument.

11.4 Compare representations against the requirements

Representation	Add syntax	Add operation	What the compiler or runtime guarantees
Closed variants (<code>Expr.t</code>)	Edit the type and affected matches	Add a new function/fold algebra	Exhaustiveness checks over the closed constructor set
Extensible variants	Declare a constructor in another module	Add dispatch/handlers for cases	Unknown constructors need a fallback; no closed-world exhaustiveness
Objects with evaluation methods	Add a class satisfying the interface	Extend classes/interfaces or introduce another abstraction	Method availability and subtyping, not an automatic new-operation solution
Closed visitors	Extend visitor interface and visitors for new cases	Add a visitor over the fixed case set	Operations are extensible while the visited shape remains fixed
Polymorphic variants and open recursion	Extend rows and compose cases	Add another recursive consumer	Row constraints describe accepted cases; composition still needs design
GADT syntax	Add indexed constructors and affected matches	Add a type-indexed interpreter	Object-language result types; extension tradeoffs remain
Tagless-final modules	Extend the signature and implementations	Instantiate another interpreter	Programs abstract over operations their signature provides

These are design tradeoffs, not a ranking of universally successful or failed encodings. Dynamic loading and static exhaustiveness are particularly different requirements. Repeating an entire evaluator for every row would hide that choice.

Open recursion with polymorphic variants

Make recursion an argument to the base cases, so an extension can decide how recursive children are dispatched:

```
let eval_base recur = function
  | `Number n -> n
  | `Add (a,b) -> let x=recur a in let y=recur b in x +. y
let rec eval = function
  | (`Number _ | `Add _) as e -> eval_base eval e
  | `Neg e -> -. eval e
let () = assert (eval (`Add (`Number 2., `Neg (`Number 3.))) = -1.)
```

The extended dispatcher recurs into itself, so a negation may occur beneath an old addition. Closing recursion over only the base evaluator would lose that property. The row inferred for this example does not admit arbitrary new tags; another extension must compose another appropriate dispatcher.

11.5 Typed syntax and final representations

A GADT can distinguish numeric and Boolean results. Here is a deliberately small extension experiment; it is a typed subset, not a second full parser/evaluator:

```
type _ expression =
  | Number : float -> float expression
  | Add : float expression * float expression -> float expression
  | Less : float expression * float expression -> bool expression
  | If : bool expression * 'a expression * 'a expression -> 'a
  ⇨ expression
let rec eval : type a. a expression -> a = function
  | Number n -> n
  | Add (a,b) -> let x=eval a in let y=eval b in x +. y
  | Less (a,b) -> let x=eval a in let y=eval b in x < y
```

```

| If (condition,yes,no) -> if eval condition then eval yes else
↪ eval no
let () = assert (eval (If (Less (Number 1., Number 2.), Number 3.,
↪ Number 4.)) = 3.)

```

Add cannot accept a Boolean expression. But all four cases are still listed in `eval`, so adding a constructor may require editing this operation. An indexed extensible variant would need a missing-case policy just like an unindexed one.

Tagless-final describes a program through the operations it uses:

```

module type ARITH = sig
  type repr
  val number : float -> repr
  val add : repr -> repr -> repr
end
module Example (S : ARITH) = struct
  let value = S.add (S.number 2.) (S.number 3.)
end
module Evaluate = struct
  type repr = float
  let number x = x
  let add = (+.)
end
module Print = struct
  type repr = string
  let number = string_of_float
  let add a b = "(+ " ^ a ^ " " ^ b ^ ")"
end
module Value = Example (Evaluate)
module Text = Example (Print)
let () = assert (Value.value = 5. && Text.value = "(+ 2. 3.)")

```

A new interpreter supplies another `ARITH` module. A negation extension can include `ARITH` and add `neg`, with implementations including their base module. Old program functors still accept those modules; new programs require the larger signature. Extracting syntax for an arbitrary rewrite is easiest when one of the interpretations reifies an AST. The final interface does not make inspection free.

11.6 A genuinely separately compiled plugin

`projects/plugins` is a complete native dynamic-loading project:

- `plugin_api.ml` defines an extensible syntax, core evaluation/printing, parsing through the shared S-expression frontend, and a registration interface.
- `negate_plugin.ml` defines a new constructor and registers its builder, evaluation and printing handlers. It compiles to `negate_plugin.cmxs`.
- `host.ml` links the API and `Dynlink`, but does **not** link the plugin. It checks behavior before and after loading the compiled plugin.

Run `dune runtest projects/plugins`. This builds separate artifacts, loads the plugin, and evaluates `(+ 2 (neg (let x 3 (* x 4))))` to `-10`. Old constructors can contain the new one and vice versa. Core expressions can also be translated with the Chapter 6 fold, retaining their evaluation result.

The failure tests cover an unavailable syntax before loading, an unknown constructor lacking an operation, a missing plugin file, wrong extension arity, duplicate registration and repeated loading. Registration rejects reserved core names. Native plugins must match their host's compilation interfaces; dynamic loading is trusted code execution, not a sandbox. Runtime fallback is explicit: unknown operations raise `Missing_operation` instead of silently returning a fabricated value.

Adding a new operation still requires appropriate handlers for every extension one wants to support. This project meets separate syntax loading with declared failure behavior; it does not claim the static exhaustive solution to every expression-problem requirement.

11.7 Exercises

1. **Practice.** Substitute `x := y` under two nested binders named `y` and `y'`. Check the free-variable set and evaluate the result with an outer value for `y`.
2. **Proof.** Prove the free-variable formula for `Let`. Explain why removing its name from the value expression's free set is wrong.
3. **Experiment.** Give `many` a zero-consuming parser and then a consuming one. Explain why the progress check cannot diagnose every divergent parser.
4. **Project.** Add a separately compiled absolute-value plugin. Pass nested old/new syntax, print/parse round trips, wrong-arity and missing-handler tests.
5. **Proof / design.** For each representation in the table, identify which files change when adding a constructor and when adding an operation.

State whether recompilation, runtime failure, or interface changes are allowed in your goal.

Selected answer (2). $FV(\text{Let}(x,v,b)) = FV(v) \text{ union } (FV(b) \text{ minus } \{x\})$. In `let x = x in x`, the occurrence in the value is free; the occurrence in the body is bound. The executable `shadow` example above checks exactly that boundary.

Part III: Computation over time and choices

Chapter 7: Streams, demand, and sharing

Prerequisites: lists, folds and the cost discussion in Chapters 3 and 6.

Route: Part III begins here. Chapter 8 adds choice; Chapter 10 consumes a recorded input stream. Numerical analysis and pretty printing are optional projects.

A delayed computation is a promise to do work later. It is not a promise to do less work, or to retain less memory. We will separate four questions: what demands a value, whether repeated demands share work, what history remains reachable, and who closes a resource when demand stops early.

7.1 Call by name and call by need in a strict language

OCaml evaluates an argument to a value before entering a function. A closure is already a value, so `fun () -> work ()` delays `work`; calling the closure twice runs `work` twice. A lazy value `lazy (work ())` instead caches the result when forced. This models call by need in this single-threaded example.

```
let calls = ref 0
let work () = incr calls; 42
let thunk = fun () -> work ()
let shared = lazy (work ())
let () =
  assert (!calls = 0);
  ignore (thunk ()); ignore (thunk ());
  assert (!calls = 2);
```




Figure 9.1: A camel opens a sluice to draw one cup of water

```
ignore (Lazy.force shared); ignore (Lazy.force shared);
assert (!calls = 3)
```

Passing an OCaml reference cell still passes a value. A callee can mutate that shared cell, but cannot rebind the caller's variable. This differs from a call-by-reference parameter that aliases the variable itself. Likewise, normal order in Chapter 4 reduces under lambdas; a thunk here only runs when called.

7.2 Streams expose demand

A stream has an immediate head and a delayed tail:

```
type 'a stream = Nil | Cons of 'a * (unit -> 'a stream)
let rec from n = Cons (n, fun () -> from (n + 1))
let rec map f = function
  | Nil -> Nil
  | Cons (x, tail) -> Cons (f x, fun () -> map f (tail ()))
let rec take n = function
  | _ when n <= 0 -> []
  | Nil -> []
```

```

| Cons (x, _) when n = 1 -> [x]
| Cons (x, tail) -> x :: take (n - 1) (tail ())
let () =
  assert (take 4 (map (( * ) 2) (from 0)) = [0;2;4;6]);
  assert (take 1 (Cons (7, fun () -> failwith "not demanded")) =
    ⇨ [7])

```

The `n = 1` case matters: a consumer of one element must not demand a second one. However, the head is strict. Constructing `map f s` applies `f` to the first head immediately. A fully suspended sequence would put the first node behind a thunk too, as `Seq.t` does. Demand contracts belong to representations and operations, not to the word “stream”.

A stream may be finite, infinite, or fail while producing a tail. Arithmetic in `from` uses bounded machine integers; “infinite” describes how long it can keep producing nodes, not an unbounded integer representation.

7.3 Sharing saves recomputation and retains history

```

type 'a lazy_list = End | More of 'a * 'a lazy_list Lazy.t
let rec memoize = function
  | Nil -> End
  | Cons (x, tail) -> More (x, lazy (memoize (tail ())))
let rec take_lazy n = function
  | _ when n <= 0 -> []
  | End -> []
  | More (x, _) when n = 1 -> [x]
  | More (x, tail) -> x :: take_lazy (n - 1) (Lazy.force tail)
let () =
  let calls = ref 0 in
  let s = Cons (1, fun () -> incr calls; Cons (2, fun () -> Nil))
    ⇨ in
  ignore (take 2 s); ignore (take 2 s);
  assert (!calls = 2);
  let cached = memoize s in
  ignore (take_lazy 2 cached); ignore (take_lazy 2 cached);
  assert (!calls = 3)

```

Keeping `cached` retains every forced node reachable through its memoized tails. A consumer that advances and drops old roots may allow old nodes to be collected; a closure that still captures the original head prevents that. Conversely, a nonmemoized stream can recompute a prefix each time it is revisited. Neither representation is uniformly faster or smaller.

Representation	Repeated tail demand	Potential retained data
Thunk tail	Recomputes, or repeats effects	Whatever the closure captures
Lazy tail	Shares the result (or cached failure)	Forced prefix reachable from old roots
Mutable reader	Advances one external cursor	Open resource until its owner closes it

A self-referential definition must be *productive*: each requested node needs to be produced after finite work. `lazy` (`Lazy.force itself`) is not made productive by the `lazy` keyword. A recurrence with a known first node and delayed recursion can be productive, but sharing and arithmetic costs still need analysis.

7.4 Resource ownership is a scope

An EOF-only close leaks the resource when the consumer takes one line and stops, or raises before EOF. Returning a lazy list does not tell us when that consumer is finished. Instead make a callback own the whole reading scope:

```
let with_lines filename consume =
  Scoped.with_lines filename consume
```

Scoped is the compiled module in `projects/streams/scoped.ml`. Its implementation is short enough to inspect in full:

```
(* The callback owns the resource scope; an escaped reader is
   ↪ invalidated. *)
let with_source ~acquire ~read ~close consume =
  let resource = acquire () in
  let active = ref true in
  let next () =
    if not !active then invalid_arg "reader used outside its
    ↪ scope";
    read resource in
  Fun.protect
    ~finally:(fun () -> active := false; close resource)
    (fun () -> consume next)

let with_lines filename consume =
```

```

with_source
  ~acquire:(fun () -> open_in filename)
  ~read:(fun ch -> try Some (input_line ch) with End_of_file ->
    ↪ None)
  ~close:close_in_noerr consume

```

`Fun.protect` closes the source on success, early return, or exception. It also marks an escaped `next` function inactive, so an attempted read after the callback returns raises an error. OCaml's type here does not prevent escape statically; the runtime check enforces the scope. Read values such as strings may safely escape. `close_in_noerr` avoids replacing a consumer failure with a close failure. For a general resource whose close can fail, choose and document an error policy.

The project tests an early stop, a consumer exception, an escaped reader, and an actual file read. Reading twice from the same cursor is intentionally different from forcing the same memoized lazy node twice.

7.5 Separate formal coefficients from approximation

A stream of coefficients is a representation of a formal series. A finite prefix can be manipulated without claiming that its infinite sum converges anywhere. To evaluate numerically, specify the domain, truncation rule and error criterion. Small coefficients alone prove nothing about an unseen tail: at $x=1$, the polynomial $1 + x^{100}$ has a long stretch of unchanged partial sums before its answer changes from one to two.

The numerical project, `projects/numerical/README.md`, provides finite polynomial operations, a requested-length formal quotient with explicit zero-denominator rules, and a fixed-degree approximation of `exp x` on $[0,1]$. It derives a real truncation bound and distinguishes it from floating-point roundoff. It contains no function called `exact` that merely checks repeated rounded values.

The larger pipe-based pretty-printer and circular-list construction are preserved in `projects/pipes/README.md`. Use them to investigate buffering and delayed references after the demand model above is familiar.

7.6 Exercises

1. **Practice.** Instrument both `map` and `take`. Count the calls for requesting zero, one and three outputs, including the work performed during construction.

2. **Proof.** State `take`'s demand contract and prove that a request for `n` > 0 nodes demands at most `n-1` tails, or fewer if the stream ends.
3. **Experiment.** Traverse a memoized prefix twice. Then keep only an advanced tail. Explain which old nodes are unreachable; do not infer garbage collection from a wall-clock speed comparison.
4. **Practice.** Change the reader callback to raise after its first read and verify one acquisition and one close. The selected solution is the exception-path check in `projects/streams/laws.ml`.
5. **Project.** Implement an approximation with a domain and remainder estimate, following the numerical project's acceptance criteria. Explicitly account for rounding or limit the claim to a real-arithmetic truncation bound.

Chapter 8: Choices and their interpreters

Prerequisites: Chapters 5–7: modules, folds, finite search and delayed work. **Route:** Part III. Chapter 9 changes how operations are represented, using OCaml effects; the shared probability project spans both chapters.

A search program asks for alternatives and rejects some results. Does it want all answers, the first successful answer, or only their count? Begin with those operations, then choose their interpretation. A type signature alone does not make different interpretations equivalent.

8.1 A small language of search

```
type 'a t = Return of 'a | Fail | Choice of 'a t list
let return x = Return x
let rec bind m f = match m with
  | Return x -> f x
  | Fail -> Fail
  | Choice branches -> Choice (List.map (fun m -> bind m f)
    ↪ branches)
let ( let* ) = bind
let choose xs = Choice (List.map return xs)
let guard b = if b then return () else Fail
```

`Return` supplies an answer, `Fail` supplies none, and `Choice` records alternatives. `bind m f` replaces each successful leaf of `m` by the next search `f x`. The syntax is a finite tree; it eagerly constructs branches. We make no fairness claim for an infinite or diverging branch. A delayed representation could change that demand policy, as Chapter 7 suggests.



Figure 10.1: Chapter 8 illustration

```
let pairs target =
  let* x = choose [1;2;3] in
  let* y = choose [1;2;3] in
  let* () = guard (x + y = target) in
  return (x,y)
```

Read `let*` as sequencing through the language's `bind`. It is a binding operator, not built-in backtracking. Changing the definition of `let*` changes how the right-hand computation supplies values to the body. Ordinary `let` continues to mean ordinary OCaml evaluation.

8.2 Three interpretations of the same tree

```
let rec all = function
  | Return x -> [x]
  | Fail -> []
  | Choice branches -> List.concat_map all branches

let rec first = function
  | Return x -> Some x
  | Fail -> None
```

```

| Choice branches ->
  let rec loop = function
    | [] -> None
    | m::ms -> match first m with None -> loop ms | answer ->
  ↪ answer in
    loop branches

let rec count = function
| Return _ -> 1
| Fail -> 0
| Choice branches -> List.fold_left (fun n m -> n + count m) 0
  ↪ branches

```

```

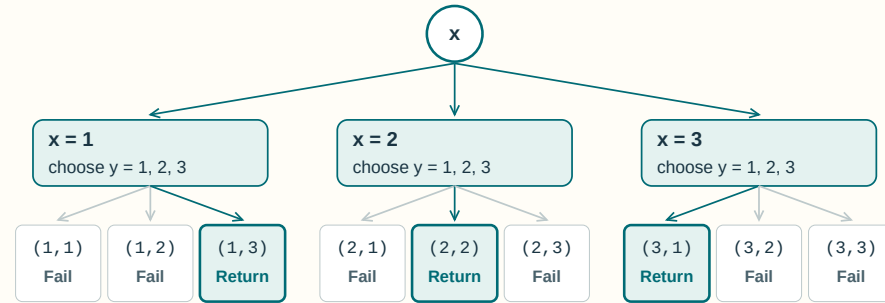
let () =
  assert (Search.all (Search.pairs 4) = [1,3;2,2;3,1]);
  assert (Search.first (Search.pairs 4) = Some (1,3));
  assert (Search.count (Search.pairs 4) = 3);
  assert (Search.first (Search.pairs 9) = None)

```

These interpreters are folds over the search representation. `all` preserves the left-to-right order and multiplicity of leaves. `first` chooses the first *successful leaf after downstream failures*. `count` counts leaves, including repeated equal answers; its machine integer can overflow on a very large tree. The agreement laws for finite searches are `first m = List.nth_opt (all m) 0` and `count m = List.length (all m)`, when that count fits. These are weaker than saying the three results are identical.

One search tree, three observations

pairs 4: choose x, choose y, then keep $x + y = 4$.



Read the surviving leaves from left to right:

all
[(1,3); (2,2); (3,1)]

first
Some (1,3)

count
3

Figure 10.2: The nine leaves of pairs 4, with three successes. All, first, and count observe the same tree in different ways.

The source and tests are `projects/choices/search.ml` and `laws.ml`. The Honey Islands project also expresses its choices through a module interface, so its direct and monadic implementations can share a reference specification.

8.3 Monad laws say how sequencing associates

A monad interface supplies a type constructor `'a t`, `return`, and `bind`:

```
module type MONAD = sig
  type 'a t
  val return : 'a -> 'a t
  val bind : 'a t -> ('a -> 'b t) -> 'b t
end
```

For the chosen notion of observational equality, the laws are:

- `bind (return x) f = f x` (left identity).
- `bind m return = m` (right identity).

- `bind (bind m f) g = bind m (fun x -> bind (f x) g)` (associativity).

For our search language, compare **all** results as lists: order and duplicates matter. Some syntactically different trees then count as equal. The laws follow by induction on the tree and list concatenation's associativity. They are not claims about identical allocation, termination on infinite structures, or equal performance. `projects/choices/laws.ml` gives bounded executable instances.

Options, lists and state computations can all support lawful sequencing, but they do different things. `Option.bind None f` does not run `f`; list `bind` runs it for each element. A state computation receives an initial state and returns a result together with a new state. The interface organizes composition without identifying these behaviors.

8.4 Choice needs additional laws

Let **zero** be failure and **plus** combine searches. List interpretation uses `[]` and `(@)`. It satisfies associative choice with failure as identity and, for finite pure computations, left distribution through `bind`:

```
bind (plus a b) f = plus (bind a f) (bind b f)
bind zero f = zero
bind m (fun _ -> zero) = zero
```

These equations justify exploring alternatives and then applying a common continuation. They do not follow from monad laws plus monoid laws alone. Left-biased option has a lawful monad and associative choice, but fails the first:

```
let plus a b = match a with Some _ -> a | None -> b
let f x = if x = 2 then Some x else None
let () =
  let left = Option.bind (plus (Some 1) (Some 2)) f in
  let right = plus (Option.bind (Some 1) f) (Option.bind (Some 2)
    ↪ f) in
  assert (left = None && right = Some 2)
```

Committing to `Some 1` discarded the alternative before `f` rejected it. In contrast, `Search.first` sees the completed search tree and can try the second branch. This is why a `MONAD_PLUS` signature is not enough to certify a backtracking algorithm.

Other tempting laws also depend on equality. List choice is not commutative or idempotent when order and multiplicity are observed. Right distribution can reorder results: branching inside each value produces a different order from collecting one branch's answers and then the other's. A set interpreter may forget those differences; a probability interpreter may need their multiplicities. State each law the optimization actually uses.

8.5 A concrete reason to combine effects

Suppose a branch increments a counter and then fails. Should the alternative branch see the increment? There are two useful answers:

- **Branch-local state:** a failed branch rolls back its state. Represent a computation as `state -> (answer * state) list`, conventionally `StateT(List)`.
- **Shared execution state:** exploration updates one state even when a branch produces no answers. A possible representation is `state -> answer list * state`; its sequencing and traversal order must be specified, not inferred from its type.

Here is the first semantics in a small example:

```
type 'a local = int -> ('a * int) list
let return x s = [x,s]
let bind m f s = List.concat_map (fun (x,s') -> f x s') (m s)
let ( let* ) = bind
let get s = [s,s]
let put s _ = [(),s]
let fail _ = []
let plus a b s = a s @ b s
let losing = let* () = put 1 in fail
let () = assert (plus losing get 0 = [0,0])
```

Both alternatives receive the original state zero. Compare an explicitly shared state interpretation of that same policy question:

```
let shared_counter = ref 0
let losing_shared () = shared_counter := 1; []
let alternative_shared () = [!shared_counter]
let () =
  let a = losing_shared () in
  let b = alternative_shared () in
```

```
assert (a @ b = [1])
```

A transformer is useful once this semantic choice is concrete. It systematically builds one interface from another; it does not make the order irrelevant. The Honey Islands transformer uses branch-local removal lists, and its tests compare all resulting removal sets against exhaustive subsets. Accidental shared state would change that solver’s meaning.

8.6 Probability is weighted choice

A finite distribution assigns nonnegative, finite weights to outcomes. To sample, the support must contain positive total mass; normalize the weights before selecting an interval. Zero-weight outcomes must never be selected. Conditioning multiplies weights by likelihoods and normalizes at the end; impossible evidence has zero total mass and cannot yield a posterior distribution.

For a fair Boolean `b`, observing likelihood 0.8 if true and 0.2 otherwise gives unnormalized masses 0.4 and 0.1, hence posterior $P(b=\text{true})=0.8$. A subsequent irrelevant fair draw must not change that posterior. Recounting the observation on each replay would incorrectly strengthen the evidence.

`projects/probability/README.md` uses the same tiny models with finite enumeration, likelihood weighting, and replay with or without resampling. “Exact enumeration” means the entire finite support is explored; its numerical weights are still floats. It is a reference for small models, not a cure for underflow or a proof that a sampling estimator is consistent. The larger sensor-fusion application uses the same compiled inference module.

8.7 Exercises

1. **Practice.** Add an interpreter returning the last successful result. State its relation to `all`, including the empty case, and check it on `pairs`.
2. **Proof.** Prove left distribution for list bind. Exhibit the order difference in a proposed right-distribution law using two inputs and two outcomes.
3. **Experiment.** Run the option counterexample after changing `f` so both inputs succeed. Explain why that passing case does not establish the law.
4. **Practice.** Count *distinct* answers by interpreting to a set. Give a search where this differs from `count`; specify a comparison function for the answers.

5. **Project.** Port a new Honey Islands pruning rule through its direct and monadic implementations, preserving its exhaustive reference comparisons.
6. **Proof / experiment.** Derive the posterior above by hand, then compare the inference project outputs. Explain why another draw must not square the likelihood, and why a fixed-seed test is weaker than a statistical theorem.

Selected answer (2). With inputs $[1;2]$, branches $\mathbf{f} \ x = [x]$ and $\mathbf{g} \ x = [10*x]$, branching inside bind gives $[1;10;2;20]$; concatenating the two bound computations gives $[1;2;10;20]$. They are equal as multisets, not as lists.

Chapter 9: Effects, ownership, and cancellation

Prerequisites: Chapter 3’s continuations, Chapter 5’s interfaces, and Chapter 8’s interpreters. **Route:** Part III. The executable project is `projects/effects`; probability shares `projects/probability` with Chapter 8.

An effect operation transfers control to a handler. The handler receives a continuation: the suspended rest of the computation. It can resume that continuation or discontinue it with an exception. Once suspension is possible, “who owns the continuation?” becomes as important as “what value does it return?”. We establish that ownership before building a scheduler.

9.1 Operations have result types

An extensible GADT describes the type of each operation’s result:

```
type _ Effect.t += Ask : string Effect.t
let greeting () = "Hello, " ^ Effect.perform Ask
let answer () =
  match greeting () with
  | text -> text
  | effect Ask, k -> Effect.Deep.continue k "reader"
let () = assert (answer () = "Hello, reader")
```

`Ask` returns a string, so its continuation expects a string. By contrast a `Yield : unit Effect.t` operation expects `()`. A GADT constructor refines its result index; it is not merely a tag in an untyped message channel.

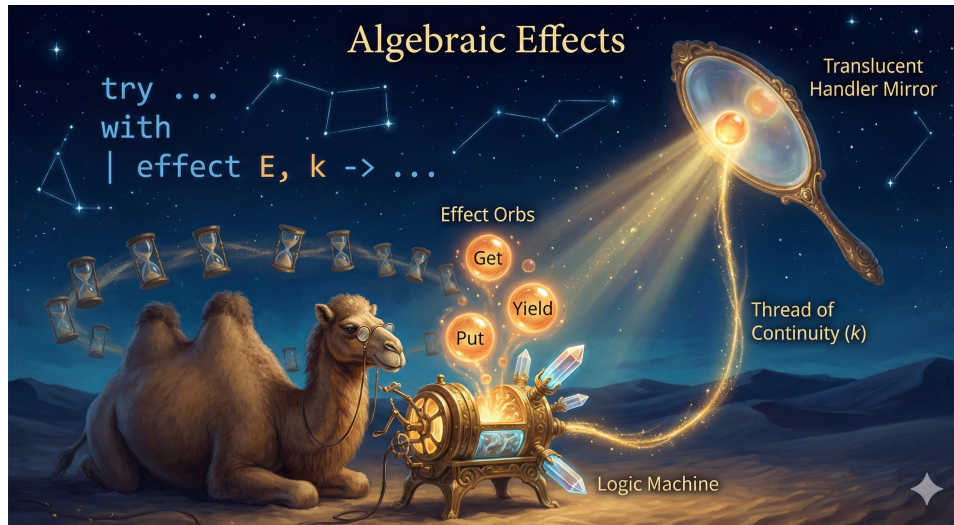


Figure 11.1: Chapter 9 illustration

Chapter 11 uses the same idea to index expression syntax by its evaluation result.

The syntax here requires OCaml 5.3 or later. The [OCaml effect-handler reference](#) describes deep handlers, one-shot continuations and discontinuation. A deep handler remains installed when its continuation resumes, so later operations can be handled by the same interpretation.

9.2 Consume a continuation exactly once

A captured continuation is one-shot. Calling `continue` twice, or calling `discontinue` after continuing, is an error. An abandoned continuation can retain resources and skip cleanup that would have run during stack unwinding. A handler must therefore own it until either a resumption or a discontinuation consumes it.

```
type _ Effect.t += Pause : unit Effect.t
exception Stop
let saved : (unit, unit) Effect.Deep.continuation option ref = ref
  ↪ None
let released = ref 0
```

```

let () =
  (match Fun.protect ~finally:(fun () -> incr released)
    (fun () -> Effect.perform Pause) with
  | () -> ()
  | effect Pause, k -> saved := Some k);
  assert (!released = 0);
  let k = Option.get !saved in
  saved := None;  (* Consume the ownership slot before transferring
  ↪ control. *)
  (try Effect.Deep.discontinue k Stop with Stop -> ());
  assert (!released = 1)

```

The resource is released on discontinuation because the exception travels through the suspended `Fun.protect`. Removing the only reference to `k` without that step would not express this cleanup policy. In a real resource scope, acquire before the protected computation, and decide how a cleanup failure interacts with an earlier exception.

Nesting also matters. The innermost matching handler handles an operation; an unmatched operation can propagate outward. A task created by one scheduler must not be awaited or cancelled through another scheduler's queue. Our handles carry an owner identity to reject that mistake, while nested independent runs work.

9.3 Define the scheduling policy first

Our teaching runtime has these explicit rules:

Event	Policy
Spawn	Enqueue a new child; return its handle without running the child
Yield	Put the current continuation at the back of the FIFO ready queue
Await	Suspend until the target finishes; propagate its result or exception
Cancel	Discontinue suspended work with Cancelled ; never start a new cancelled child
Child failure	Stop the run, cancel unfinished tasks, then propagate the first failure
Root completion	Cancel unfinished children before returning

Event	Policy
No runnable tasks with unfinished root	Raise Deadlock and release suspended work
Nested run	Own a separate queue and reject handles from another run

Tasks cooperate: a computation that never performs a scheduler operation prevents others from running. Cancellation exceptions must not be swallowed indefinitely, and cleanup functions must not suspend. These are program contracts, not facts proved by the interface. This is concurrency on one domain: tasks interleave. Parallel execution would run work simultaneously on multiple domains and needs synchronization policies absent from this runtime.

```

let events = ref []
let worker name () =
  events := (name ^ "1") :: !events;
  Runtime.yield ();
  events := (name ^ "2") :: !events
let () =
  Runtime.run (fun () ->
    let a = Runtime.spawn (worker "A") in
    let b = Runtime.spawn (worker "B") in
    Runtime.await a;
    Runtime.await b);
  assert (List.rev !events = ["A1"; "B1"; "A2"; "B2"])

```

There are no timing assumptions in this test. The trace follows from queue order: spawning enqueues A and B, awaiting suspends the parent, A yields behind B, and B yields behind A. The scheduler's driver owns dequeuing; spawning never recursively runs a child to completion.

9.4 The runtime and its ownership invariant

The public interface is in `projects/effects/runtime.mli`. Here is the complete implementation, so the cancellation paths are reviewable alongside normal resume:

```

open Effect
open Effect.Deep

```

```

exception Cancelled
exception Deadlock

type task = {
  owner : int;
  mutable state : state;
  mutable cancelled : bool;
  mutable queued : bool;
  mutable waiters : task list;
}
and state =
  | New of (unit -> unit)
  | Running
  | Paused of suspension
  | Finished of (unit, exn) result
and suspension = { resume : unit -> unit; abort : exn -> unit }

type _ Effect.t +=
  | Spawn : (unit -> unit) -> task Effect.t
  | Yield : unit Effect.t
  | Await : task -> unit Effect.t
  | Cancel : task -> unit Effect.t

let spawn f = perform (Spawn f)
let yield () = perform Yield
let await task = perform (Await task)
let cancel task = perform (Cancel task)
let next_owner = ref 0

let run main =
  incr next_owner;
  let owner = !next_owner in
  let queue = Queue.create () and tasks = ref [] and failure = ref
    ↪ None in
  let enqueue task =
    match task.state with
    | Finished _ | Running -> ()
    | New _ | Paused _ ->
      if not task.queued then (task.queued <- true; Queue.add task
        ↪ queue) in
  let create f =
    let task = {owner; state=New f; cancelled=false; queued=false;
      ↪ waiters=[]} in
    tasks := task :: !tasks; enqueue task; task in

```

```

let finish task result =
  task.state <- Finished result;
  (match result with
   | Error Cancelled | Ok () -> ()
   | Error exn -> if !failure = None then failure := Some exn);
  List.iter enqueue (List.rev task.waiters);
  task.waiters <- [] in
let rec stop task =
  task.cancelled <- true;
  match task.state with
  | Finished _ -> ()
  | New _ -> finish task (Error Cancelled)
  | Running -> () (* Delivered at the next scheduler operation.
↳ *)
  | Paused continuation ->
    task.state <- Running; (* Consume the ownership slot before
↳ resuming. *)
    continuation.abort Cancelled
and start task f =
  match_with f () {
    retc = (fun () -> finish task (Ok ()));
    exnc = (fun exn -> finish task (Error exn));
    effc = (fun (type a) (operation : a Effect.t) ->
      match operation with
      | Yield -> Some (fun (k : (a, unit) continuation) ->
        if task.cancelled then discontinue k Cancelled
        else begin
          task.state <- Paused {
            resume=(fun () -> continue k ());
            abort=(fun exn -> discontinue k exn)};
          enqueue task
        end)
      | Spawn f -> Some (fun (k : (a, unit) continuation) ->
        if task.cancelled then discontinue k Cancelled
        else let child = create f in continue k child)
      | Cancel target -> Some (fun (k : (a, unit) continuation)
↳ ->
        if target.owner <> owner then
          discontinue k (Invalid_argument "task belongs to
↳ another run")
        else if target == task || task.cancelled then begin
          task.cancelled <- true; discontinue k Cancelled
        end else (stop target; continue k ()))
      | Await target -> Some (fun (k : (a, unit) continuation) ->
        if target.owner <> owner then

```

```

        discontinue k (Invalid_argument "task belongs to
↳ another run")
      else if task.cancelled then discontinue k Cancelled
    else
      let resume () = match target.state with
        | Finished (Ok ()) -> continue k ()
        | Finished (Error exn) -> discontinue k exn
        | _ -> failwith "scheduler resumed an unfinished
↳ await" in
      (match target.state with
        | Finished _ -> resume ()
        | _ ->
          task.state <- Paused {resume; abort=(fun exn ->
↳ discontinue k exn)};
          target.waiters <- task :: target.waiters))
      | _ -> None)
    }
  in
  let root = create main in
  let rec drive () =
    match !failure, root.state with
    | Some _, _ | _, Finished _ -> ()
    | None, _ when Queue.is_empty queue -> failure := Some Deadlock
    | None, _ ->
      let task = Queue.take queue in
      task.queued <- false;
      (match task.state with
        | New f -> task.state <- Running; start task f
        | Paused continuation -> task.state <- Running;
↳ continuation.resume ()
        | Running | Finished _ -> ());
      drive () in
    (* Scope exit cancels children, including blocked awaiters. *)
    Fun.protect ~finally:(fun () -> List.iter stop !tasks) drive;
    match !failure, root.state with
    | Some exn, _ -> raise exn
    | None, Finished (Ok ()) -> ()
    | None, Finished (Error exn) -> raise exn
    | _ -> raise Deadlock

```

A task is new, running, paused, or finished. Only Paused owns a continuation. Both the driver and `stop` change the state to **Running** **before** invoking its resumption or abort closure. Thus a stale queue entry cannot consume that same slot again. A finished task's stale entries do nothing. The `queued` bit prevents duplicate enqueueing while awaiters are awakened.

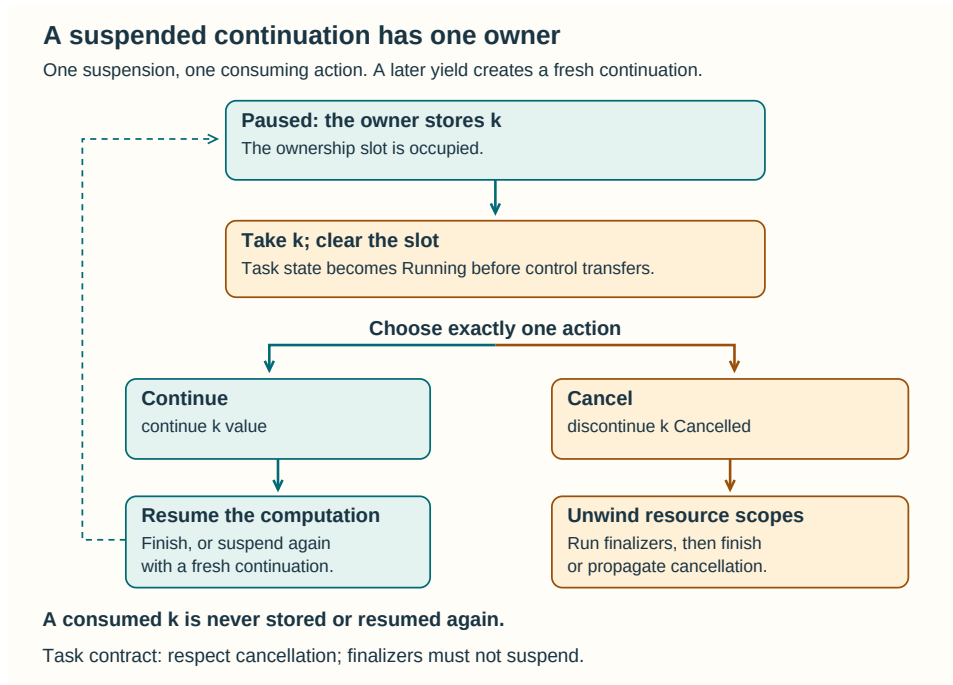


Figure 11.2: The owner clears its slot before continuing or cancelling a suspended computation. A later suspension supplies a fresh continuation.

An await suspension stores a resume closure that checks the target’s final result. It does not guess that waking means success. A cancelled awaiter can remain temporarily in its target’s waiter list, but enqueueing a finished task has no effect; scope exit clears the remaining lists while finishing the tasks.

The finalizer owns the entire run’s unfinished children. A child that never started acquired nothing; a paused child is discontinued to unwind its dynamic resource scopes. If a child failed, its exception is recorded before siblings are cancelled, preserving that original failure. The tests cover these separate paths rather than only a happy scheduling trace.

9.5 A monadic program over the same operations

Chapter 8 represented a computation as data. We can do that here too:

```

(* A monadic syntax for the same operations, interpreted by the
   ↪ teaching runtime.
   Bind builds a program; it does not run an action during
   ↪ construction. *)
type 'a t =
  | Return : 'a -> 'a t
  | Bind : 'b t * ('b -> 'a t) -> 'a t
  | Action : (unit -> 'a) -> 'a t
  | Yield : unit t
  | Spawn : unit t -> Runtime.task t
  | Await : Runtime.task -> unit t
  | Cancel : Runtime.task -> unit t
  | Protect : 'a t * (unit -> unit) -> 'a t

let return x = Return x
let ( let* ) m f = Bind (m,f)
let rec interpret : type a. a t -> a = function
  | Return x -> x
  | Bind (m,f) -> let x = interpret m in interpret (f x)
  | Action f -> f ()
  | Yield -> Runtime.yield ()
  | Spawn m -> Runtime.spawn (fun () -> interpret m)
  | Await task -> Runtime.await task
  | Cancel task -> Runtime.cancel task
  | Protect (m, release) -> Fun.protect ~finally:release (fun () ->
    ↪ interpret m)
let run m = Runtime.run (fun () -> interpret m)

```

Action delays host work; constructing a Bind does not execute it. Protect records a cleanup scope. The interpreter folds the syntax into the direct-style runtime. This shares the scheduler policy, so it tests equivalence of two program representations, not independence of two scheduler implementations.

The same test functor in `projects/effects/laws.ml` runs both representations. It asserts the A/B trace, cleanup after scope exit, explicit repeated cancellation, no acquisition for a cancelled new child, child-failure propagation and nested runs. Additional tests reject foreign handles and clean up a deadlocked await.

The explicit syntax makes operations available for inspection and alternative interpretation. Direct style uses the host stack for continuations and makes ordinary function calls natural. Either representation still owes an ownership policy. A `let*` does not by itself make resource use safe, and an effect handler does not by itself make scheduling structured.

9.6 Inference is another interpretation

The probability project exposes typed `Choose`, `Gaussian` and `GObserve` operations. Its finite reference enumerator only accepts finite choices; a Gaussian draw cannot be enumerated as a finite support. Likelihood weighting samples each draw and multiplies likelihoods along that run. The replay filter records draws and restarts a pure model at the next choice boundary.

On replay, observations already accounted for must not be multiplied again. When resampling active particles, preserve their **total active mass**, including when other particles have already finished; resetting every active weight to one would change their mass relative to finished results. Zero-mass particles must not be resurrected. Paused continuations are discontinued before restarting.

```
let model () =
  let b = Probability.GProb.choose [false; true] in
  Probability.GProb.observe (if b then 0.8 else 0.2);
  ignore (Probability.GProb.choose [0;1]);
  b
let () =
  let exact = Probability.Enumerate.infer model in
  assert (abs_float (List.assoc true exact -. 0.8) < 1e-12)
```

`projects/probability/laws.ml` compares this model, an early-completion model, and a zero-mass model across enumeration, importance sampling and replay with and without resampling. Impossible evidence returns an empty result; it is not a posterior assigning equal probabilities to everything. Supports, weights and sample counts are validated before interpretation.

Replay has a stricter contract than ordinary effect handling: the model must be deterministic apart from its handled draws, terminate on each explored trace, and not catch the private exceptions used to pause or reject it. Arbitrary I/O, mutation observed across runs, or changed choice support can invalidate replay. Long traces and very small likelihoods need log-domain or otherwise stabilized weights; the current short-model float implementation does not solve underflow. The sensor-fusion executable is an application experiment, not a validated physical estimator.

9.7 Exercises

1. **Practice.** Change the A/B program so the parent yields after spawning only A. Predict the trace before running it; explain the queue after each operation.
2. **Proof.** Audit the continuation ownership invariant. List every transition out of `Paused` and explain why repeated cancellation cannot resume twice.
3. **Experiment.** Make a child raise after its first yield. Check both the propagated exception and the sibling's release count.
4. **Project.** Add a timeout expressed in logical scheduler ticks. Specify which side wins when completion and timeout happen at the same tick. Test the policy in both program representations without wall-clock sleeps.
5. **Proof / experiment.** Derive the early-completion posterior in the probability project. Explain why normalizing only active particles changes its answer.

Boundary of this project. There is no OS I/O polling, multicore synchronization, preemption, priority system or production cancellation protocol here. Adding those requires new contracts and tests. The complete project demonstrates a small, explicit scope policy; it is not an application runtime recommendation.

Chapter 10: One game, three interpretations

Prerequisites: Chapters 7–9: streams, interpreters and owned continuations. **Route:** the final chapter of Part III. Chapter 12 returns to the laws connecting representations. Zipper navigation and rewriting are in `projects/zippers`.

A moving ball on a screen is not a specification. Before comparing reactive libraries, define one transition, a logical clock, an input log and its expected output. Then ask whether streams, incremental signals and direct-style effects interpret that same program. Drawing becomes a consumer of the verified states.

10.1 Define a discrete world

The playfield has integer x-coordinates 0–10 and y-coordinates 0–6. The ball moves one cell per axis per tick. A paddle centered at `paddle` covers its center and one cell on either side. Each input moves it by at most one cell, clamped to centers 1–9. Move the paddle first, then move the ball, then resolve collisions. A ball arriving at `y=0` bounces if within the paddle; otherwise the game is lost. At `x=0` or `x=10` its horizontal velocity reverses; at `y=6` its vertical velocity reverses. Simultaneous collisions emit events in wall, ceiling, paddle/miss order.

```
type status = Playing | Lost
type state = { tick:int; x:int; y:int; vx:int; vy:int; paddle:int;
  ↪ status:status }
```



Figure 12.1: A camel studies three views of a tabletop paddle game

```

type input = { move:int }
type event = Paddle | Wall | Ceiling | Miss
let initial = {tick=0; x=5; y=2; vx=1; vy=(-1); paddle=5;
  ↪ status=Playing}
let clamp lo hi x = max lo (min hi x)
let step s input =
  if input.move < -1 || input.move > 1 then invalid_arg "move must
    ↪ be -1, 0 or 1";
  let paddle = clamp 1 9 (s.paddle + input.move) in
  let tick = s.tick + 1 in
  if s.status = Lost then {s with tick; paddle}, []
  else
    let x = s.x + s.vx and y = s.y + s.vy in
    let wall = x = 0 || x = 10 in
    let ceiling = y = 6 in
    let bottom = y = 0 in
    let hit = bottom && abs (x - paddle) <= 1 in
    let missed = bottom && not hit in
    let state = {tick; x; y; paddle;
      vx=(if wall then -s.vx else s.vx);
      vy=(if ceiling || hit then -s.vy else s.vy);
      status=(if missed then Lost else Playing)} in
    let events =

```

```

      (if wall then [Wall] else []) @ (if ceiling then [Ceiling]
↪ else []) @
      (if hit then [Paddle] else []) @ (if missed then [Miss] else
↪ []) in
      state, events

```

The transition's state precondition is a state reachable from `initial` by valid inputs. It is not a continuous-physics solver: there is no variable time step, subpixel velocity or swept collision detection. After loss, ticks and paddle inputs continue, while ball position and velocity remain fixed and no new collision event is emitted. The simple integer model makes the semantics exact until the machine tick counter overflows.

10.2 Establish the expected trace

The recorded paddle movements are `[1;1;0;-1;0;0;0;0;0;0;0;0]`. Starting at `(5,2)` with paddle 5, the first input moves the paddle to 6 and ball to `(6,1)`. The second moves the paddle to 7, so the ball hits it at `(7,0)` and reverses vertically.

Tick	Ball position	Event
1	(6,1)	none
2	(7,0)	paddle
3	(8,1)	none
4	(9,2)	none
5	(10,3)	wall
6	(9,4)	none
7	(8,5)	none
8	(7,6)	ceiling
9	(6,5)	none
10	(5,4)	none
11	(4,3)	none
12	(3,2)	none
13	(2,1)	none
14	(1,0)	miss

At the last tick the paddle is still centered at 6, so the ball misses. This trace exercises the paddle rule that a picture alone would not establish.

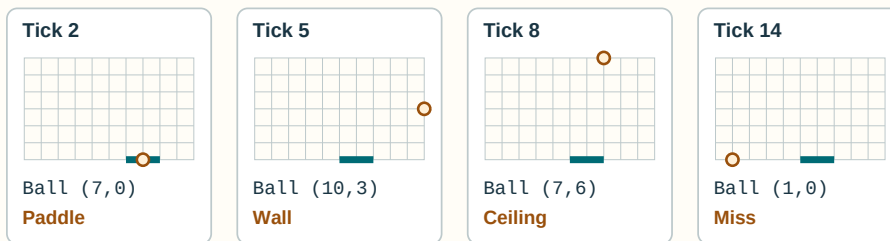
```

let trace = Game.through_stream Game.recorded
let () =
  assert (List.map (fun (s,_) -> s.Game.x,s.Game.y) trace =
    ↪ Game.expected_positions);
  assert (List.filter_map (fun (s,e) ->
    if e=[] then None else Some (s.Game.tick,e)) trace =
    [2,[Game.Paddle];5,[Game.Wall];8,[Game.Ceiling];14,
    ↪ ,[Game.Miss]])

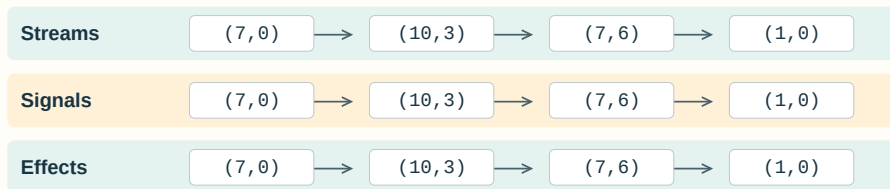
```

One recorded input, the same observable trace

Input moves: 1, 1, 0, -1, then ten zeros. Start: ball (5,2), paddle 5.



The same step function is used in each interpretation:



Selected event ticks are shown; the tests compare all fourteen complete states and event lists. Drawing is a consumer of the trace.

Figure 12.2: Four event ticks from the common game trace. Streams, signals, and effects produce the same positions and events.

10.3 A stream is a sequence of transitions

`Game.stream state inputs` yields the next `(state, events)` pair and delays the rest behind a `Seq.t` thunk. It applies `step` once per demanded node. It is not memoized: traversing the same sequence twice recomputes its pure transitions. The input list is immutable, so recomputation gives the same trace.

```
let () = assert (Game.through_stream Game.recorded = trace)
```

This interpretation is a scan: a fold that exposes each intermediate state. With a live reader, re-traversal would be a different contract; Chapter 7’s resource scope would still be required. A finite input log separates that concern from the game’s state semantics.

10.4 Incremental signals cache dependencies

The project’s **Signal** module implements a small static dependency graph. A variable has a revision; setting it to an equal value leaves that revision alone. A **map2** node samples its two dependencies and caches its output under their pair of revisions. Re-reading unchanged dependencies returns the cached value.

The game scan depends on **both logical tick and input value**. Holding the same movement key for two ticks must still move the ball twice. The driver delivers every consecutive tick and samples it before delivering the next; an attempted skip raises an error. The scan’s stateful update is confined to that node, while ordinary derived nodes can be pure functions of the resulting snapshot.

```
let () =
  let update,sample,cost = Game.incremental () in
  update {Game.move=1};
  let first = sample () in
  assert (sample () = first && cost () = 1);
  update {Game.move=1};
  ignore (sample ());
  assert (cost () = 2);
  assert (Game.through_incremental Game.recorded = trace)
```

The input value is unchanged on the second update, but the tick revision changes. This is the same issue an event edge detector faces: a rising edge belongs to one tick, not every future sampling of a cached **Some event**. The **rising_edge** example explicitly invalidates by tick. Repeated samples within one tick return the same occurrence, and an unchanged true input on the next tick produces none.

```
let () =
  let edge = Game.rising_edge () in
```

```

assert (edge ~tick:1 true);
assert (edge ~tick:1 true);
assert (not (edge ~tick:2 true));
assert (not (edge ~tick:3 false));
assert (edge ~tick:4 true)

```

Caching a result and consuming an event are separate operations. `Game.consumer` remembers the most recently consumed tick for one monotone consumer, returning no events on a second poll. Another consumer owns its own cursor. No global clearing of the event is needed, so one observer cannot silently steal it from another.

This small graph makes dependency invalidation visible. It omits dynamic graph rewiring, disposal of observers, scheduling priorities and equality cutoffs at derived nodes. The optional GUI laboratory describes what a port to a library must verify; it does not infer equivalence from similar APIs.

10.5 Direct-style scripts suspend for input

A direct-style script asks for an input, calls `step`, publishes the snapshot, and repeats. `Game.start` handles its input effect by owning the suspended continuation. `push` consumes that continuation and supplies one input; the script then publishes and suspends again. `close` discontinues the pending continuation and runs its resource finalizer. Pushing after close is an error; closing twice is harmless.

```

let () =
  let released = ref 0 and output = ref [] in
  let session = Game.start
    ~publish:(fun snapshot -> output := snapshot :: !output)
    ~release:(fun () -> incr released) in
  Fun.protect ~finally:session.close
    (fun () -> List.iter session.push Game.recorded);
  assert (List.rev !output = trace);
  assert (!released = 1);
  assert (Game.through_effects Game.recorded = trace)

```

The ownership transfer is the same as Chapter 9: remove the continuation from its slot *before* resuming it. If publication raises, stack unwinding releases the resource and closes the script. The tests cover that failure as well as normal closure and closure while waiting for another input.

10.6 Equivalence before drawing and timing

`projects/reactive/laws.ml` first checks the explicit trace above, then compares all three interpreters on 100 fixed-seed, 30-input traces. It also checks repeated sampling, repeated event consumption, unchanged Boolean edges, and abandoned or failed scripts. Equality includes every field of state and the ordered event list, not just ball coordinates.

The reason for agreement is simple enough to prove: each interpreter starts from `initial`, consumes each input once in order, and emits exactly the result of the same `step`. Induction on the input prefix establishes equal emitted traces. The incremental interpretation additionally needs its consecutive-tick contract; the effect interpretation needs successful publication and the stated ownership policy.

Only now connect a renderer. A renderer observes a snapshot; it must not advance physics just because the window redraws. A timer or input adapter determines logical ticks. If drawing takes longer than a tick, choose whether to queue, drop or coalesce inputs and state the resulting trace policy. None of those policies follows automatically from “reactive”.

The historical Bogue/Lwd/Incremental executable remains an optional laboratory in `projects/gui/README.md`, outside the maintained headless suite. It has not been certified as an implementation of this transition. No new GUI framework is required for the chapter’s behavioral comparisons.

10.7 Exercises

1. **Practice.** Record a trace in which the paddle misses at tick 2. Check all fields of the final state, not just its `Lost` status.
2. **Proof.** State the prefix invariant for the three interpreters and explain where the no-skipped-tick premise is used.
3. **Experiment.** Remove the tick dependency from the incremental scan and feed repeated equal inputs. Keep the failed trace as a regression test.
4. **Practice.** Create two independent event consumers. Check that each receives a paddle event once, regardless of the order in which they poll.
5. **Project.** Port the renderer to the shared transition. Pass the same recorded trace, verify cleanup when its window closes, and only then measure updates versus cached redraws. Specify a backlog policy for slow rendering.

Part IV: Mathematical synthesis

Chapter 12: Constructions and their laws

Prerequisites: Chapters 2, 3, 5, 6 and 11; induction on finite data and equality of functions at every argument. **Route:** Part IV. Optics and codensity are optional further study in `projects/optics/README.md`.

The previous chapters changed representations and checked what survived. We now prove a few general constructions: functor laws, the uniqueness of a fold, an adjunction and a polymorphic Yoneda representation. For each claim we name its objects, maps, equality and hypotheses. A type index can prevent mismatched endpoints; it cannot prove associativity or naturality by itself.

12.1 Choose a category and an equality

A category has objects, arrows between objects, identity arrows and composition of arrows with matching endpoints. Composition is associative and identities are units. In **Set**, objects are sets, arrows are total functions, and equality of arrows is extensional: two functions are equal when they agree at every input. For functions $f : A \rightarrow B$, $g : B \rightarrow C$, and $h : C \rightarrow D$, both bracketings of composition send a to $h (g (f a))$; either identity law reduces to $f a$.

Pure, terminating OCaml examples can illustrate this setting. General OCaml functions may diverge, mutate state or raise exceptions. Resource exhaustion is also ignored in this mathematical interpretation. To model a possibly missing lookup as a total Set arrow, return an explicit option/result value rather than silently treating a raised exception as an ordinary result.

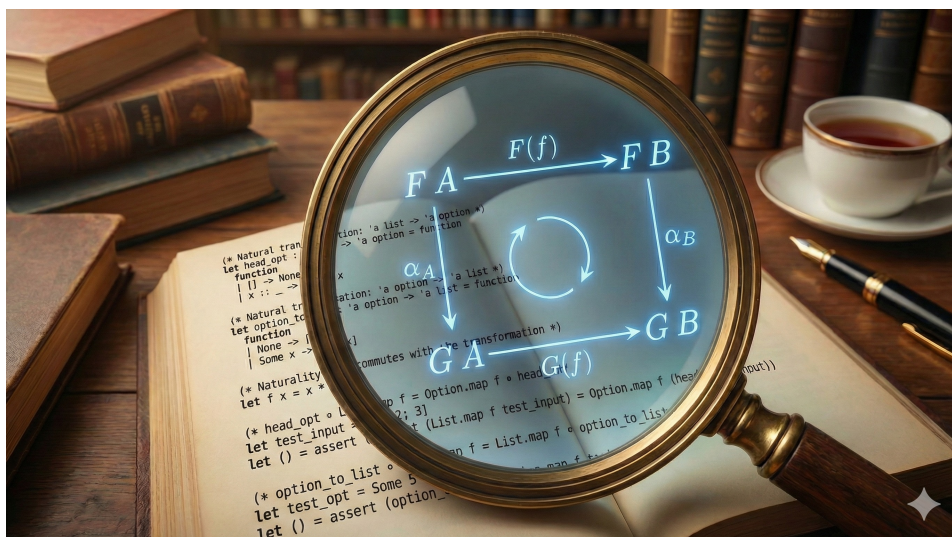


Figure 13.1: Chapter 12 illustration

```

module type CATEGORY = sig
  type ('a, 'b) hom
  val id : ('a, 'a) hom
  val compose : ('b, 'c) hom -> ('a, 'b) hom -> ('a, 'c) hom
end

module Functions = struct
  type ('a, 'b) hom = 'a -> 'b
  let id x = x
  let compose g f x = g (f x)
end

```

This interface checks that endpoints match. An implementation could still violate the laws; the interface is not a proof. Likewise, two OCaml function values cannot be compared extensionally by polymorphic `=`. We test selected arguments and prove the general equation separately.

A poset gives another category: objects are its elements and an arrow $a \rightarrow b$ exists exactly when $a \leq b$, with at most one such arrow. Reflexivity supplies identities and transitivity composition. A monoid gives a one-object category, with monoid elements as arrows, multiplication as composition and its unit as identity. The equality notion changes with the example: order witnesses in a poset are unique, whereas a monoid may have many distinct arrows.

12.2 List mapping is a functor

A functor maps objects and arrows, preserving identities and composition. Here `List : Set -> Set` maps a set **A** to its **finite** lists and a function **f** to pointwise list mapping. Its laws are

$$\text{map}(\text{id}) = \text{id}, \quad \text{map}(g \circ f) = \text{map}(g) \circ \text{map}(f).$$

Prove identity by induction. Mapping over `[]` returns `[]`. On `x::xs`, the mapped head is `x` and the mapped tail is `xs` by induction. For composition, both sides on `[]` are empty; on `x::xs`, both heads are `g (f x)` and the tails agree by induction. This proof needs total pure **f** and **g**: effect order is not part of this Set theorem.

```
let () =
  List.iter (fun xs ->
    assert (List.map Fun.id xs = xs);
    let f x = x + 1 and g x = 2 * x in
    assert (List.map (fun x -> g (f x)) xs = List.map g (List.map f
      ↪ xs)))
  [[]; [1]; [1;2;3]]
```

A rewrite that changes constructors is a fold into syntax, not this shape-preserving map. An OCaml *module functor* is a parameterized module; to call one a categorical functor requires an independently specified category and an action on arrows satisfying these laws. Similar names do not supply those data.

12.3 Why a fold is unique

Fix an element set **A** and the endofunctor $F(X) = 1 + A \times X$ on Set. An **F-algebra** consists of a carrier set **X**, a chosen $z : X$, and a function $c : A * X \rightarrow X$. An algebra morphism $h : (X, z, c) \rightarrow (Y, z', c')$ is a total function such that $h(z) = z'$ and $h(c(a, x)) = c'(a, h(x))$. Identity and composition preserve these equations, giving a category of F-algebras.

Finite lists, with `[]` and `(::)`, form an **initial** algebra: for any target algebra there is exactly one algebra morphism from lists to that target. The candidate is the fold:

```

let rec fold c z = function
| [] -> z
| x::xs -> c x (fold c z xs)

```

Existence follows from its defining equations: `fold c z [] = z` and `fold c z (x::xs) = c x (fold c z xs)`. To prove uniqueness, suppose `h` also satisfies those equations. At `[]`, `h [] = z = fold c z []`. At `x::xs`,

$$h(x :: xs) = c(x, h(xs)) = c(x, \text{fold}(c, z)(xs)) = \text{fold}(c, z)(x :: xs),$$

where the middle equality uses the induction hypothesis. Thus `h` and the `fold` are extensionally equal on every finite list. This is the universal property; a few tests of a recursive implementation would not establish uniqueness.

The expression `fold` has the same argument

For the common syntax, fix sets `K` of constant labels, `N` of names and `O` of operator labels. The syntax functor is

$$E(X) = K + N + (O \times X \times X) + (N \times X \times X).$$

Its four summands are number, variable, binary operation and `Let`. On arrows, `E(h)` applies `h` to each recursive child and leaves labels untouched. The carrier of its initial algebra is finite `Expr.t` syntax. The algebra fields in Chapter 6 implement the corresponding maps to an arbitrary carrier.

Existence is the four defining equations of `Expr.fold`. Uniqueness uses induction on the same four constructors: number and variable are base cases; for binary and binding forms use the hypotheses for both children. Names are literal labels here. We have **not** quotiented syntax by alpha equivalence, nor claimed that binding semantics itself is a polynomial functor on names.

The evaluation algebra in Chapter 6 uses a function carrier to handle binding. To apply this Set theorem literally, totalize lookup failure as an explicit result and treat arithmetic outcomes as values. The exception-raising executable is an operational illustration with a separately stated failure contract.

Fusion is a consequence, with a premise

If `h` preserves a source algebra's operations into a target algebra, then `h (fold source e) = fold target e`. Both sides are algebra morphisms from the initial syntax algebra; uniqueness gives equality. The preservation premise is essential: not every post-processing function can be pushed through a fold.

```

let () =
  let join xs = fold (^) "" xs in
  let total_length xs = fold (fun x n -> String.length x + n) 0 xs
    ↪ in
  List.iter (fun xs -> assert (String.length (join xs) =
    ↪ total_length xs))
    [[]; ["a"; "bc"]; [""; "hello"]]

```

Here `String.length` preserves the empty string and concatenation into zero and addition, for finite strings within machine limits. That is the exact premise which licenses fusion in this example.

12.4 Derive an adjunction

Fix a set B . Define $L(A) = A \times B$ and $R(C) = C^B$, the set of functions from B to C . On arrows, $L(u)(a, b) = (u(a), b)$ and $R(v)(k) = v \circ k$. These are functors Set to Set : substitution verifies both functor laws. The adjunction $L \dashv R$ is the natural family of bijections

$$\text{Set}(A \times B, C) \cong \text{Set}(A, C^B).$$

```

let curry f a b = f (a,b)
let uncurry g (a,b) = g a b

```

The first round trip is pointwise: $\text{uncurry} (\text{curry } f) (a, b) = \text{curry } f a b = f (a, b)$. The other is $\text{curry} (\text{uncurry } g) a b = \text{uncurry } g (a, b) = g a b$. Both equalities are extensional; no function comparison is used.

A bijection of hom-sets alone is not the whole claim: it must be natural in A and C . For $u : A' \rightarrow A$ and $v : C \rightarrow C'$, start with $f : A * B \rightarrow C$. Both ways around the naturality square evaluate at a' and b to $v (f (u a', b))$. Precomposition on the input and postcomposition on the output therefore commute with currying. This proves the required naturality.

```

let () =
  let f (a,b) = a + b and u a = 2*a and v c = string_of_int c in
  List.iter (fun a -> List.iter (fun b ->
    assert (uncurry (curry f) (a,b) = f (a,b));
    assert (curry (fun (a,b) -> v (f (u a,b))) a b = v (curry f (u
    ↪ a) b)))

```

```
[0;1;2]) [0;1;2]
```

A boundary-sensitive order example

View the mathematical reals and integers as poset categories ordered by $\leq$. Ceiling $c : \mathbb{R} \rightarrow \mathbb{Z}$ is left adjoint to the integer embedding $i : \mathbb{Z} \rightarrow \mathbb{R}$, because

$$c(x) \leq n \iff x \leq i(n).$$

If $c(x) \leq n$, then $x \leq c(x) \leq n$. Conversely, if $x \leq n$ and n is an integer, the least integer above x is no larger than n . Both maps are monotone. In thin poset categories, the hom-set bijection is exactly this equivalence of inequalities. Replacing ceiling by floor fails at $x=3.7, n=3$.

```
let () =
  List.iter (fun x -> List.iter (fun n ->
    assert ((int_of_float (Float.ceil x) <= n) = (x <=
      ↪ float_of_int n)))
    [-4;-3;0;3;4]) [-3.7;0.;3.7;4.]
```

This finite check uses finite floats and exactly represented small integer bounds. It is not a theorem about converting arbitrary floats, NaNs or out-of-range values into bounded OCaml integers.

12.5 Naturality and Yoneda, with both inverse laws

Let $F : \mathbf{Set} \rightarrow \mathbf{Set}$ be a functor and fix a set A . The functor $H_A(B) = \mathbf{Set}(A, B)$ acts on $g : B \rightarrow C$ by postcomposition: $H_A(g)(f) = g \circ f$. A natural transformation $\eta : H_A \Rightarrow F$ has a component $\eta_B : (A \rightarrow B) \rightarrow F(B)$ at every B , with

$$F(g)(\eta_B(f)) = \eta_C(g \circ f).$$

The covariant Yoneda bijection identifies these natural transformations with $F(A)$. We can construct and prove it directly:

- Given $x : F(A)$, define $\eta_B^x(f) = F(f)(x)$. Its naturality follows from $F(g)(F(f) x) = F(g \text{ composed with } f) x$, the composition law.
- Given a natural η , recover $x = \eta_A(\text{id}_A)$.

Recovering from the first construction gives $F(\text{id}_A)(x) = x$, by the identity law. For the other round trip, naturality at the arrow $f : A \rightarrow B$, applied to id_A , gives

$$F(f)(\eta_A(\text{id}_A)) = \eta_B(f \circ \text{id}_A) = \eta_B(f).$$

Thus every component is recovered on every input function. Equality of natural transformations means precisely that componentwise extensional equality. The proof used functor laws and naturality, not just a suggestive type abbreviation. For the general formulation in a locally small category, see Riehl, *Category Theory in Context*, Theorem 2.2.4.

The universal quantifier must be real

For the list functor, OCaml can encode the varying result type with a polymorphic record field:

```
type 'a yoneda = { run : 'b. ('a -> 'b) -> 'b list }
let to_yoneda xs = {run = fun f -> List.map f xs}
let from_yoneda phi = phi.run Fun.id
let () =
  let phi = to_yoneda [1;2;3] in
  assert (from_yoneda phi = [1;2;3]);
  assert (phi.run string_of_int = ["1";"2";"3"]);
  assert (phi.run (fun x -> x mod 2 = 0) = [false;true;false]);
  assert ((to_yoneda (from_yoneda phi)).run ((+) 1) = phi.run ((+)
    ↪ 1))
```

The same stored `phi` accepts both a string-producing and a Boolean-producing function. A monomorphic function with one fixed result type would not express this quantifier. Nevertheless, OCaml's type alone does not enforce naturality:

```
let unnatural = {run = fun f -> List.sort compare (List.map f
  ↪ [1;2])}
let () =
  let recovered = to_yoneda (from_yoneda unnatural) in
  assert (unnatural.run (fun x -> -x) = [-2;-1]);
  assert (recovered.run (fun x -> -x) = [-1;-2])
```

Polymorphic comparison inspects the result representation, breaking uniformity; it can also raise on function values. The reverse law therefore applies to natural families, for example those produced by `to_yoneda`, or to

a suitably restricted total parametric language with a justified parametricity theorem. It is false for all OCaml values of this record type.

For the identity functor, the same construction gives a polymorphic CPS value:

```
type 'a cps = { run_cps : 'b. ('a -> 'b) -> 'b }
let to_cps x = {run_cps = fun k -> k x}
let from_cps p = p.run_cps Fun.id
let () = assert (from_cps (to_cps 42) = 42)
```

This is the value-level representation behind the identity-continuation idea in Chapter 3. An arbitrary fixed-answer-type continuation $(A \rightarrow R) \rightarrow R$ has a different contract; it is not this universally quantified representation.

12.6 What carries back to the programs

Chapter 11’s plugin embedding satisfies an ordinary equation between functions: `eval_extended (embed e) = eval e`, on its stated domain and failure semantics. Its compiled test checks that equation. Calling it naturality would additionally require categories, functors and a family of embeddings; extensible variants do not supply those automatically.

Chapter 2’s context derivative also has a precise scope. For finite polynomial containers, differentiating with respect to the element parameter describes an element hole. For $T = 1 + aT^2$, $T' = T^2 + 2aTT'$ records the removed node’s children and its ancestor path. A subtree hole instead keeps just the path. See Abbott, Altenkirch, Ghani and McBride, *Derivatives of Containers*, for the container setting behind this calculation. It does not make every effectful stream or arbitrary OCaml datatype a polynomial container.

For Chapter 1, the pure simply typed product/function calculus can be interpreted in a cartesian closed category: products model pairs, a terminal object models unit, and exponentials model functions with evaluation and currying. Adding coproducts and an initial object models sums and the empty type. General recursion and effects need further semantics. A correspondence restricted to that fragment is useful precisely because its assumptions say when it applies.

12.7 Exercises

1. **Proof.** Prove `Option.map` preserves identities and composition. Specify the objects and arrows before writing the two constructor cases.
2. **Proof.** Complete the uniqueness proof for the four-constructor expression fold. Explain why it treats a binder name as a label, not as a quotient by renaming.
3. **Practice.** Test both currying round trips on a function returning a pair. Write the equalities pointwise; do not compare function values with `=`.
4. **Proof.** Reproduce the reverse Yoneda round trip, explicitly naming the naturality arrow and the argument to the component at `A`.
5. **Experiment.** Use `unnatural` to find a failed naturality square with integer negation. Selected answer: map negation after sorting `[1;2]` gives `[-1;-2]`, while sorting after mapping gives `[-2;-1]`.
6. **Project.** Continue with the optional optics route. Prove the three lens laws for a composed record lens before considering its higher-rank encoding.

The payoff is not a name for every program. It is a reusable proof: identify the representation, state its laws, and know which changes those laws justify.